

Highlights

Local skill, unstable portability: marginal diagnostics do not order the cross-region transfer of pre-fire thermal dryness in Mediterranean wildfire regions

Yunus Emre Cogurcu, Emrehan Metin

- Pre-fire thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- The same block adds +0.004 AUC across 20 transfer directions, interval spans zero
- Highest niche-overlap pair fails both ways; lowest transfers both ways (points)
- Of 20 diagnostics, only conditional direction agreement tracks transfer (ρ 0.84)
- Label-blind adaptation compresses transfer toward chance; only target labels help

Local skill, unstable portability: marginal diagnostics do not order the cross-region transfer of pre-fire thermal dryness in Mediterranean wildfire regions

Yunus Emre Cogurcu^{a,*}, Emrehan Metin^a

^a*Çukurova University, Adana, Türkiye*

Abstract

Pre-fire thermal dryness, measured by land surface temperature, its anomalies and dryness indices, separates a fire year from a normal year. Models built on it are rarely tested outside their training region.

Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation. Six pre-fire thermal predictors were added to a static and near-static terrain, fuel and greenness baseline. Within every region, ROC-AUC rose by +0.06 to +0.15, and the gain survived coarser blocks and an earlier predictor window.

The same predictors were then transferred between regions. Their mean contribution over twenty ordered directions was +0.004, with an interval spanning zero. Removing the two reversing predictors cost -0.081 of within-region skill, supported in every region, and changed mean transfer by +0.014, whose pair-clustered interval spans zero. The debit is measured and the credit is not.

The failure is conditional. Inside one study area, two fires eleven months

*Corresponding author.

Email address: `ycogurcu@cu.edu.tr` (Yunus Emre Cogurcu)

apart reversed the elevation-burning link, with disjoint bootstrap intervals. Season, year and population all differ there, so place alone is what the design holds fixed.

Twenty candidate diagnostics were rank-correlated against observed transfer. Only two had bootstrap intervals excluding zero, both conditional, the stronger being agreement in the sign of each predictor’s association (Spearman $\rho = 0.84$ over eight pairs). No marginal, niche-overlap or regime measure ordered the matrix; at the point estimate the highest-overlap pair failed both ways and the lowest transferred both ways. Label-blind adaptation by standardisation and CORAL pushed transfer towards chance in fourteen of the twenty directions. Niche and regime measures need labels in both regions too, so only the marginal family can run before deployment, and it fails.

Transfer skill therefore has to be measured, not inferred from similarity. What carries the information is conditional, so the price is target labels, not better unsupervised alignment: thirty-two labelled target blocks recovered 85 to 89 % of the ceiling in three of six directions and 30 to 57 % in the rest. *Keywords:* wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [30]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from
5 erature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [42, 15]. The domi-

nant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [4]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a cost that the pattern does not price. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also less portable, as we argue and then show. The relationship they carry is reparameterised locally. It exists everywhere, but it points in different directions in different places. The claim advanced here is a trade-off and not a dismissal: **dynamic state predictors buy local skill at the cost of portability.** Two measurements carry that claim. The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region and contributes nothing distinguishable from zero between them, with an unstable sign. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, while changing mean transfer by +0.014, an estimate whose own interval spans zero. Whether the block helps or harms on transfer is a property of the source-target pair, not of the block.

1.1. The trade-off goes unpriced because portability goes unmeasured

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, and often from the same fire event. Where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [34, 32] and is addressed

by spatially blocked cross-validation [40, 26]. Controlling autocorrelation makes the within-region estimate honest. It says nothing about whether the fitted relationship holds elsewhere.

35 Because only that side of the ledger is routinely reported, the trade-off is structurally invisible. A predictor block is adopted on the strength of the increment it delivers inside its training footprint. The portability it consumes is never entered as a debit. Yet any regional product built from locally trained models implicitly promises generalisation beyond that footprint.

40 For fire specifically, the transfer question is asked far less often than in adjacent fields. Meteorologically derived fire-danger indices are known not to port cleanly between fire environments [33]. Two recent studies test model transfer systematically, across Mediterranean countries [8] and US counties [17], and both report that it largely succeeds between similar regions. Both,
45 crucially, transfer models whose dominant predictors are *spatially stationary*: terrain, human modification, night-time lights, population density and long-term reanalysis climatologies. Each of these describes a place rather than a season, which is exactly the class our thesis expects to travel well. In species distribution modelling, transferability has been examined far more system-
50 atically [46]. Two studies there report that geographic and environmental similarity do *not* reliably predict transfer success [41, 35].

1.2. The pre-fire thermal block is the natural test case

The dynamic predictor class used here is pre-fire thermal dryness. It is under-exploited relative to static terrain and fuel, and it is the class most
55 plausibly *expected* to transfer. The standard susceptibility predictors are terrain, land cover or fuel type, vegetation greenness, long-term climatology,

and proximity to roads and settlements. Most of them are static or near-static over the timescale at which fire danger varies. The vegetation-index composite is the one time-varying member, and it responds slowly. Such a
60 set explains poorly why a particular summer burned and the preceding one did not.

What changes between those summers is the state of the surface: fuel dryness, vegetation moisture stress, and how anomalously hot the land surface has become relative to its own climatological baseline. Satellite thermal
65 observation gives direct and repeated access to part of that state through fuel moisture content [47]. The pairing used here is surface temperature with a vegetation index. It was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [6]. The Temperature-Vegetation Dryness Index [36] formalises that feature space into an internally normalised mea-
70 sure. In principle it is less exposed to absolute-temperature offsets between regions than raw land surface temperature. That theoretical portability advantage is tested empirically here.

That pre-fire thermal state carries genuine information about subsequent fire is established [19, 21, 20]. Gelabert et al. [9] found dead fine fuel moisture
75 and its yearly anomalies to be the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been established is whether the *predictive skill* of a pre-fire thermal classifier trained in one fire region survives strict, label-free application to another.

80 The physics linking moisture stress to combustion is universal. Portability should therefore be *most* expected for this predictor class, which is what

makes it diagnostic. A loss here cannot be dismissed as a peculiarity of a locally defined covariate. It is also the class on which a self-calibrating satellite thermal monitoring system would have to be built. That system is
85 the long-run motivation for this work rather than anything it demonstrates.

1.3. *Why the loss is invisible: marginal distance versus conditional reversal*

If a model trained in one region underperforms in another, the deficit can arise from two very different mechanisms. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response re-
90 lationship is preserved. This is in principle correctable without target labels, by per-region standardisation or by covariance alignment such as CORAL [38]. Such adaptation has an established remote-sensing literature [39]. Under **concept shift** the conditional relationship itself changes. A predictor positively associated with burning in one region is negatively associated with
95 it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [28], and it has begun to be operationalised in applied remote sensing. Huang et al. [12] decompose domain shift for hyperspectral foliar-trait retrieval and find concept shift
100 dominant.

This distinction is what makes the trade-off hard to see. The field’s principal transferability diagnostic is the area of applicability of Meyer and Pebesma [24]. The assessment critique [25] and the global-model audit of Ludwig et al. [18] are built on the same construction. Transferability is
105 judged by *dissimilarity in predictor space* between the prediction location and the training distribution. That is a **marginal** quantity, a statement

about where the predictors live, computed without reference to the response. It is a well-founded instrument for the failure mode it was designed for, and it is used as such here. By construction, however, it cannot register a
 110 failure in which the predictor distributions overlap acceptably while the conditional predictor-response relationship has changed sign. Two regions can be mutually inside each other’s area of applicability and still be mutually unpredictable. This line of work is therefore extended rather than contradicted. The marginal diagnostics are computed, what they do and do not order is
 115 reported, and a **conditional** diagnostic is supplied for what they miss.

One qualification belongs here rather than deep in the discussion. The single sharpest bootstrap-supported reversal found in this study belongs to elevation, a static and perfectly measured variable. The thermal block is where the trade-off is costly, because that is where the within-region gain
 120 sits. It is not where the instability is worst. The portability problem is therefore a property of the mapping from any landscape variable to burning, not of thermal predictors specifically.

1.4. Objectives

This study evaluates the local-skill and portability trade-off for dynamic
 125 pre-fire thermal predictors in Mediterranean wildfire regions, under a strict train-in-A, apply-to-B protocol. Four questions follow.

Q1. The trade-off. How much within-region skill does the dynamic pre-fire thermal block buy, and what does it contribute between regions? The answer comes from a paired transfer contrast across all ordered region
 130 pairs. The contrast sets a matched baseline against the same baseline plus the thermal block. That baseline is a static and near-static set of terrain, fuel

and greenness predictors, whose only time-varying member is the vegetation-index composite. It is called the baseline from here on. Two subordinate questions supply the first half of the trade-off, and they are supporting evidence rather than findings in their own right. The first is whether the thermal
135 block improves burned-area discrimination beyond that baseline within a region. The second is whether that improvement survives progressively coarser spatial blocking.

Q2. Sufficiency of similarity. Are geographic and bioclimatic similarity
140 *sufficient* for transfer? This is posed as a sufficiency question, which a single strong counterexample settles. It is not posed as a claim about correlation across pairs, which our number of pairs would not support.

Q3. Recoverability without labels. Where transfer fails, how much of the deficit is label-correctable covariate shift and how much is residual
145 concept shift? The recoverable fraction is defined operationally, as what the best available label-blind alignment actually recovers. The residual is what that alignment demonstrably cannot recover. The resulting method dependence is reported rather than concealed.

Q4. Diagnosis. Do marginal, predictor-space-distance diagnostics of
150 the area-of-applicability family order the observed transfer performance? And does a conditional diagnostic track it where the marginal ones do not? The conditional diagnostic is the signed reversal, between source and target, in the univariate association between a predictor and burning.

The evaluation standard throughout is deliberately conservative. The
155 unit of analysis is the native cell of the burned-area product. Cross-validation is spatially blocked and repeated at several block sizes. Uncertainty comes

from a spatial-block bootstrap. The primary population excludes cropland and bare surfaces, so that stubble burning cannot masquerade as wildfire skill. Every column carrying label information is hard-excluded from every
160 feature set. These choices lower the headline numbers, which is the point.

1.5. Contributions

The claims below are ordered by weight and stated at the strength the evidence supports. The nearest prior work is named inside each claim rather than omitted.

165 **Contribution 1. The local-skill and portability trade-off, quantified per direction.** For every ordered region pair, the change in transfer skill from adding the dynamic pre-fire thermal block to the matched baseline is quantified and set against the within-region increment the same block delivers. The block is worth +0.056 to +0.153 ROC-AUC inside every re-
170 gion. Across the twenty ordered transfer directions it contributes +0.004 on average, an estimate whose interval spans zero under every resampling unit the design permits, and its sign is unstable. The paired deltas run from -0.148 to +0.132, twelve of them positive and eight negative. Neither the point estimates nor those signs depend on the blocking scale of the bootstrap.
175 The interval verdicts do. At the conservative 5 km blocking the contribution is CI-supported positive in five to six directions and negative in three to four, with ten or eleven carrying no verdict. The finer 1 km blocking gives ten, seven and three, and even that split turns on a single bound within a thousandth of zero. The block is also the swing factor at the chance line,
180 dragging three directions below chance and lifting one above it. Feature removal measures the cost side directly: removing the two reversing predic-

tors costs -0.081 mean within-region AUC, supported in every region, and changes mean transfer AUC by $+0.014$, an estimate whose interval spans zero (Section 4.6b). This is the paper’s thesis: the dynamic block is where
185 local skill is bought and where portability is spent. Dimarco et al. [8] transfer a predominantly spatially stationary predictor set successfully across a comparable Mediterranean design. Our result is the complementary half of that picture rather than a contradiction of it.

Contribution 2. Label-free alignment does not recover transfer.
190 **It compresses it.** Region-wise standardisation and covariance alignment [38] are tested as label-blind remedies on all twenty directions. Adaptation compresses the transfer matrix towards chance. It recovers at most 34 % of the deficit where transfer fails, and it produces *negative* recovery in seven of twelve decomposed directions. Of the twelve directions that already
195 transferred above chance, nine are degraded and three are raised, all three with Montiferru as their source. Inside the four-AOI decomposition, which contains no Montiferru direction, all six of the above-chance directions are degraded without exception. The worst case is a degradation of -0.86 of the gap. We found no prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction. This is a negative result
200 reported with a mechanism, not an absence.

Contribution 3. Similarity is not sufficient for transfer, whether geographic, climatic or environmental. Manavgat and Muğla lie in the same country and fire year, roughly 200 km apart, and have the highest
205 burned-niche overlap of any pair in the matrix. Both directions fall below chance at the point estimate, and Muğla to Manavgat stays below chance

with interval support at 5 km blocking. Bejís and Muğla lie in different countries and years, about 2500 km apart, and transfer above chance in both directions, and that holds with interval support at both blocking scales. Be-
210 jís and Montiferru, the pair with the *lowest* niche overlap of all, also transfer above chance in both directions, but there the claim rests on the point estimates, since neither direction carries a verdict at 5 km blocking. The claim is one of *sufficiency*. It is established by coexisting counterexamples and is independent of the number of pairs available. It is presented as a contribu-
215 tion to a live disagreement rather than as a settled general law. The fire literature expects similarity to predict transfer [8, 17], while the species distribution modelling literature has found that it does not [41, 35]. Dimarco et al. reach a compatible conclusion from the opposite direction, attributing their weakest transfer to non-climatic factors rather than to bioclimatic
220 distance.

Contribution 4. The failure is conditional, and only a conditional diagnostic sees it. Twenty candidate transferability diagnostics from four families are evaluated against the observed transfer outcomes under one bootstrap framework. Area-of-applicability-style predictor-space dissim-
225 ilarity [24, 25, 18], climatic and geographic distance, learned domain separability, canonical niche-overlap statistics and burn-pattern regime distances all fail to order the matrix. The domain classifier is at ceiling, with $AUC \geq 0.96$ for every pair. Only two diagnostics have intervals that exclude zero, and both measure conditional direction agreement: whether each predictor’s
230 signed association with burning points the same way in both regions (Spearman $\rho = +0.84$ and $+0.81$). Signed associations require burned labels in

both regions, so this is a mechanism diagnosis and not a label-free screening tool. The failure that is invisible to marginal diagnostics is visible in the conditional structure. Paired with label-blind adaptation performance as an instrument for splitting recoverable from irreducible shift, it is the constructive contribution that makes the paper more than a negative result. Shift decomposition in applied remote sensing is not itself new [12]. What we add is the fire application, the adaptation-as-instrument formulation, and the explicit head-to-head of marginal, niche-overlap and conditional diagnostics on the same pairs.

Contribution 5. Within-region replication of the thermal increment. The increment is replicated across five independent Mediterranean regions under spatially blocked cross-validation. ΔAUC runs from +0.056 to +0.153, every bootstrap interval lies above zero, and the result holds in both analysis populations. It survives spatial blocks coarsened to ~ 5 km, the coarsest scale at which this design supports an interval, and a predictor window closed up to 14 days earlier. At ~ 10 km the point estimates hold but the intervals rest on too few positive-carrying blocks to be read as intervals (Section 4.2). The finding is not itself novel, since comparable within-region results exist for these landscapes [1, 14]. It is included as the evidence for the first half of Contribution 1. Two properties of the increment that are not usually reported are given with it: a single subgroup of the block recovers 71 % to 99 % of it in every region, so the six thermal predictors are not six independent measurements, and three defensible compositing choices move it by about ± 0.02 AUC on an identical cohort.

Contribution 6. The price of the failure, measured. Because the

residual gap is conditional, the resource that closes it is target labels, and their cost is measurable. For three regions in all six ordered directions, thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s own ceiling in three directions, two of which started below chance, only 51 to 57 % in the two where the concept gap is widest, and 30 % in the sixth. At small budgets the same intervention damages the direction that transfers best without any labels. This converts the negative result into a priced one. It does not license a label budget, since six directions in three regions cannot support one, and Section 4.10 says so.

Alongside these, a leakage-audited, spatially blocked evaluation and transfer protocol is released with code, configuration and frozen outputs, so that most of this negative transfer result can be re-run rather than taken on trust. Two components are exceptions and are named as such in the data-and-code availability statement: the reproduction-check code and the few-shot export’s exact commit are not in the released repository at the commit of record. This matters given evidence that wildfire transfer conclusions are sensitive to evaluation design and task formulation [45].

Section 2 reviews the relevant literature. Section 3 describes the study regions, the data, and the full modelling, cross-validation, transfer and adaptation protocol. Section 4 reports results. Section 5 discusses them, including limitations. Section 6 concludes.

2. Related work

2.1. Fire susceptibility mapping with machine learning

280 Fire susceptibility mapping estimates the spatial propensity of the land surface to burn, as distinct from short-horizon fire-danger forecasting. It has converged on supervised machine learning applied to gridded geospatial predictors, most commonly tree ensembles such as random forests [4]. Two recent syntheses document the state of the field and its methodological
285 conventions [42, 15]. Reported discrimination is consistently high, typically in the 0.85 to 0.95 AUC range, and the predictor sets that produce those numbers are remarkably uniform across studies: terrain from a digital elevation model, a land-cover or fuel-type classification, a vegetation-greenness composite, climatological summaries, and anthropogenic proximity variables.

290 The Mediterranean basin, and Türkiye within it, is one of the most intensively mapped fire environments, which is fortunate for the present study because two of its regions have direct precedents. Iban and Sekertekin [14] mapped wildfire susceptibility across Adana and Mersin provinces using remotely sensed fire data and machine learning. That is the same landscape
295 as the Kozan district that serves as our cropland-dominated negative control. Alkan Alkan Akıncı and Akıncı [1] produced a machine-learning susceptibility assessment for the **Manavgat district of Antalya**, which is our anchor study region, making it the closest published comparison to our within-region analysis. Iban and Aksu [13] extended this line to İzmir with an
300 explainability-driven framework, using SHAP to interpret the contribution of individual predictors. That study used MODIS active-fire pixels rather than a burned-area product as the target, which is a distinction that mat-

ters when comparing reported skill. The broader ecological framing of why Mediterranean fire regimes are changing is set out by Pausas and Keeley [30],
305 and the comparative structure of global fire regimes by Archibald et al. [2]. That is the notion that fire behaves as a small number of distinguishable syndromes rather than a continuum.

Where this literature stops. These studies establish that fire susceptibility is learnable within a region, and the Manavgat and Adana-Mersin
310 precedents establish it for the specific landscapes we analyse. What none of them establishes is whether a model fitted in one of these landscapes retains any skill in another. Each is validated within its own study area, so the reported AUC is an estimate of interpolation performance inside a fixed footprint, not of generalisation beyond it. Two further gaps follow from the
315 shared predictor design. First, the predictor sets are dominated by static and near-static variables, so they explain poorly why a particular summer burned and the preceding one did not. The dynamic pre-fire state of the surface is largely absent. Second, evaluation rarely reports uncertainty at all, and where cross-validation is used it is frequently random over cells.
320 For spatially autocorrelated data, that is known to be optimistically biased (Section 2.3).

Read together, these two gaps are not independent, and their conjunction is what motivates the present study. Because the field’s predictor sets are near-static and its evaluations are within-region, the natural remedy for the
325 first gap is proposed and assessed under precisely the evaluation regime that cannot reveal its cost. That remedy is to add dynamic pre-fire state, so that the model can distinguish one season from another. A predictor block

can raise within-region skill while simultaneously binding the fitted model more tightly to the place and season in which it was fitted, and a within-
 330 region AUC records only the first half of that exchange. The present study is designed to record both halves. It adds a dynamic pre-fire block to a comparable baseline of terrain, fuel and greenness predictors. That baseline is static and near-static, its only time-varying member being the vegetation-index composite. Both feature sets are evaluated under spatially blocked
 335 cross-validation with bootstrap intervals. The same paired contrast is then carried into cross-region transfer. The local gain and the portability cost attributable to the same block are therefore measured on the same footing.

2.2. Pre-fire thermal dryness from satellite observation

The physical rationale for thermal predictors runs through fuel moisture.
 340 Live fuel moisture content governs ignition and spread, and its retrieval from satellite observation for fire-danger assessment is a mature research area with a substantial operational literature [47]. Chuvieco et al. [6] established the specific combination used here, surface temperature together with a vegetation index, as an estimator of live fuel moisture content for fire-danger
 345 rating.

The index most directly relevant to our feature block is the Temperature-Vegetation Dryness Index. It was introduced by Sandholt et al. [36] as a simple interpretation of the LST-NDVI feature space: a pixel's relative position between the moisture-unlimited "wet edge" and the moisture-limited
 350 "dry edge" at its own vegetation-index level. TVDI is attractive for cross-region work precisely because it is internally normalised. It is defined relative to edges fitted within the scene, and therefore, in principle, less exposed to

absolute-temperature offsets between regions than raw land surface temperature. Whether that theoretical advantage survives contact with real between-
355 region differences is one of the questions this study answers empirically.

The most directly comparable body of work is the series by Maffei and colleagues. Maffei et al. [19] related spatiotemporal patterns of burned area and fire duration to diurnal LST anomalies computed against a climatological baseline, which is the same anomaly construction we use. Maffei and Menenti
360 [21] predicted burned area and rate of spread from pre-fire multispectral measurements, and Maffei et al. [20] extended this to a combined multispectral-and-thermal prediction of fire characteristics, benchmarked against the Fire Weather Index. Together these establish that pre-fire thermal state carries genuine information about subsequent fire.

365 Underpinning all of it are the source products. The Landsat Level-2 surface temperature used here derives from the operational retrieval of Malakar et al. [22], built on the calibration methodology of Cook et al. [7]. Burned-area labels come from the Collection 6.1 MODIS burned-area algorithm [10], whose global validation [3] documents the omission and commission characteristics that bound any model trained against it. Land cover is ESA
370 WorldCover v200 [48], and all satellite retrieval and compositing runs on Google Earth Engine [11].

Where this literature stops. The Maffei series demonstrates a relationship between pre-fire thermal state and fire outcome, but always within
375 a single region and within a probabilistic or correlational frame, using exceedance probabilities and decile-binned relationships rather than a spatially validated classifier, and never with a transfer test. The question of whether

the relationship they identify is a *regionally stable* one is not posed. Two recent studies come closer and must be acknowledged directly. Marino et al. [23] explicitly test the transferability of empirical satellite-derived live fuel moisture models to an uncalibrated region, but what transfers there is a *moisture-retrieval* model, not a burned-versus-unburned classifier. Gelabert et al. [9] find dead fine fuel moisture content and its yearly anomalies to be the most influential predictor of human-caused ignition likelihood across Europe. That variable is a dynamic dryness anomaly, structurally analogous to our LST anomaly. They do test spatial generalisation by fitting a pooled model across five pilot sites and evaluating it per site, with an AUC drop of roughly 0.10 at the weakest sites. That is a genuine cross-site generalisation test of a dryness-dominated model, and it is why we do not claim that the transferability of pre-fire dryness has "never been examined". What remains untested is narrower and, we argue, more decisive: whether the *predictive skill* of a pre-fire thermal model trained in one fire region survives strict train-in-A, apply-to-B transfer with no target labels, and whether label-free adaptation can recover it if it does not. Gelabert et al.'s design pools regions during fitting and evaluates locally, which measures something different. It measures how well one shared model serves several places, not whether the relationship learned in one place holds in another.

There is a further point this literature leaves implicit, and it is the one our thesis turns on. The case for thermal dryness rests on a mechanism that is universal: moisture stress raises flammability wherever vegetation burns, so a predictor built on that mechanism ought to be the *most* portable kind of predictor available, not the least. But the quantity a satellite measures

is not fuel moisture. It is a surface state whose mapping onto flammability runs through fuel type, canopy structure, terrain, seasonal phenology and the
405 local relation between a thermal anomaly and the dryness it stands for. Each of those mediating factors is regionally specific, so the *relationship* between an observed thermal anomaly and burning may be locally reparameterised even where the underlying physics is not. The work reviewed above cannot distinguish these two possibilities. A single-region study never places the
410 relationship learned in one region in contact with the data of another, and neither does a pooled model evaluated locally. Distinguishing them requires strict train-in-A, apply-to-B transfer, and it is that test, rather than a further demonstration of within-region skill, that the present study takes as its object.

415 2.3. *Spatial validation and the transferability of ecological models*

That randomly held-out spatial data give optimistically biased performance estimates when the data are spatially autocorrelated is now a standard result. Roberts et al. [34] set out the cross-validation strategies appropriate to temporally, spatially, hierarchically and phylogenetically structured data;
420 Ploton et al. [32] showed that spatial validation reveals substantially poorer predictive performance in large-scale ecological mapping models than random validation suggests. Blocked cross-validation, in which contiguous spatial units rather than individual cells are assigned to folds, is the standard correction, with an established implementation [40] and a target-oriented
425 variant for spatio-temporal data [26]. It is not uncontested. Wadoux et al. [43] argue that spatial cross-validation can bias map-accuracy estimates, Milà et al. [27] propose nearest-neighbour distance matching as an alternative to

fixed blocking, and de de Bruin et al. [5] compare the available strategies and find that the right choice depends on the sampling design and on the
430 inference intended. That dispute bears directly on this paper, whose transfer verdicts change in number, though not in pattern, between 1 km and 5 km blocking (Section 4.7g). It is one reason both blockings are reported rather than one.

A parallel line asks not how well a model performs but *where* it can
435 be trusted at all. Meyer and Pebesma [24] formalised this as the area of applicability of a spatial prediction model, defined in terms of dissimilarity in *predictor space* rather than geographic distance. They subsequently argued that many published global maps of ecological variables are assessed in ways that do not support their claimed coverage [25]. Ludwig et al. [18] applied
440 this apparatus to existing global spatial prediction models and found their effective transferability considerably narrower than advertised. This line of work is the field's principal answer to the question "will this model travel?", and it is the standard we adopt. Predictor-space dissimilarity is a substantial advance on geographic distance. It is computable without target labels. It
445 also correctly flags the common case in which a model is asked to predict on inputs unlike anything it was fitted to.

It is worth being precise, however, about what kind of quantity such an index is. The area-of-applicability construction compares the *marginal* distribution of predictors in the target domain against the marginal distribution
450 in the source training sample; distance is measured in scaled, importance-weighted predictor space, and the label plays no part in it beyond weighting the axes. Predictor-space dissimilarity is therefore a statement about where

the target data lie, not about what the response does there.

In species distribution modelling, transferability has been examined far
455 more systematically than in fire science, and the conclusions are sobering.
Yates et al. [46] catalogue the outstanding challenges. More pointedly for
our purposes, two studies have already tested whether proximity or similar-
ity predicts transfer success. Both found that it does not. Vesk et al. [41]
report that predictive performance in target regions did not deteriorate with
460 increasing geographic, environmental or community-compositional distance
from the reference region. Rousseau and Betts [35] find environmental sim-
ilarity to be a non-significant predictor of transferability, with species and
range traits mattering more.

Where this literature stops. This block supplies our evaluation stan-
465 dard. In Vesk et al. and Rousseau and Betts it also supplies an important
precedent. The intuition that nearby or similar domains transfer better has
already been questioned outside fire science, so our corresponding claim is a
confirmation in a new domain rather than a discovery. That point is stated
explicitly rather than letting a reviewer make it for us. It is also why we
470 frame our version as a claim about *sufficiency* (Section 2.5) rather than re-
stating theirs. Two gaps remain. First, none of this work has been applied
to fire susceptibility, where the assumption of transferability is embedded in
every regional or pan-Mediterranean product but almost never tested.

Second, and more substantively, there is a structural blind spot in the di-
475 agnostic apparatus itself, and it is the constructive opening this paper works
from. Predictor-space dissimilarity is a *marginal* quantity: it is computed
from the distribution of the inputs alone. The failure mode we document

is *conditional*: the predictor distributions of two regions can overlap to an unremarkable degree, with target cells sitting comfortably inside the source’s predictor envelope, so that an applicability index raises no objection. Meanwhile the association between a predictor and the response differs, and in the sharpest case reverses sign, between them. In the taxonomy of Section 2.4 this is concept shift rather than covariate shift, and no index built from the marginal predictor distribution can register it, because the two situations are indistinguishable in the data the index consumes. This is a statement about what the construction can see, not a criticism of how it has been applied: the area-of-applicability framework does exactly what it claims, and the regime it was designed for, extrapolation beyond the training envelope, is real and widespread. We therefore read our contribution as an extension rather than a correction. The blind spot has a practical consequence worth stating plainly: for models built on dynamic state predictors, a reassuring marginal diagnostic is not evidence of portability, and a model may be well inside its nominal area of applicability and still perform at or below chance. Establishing this requires more than an argument from construction, so we compute a predictor-space dissimilarity index alongside climatic and geographic distances, test whether they order the observed transfer outcomes, and then ask whether a conditional diagnostic tracks what the marginal ones miss. That diagnostic is the signed univariate association reversal between source and target.

2.4. Dataset shift and label-free domain adaptation

The machine-learning framing separates the ways a training and a deployment domain can differ. Moreno-Torres et al. [28] give the canonical

taxonomy: covariate shift, in which the input distribution changes while the conditional label distribution is preserved; prior or label shift, in which class
505 prevalence changes; and concept shift, in which the conditional relationship between inputs and label changes. The distinction is the same marginal-versus-conditional distinction that organises Section 2.3, and it has direct operational consequences, because only the first is correctable without target labels: aligning input distributions cannot repair a relationship that has
510 itself changed.

That limitation is worth stating as a property of the methods rather than as an empirical finding about any particular dataset. Both of the label-free alignment operations used here act on marginal summary statistics of the predictors. Region-wise standardisation subtracts a per-region mean and divides
515 by a per-region standard deviation, so it equalises first and second marginal moments coordinate by coordinate. Correlation alignment (CORAL) is the canonical simple, deterministic extension of the same idea. It is a whitening-and-recolouring linear map that aligns the source feature covariance to the target covariance [38], so it additionally equalises the second-order cross-
520 coordinate structure. Neither transformation consults a target label, and neither could: the label is by construction unavailable. It follows that whatever either method achieves, it achieves by making the source and target *input* distributions resemble one another more closely. If the conditional relationship between inputs and label differs between the domains, then any
525 input-space map leaves that difference untouched. That includes these two, and any other method whose objective is defined on the marginals alone. Where a predictor’s association with the response has opposite signs in the

two regions, aligning the marginals can only bring the source model’s error into sharper focus, since it delivers target inputs to a decision rule fitted
530 under the opposite relationship. This is a logical consequence of what the methods optimise, not a result we report; what remains genuinely empirical is how large the irreparable part is in a given application, and that is what our experiments measure.

In remote sensing specifically, domain adaptation has an established literature surveyed by Tuia et al. [39], and where unsupervised adaptation is
535 insufficient, active learning has been used to acquire a small number of maximally informative target labels [31]. That is the standard response once the conditional component is recognised as the binding constraint.

Very recently, shift decomposition has begun to appear in applied remote
540 sensing. Huang et al. [12] compare transfer-learning strategies for hyper-spectral foliar-trait retrieval and quantitatively decompose domain shift into components across four transfer scenarios, reporting concept shift as the dominant limitation on transferability. No effect coefficient or numerical result of theirs is quoted here. The citation supports only the existence and direction
545 of that decomposition.

Where this literature stops. Huang et al. establish that decomposing transfer failure into shift components is both possible and informative in an applied remote-sensing setting, and we do not claim that idea as ours. Two differences remain, and they are the ones that matter. Theirs is a hyper-
550 spectral regression problem, not a fire problem, and their instrument is a set of statistical shift metrics regressed against the observed transferability gap. Ours is an *operational* instrument: we use the performance of label-free

adaptation itself as the measuring device, so that the recoverable fraction of the gap is defined as what the best available label-free method actually
555 recovers, and the residual is what it demonstrably cannot. That definition has the advantage of being decision-relevant, because it answers "would unsupervised alignment fix this?" by trying it. It also has the disadvantage of being method-dependent, which we report rather than conceal. Beyond that, we found no application of CORAL or of region-wise feature align-
560 ment to fire susceptibility, fire occurrence or burned-area prediction at all. Work labelled "domain adaptation" in the fire literature addresses image-level post-fire tasks: burned-area segmentation, burn-severity mapping and smoke detection. These share the vocabulary but not the problem.

What this block supplies to the present argument is the vocabulary in
565 which the thesis can be stated exactly. The portability cost of dynamic state predictors is not a covariate-shift problem that better alignment would solve. It is a conditional change. The marginal instruments the field uses to anticipate transfer failure (Section 2.3), and the marginal operations it uses to repair it, are blind to it for the same reason. What the shift literature does
570 not yet supply is a diagnostic that operates on the conditional side and can be computed before deployment. Attempting the repair and observing what it fails to recover is one such instrument, and pairing it with a feature-level test for association reversal is the constructive part of our contribution.

2.5. Cross-region generalisation of fire models

575 A small number of studies test fire-model transfer directly, and they are the nearest neighbours of the present work.

Podschwit et al. [33] examined the reliability of applying global fire danger models across regions in a Peruvian case study, and provide the long-standing reference point that meteorologically derived danger indices do not
580 port cleanly between fire environments.

Dimarco et al. [8] is the closest Mediterranean analogue and the most important comparison for this paper. They harmonise 500 m predictors across four Mediterranean countries, fit random forest and gradient boosting models, select hyperparameters under five-fold spatial cross-validation, evaluate
585 on an 80/20 hold-out, and test transfer both leave-one-country-out and as a full 4×4 transfer matrix. Their predictor set comprises NDVI, and slope derived from the ASTER GDEM. It adds 2 m air temperature, 10 m wind speed and relative humidity, all taken from ERA5-Land [29] as **long-term seasonal means**. It also adds the global human modification index, VIIRS
590 night-time lights, a log1p transform of those lights, and population density.¹ Their target is burned-pixel centroids from MCD64A1, treated as an ignition proxy and matched against a 1:1 balanced background sample. They report that every transfer exceeds AUC 0.80. Transfers between bioclimatically similar countries score higher, and transfers to Morocco are systematically
595 lower. They attribute the drop not to climate alone but to differences in anthropogenic drivers, fire management practice and data reporting. They apply no domain adaptation.

Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatol-

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

ogy. We follow their Methods, which we take to be authoritative, and note
the discrepancy here so that the comparison drawn below is transparent.
Every statement in this paragraph was checked against the full text (read
2026-08-08), not the abstract alone.

Outside the Mediterranean, WildfireGenome [17] performs the most sys-
tematic transfer experiment we are aware of: a leave-one-county-out transfer
matrix across seven ecologically diverse US counties, with random forests and
SHAP-based interpretation. It reports strong within-county performance
and highly variable off-diagonal transfer, with ecologically similar county
pairs transferring well and dissimilar pairs collapsing below chance. Its la-
bel, however, is a principal-component composite of federal hazard *indicators*
rather than observed burned area, and it applies no domain adaptation and
performs no shift decomposition. Finally, Xu et al. [45] argue that wildfire
transfer conclusions depend strongly on evaluation design and task formula-
tion. That caution applies to our results as much as to anyone’s. We address
it by fixing the analysis protocol in a project log before the diagnostics were
computed, and by reporting every sensitivity axis. That log is not a for-
mal pre-registration, and no independent timestamped registration exists.
Section 3.14.1 states exactly what was fixed and when.

For scaling this kind of work, Kondylatos et al. [16] provide Mesogeos,
a multi-purpose 1 km Mediterranean datacube built for data-driven wildfire
modelling.

Where this literature stops, and what this study adds. These
studies establish that fire-model transfer is testable and that its outcome
varies. What none of them asks is *what property of a model* determines the

625 outcome. Transfer performance is reported per pair and explained, where it
is explained at all, by how similar the domains are. Read against ours, they
set up an empirical contrast that points to a different explanation, and that
contrast is the most informative result of the present work.

Dimarco et al. and this study share the essential experimental design:
630 Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensem-
bles, spatially aware validation, and an explicit transfer matrix. Theirs covers
four countries and twelve ordered pairs, ours five regions and twenty ordered
directions. What differs is the **predictor class**. Their model is built on
quantities that are attributes of a *place*: terrain slope, night-time lights, hu-
635 man modification, population density, and temperature, wind and humidity
as long-term seasonal climatologies. Every one of these is spatially stationary,
describing a region rather than a season. Their model transfers everywhere
above AUC 0.80. Our model is built on the *state of a particular pre-fire*
window: land surface temperature anomalies against a multi-year baseline,
640 thermal-optical dryness indices, and downscaled and fused thermal channels,
all composited over the weeks preceding a specific fire. Because the physics
linking moisture stress to combustion is universal, this predictor class is the
one for which transfer should be *most* expected. It is instead the one that
fails.

645 The contrast, then, is not the one we initially expected to draw. It would
be easy but wrong to reduce it to target definition, their ignition proxy being
human-driven and ours burned area. It would be easier still, and equally
wrong, to reduce it to their finding that anthropogenic pressure dominates.
That finding is a result of their analysis rather than a feature of their design,

650 and their predictor set explicitly includes vegetation condition, topography
 and climatic context. The sharper reading is about predictor class. Every
 predictor in their model is an attribute of a *place*, stable across seasons and
 years, so a model fitted on them is in effect learning a spatial ordering of sus-
 ceptibility that a neighbouring country can inherit. Every thermal predictor
 655 in ours describes the *state of a particular window* preceding a particular fire,
 so the fitted model encodes not only which places are prone to burn but how
 a given degree of anomalous dryness translates into burning in that region,
 in that season. The first kind of knowledge travels. The second is exactly the
 kind that need not. The deeper point this suggests is that predictor class,
 660 rather than domain similarity, is what governs portability. It is not that simi-
 lar regions transfer and dissimilar ones do not. It is that stationary-attribute
 models track similarity, while dynamic-state models are reparameterised lo-
 cally and therefore need not. Our data support this reading and cannot prove
 it, and we return to it below.

665 Three further differences must be stated precisely, because they bound
 how far the contrast can be pushed. **Target:** they treat burned-pixel cen-
 troids as an ignition proxy against a 1:1 balanced background, whereas we
 perform burned-area classification at the true, heavily imbalanced base rate.
 This is a different problem with different achievable ceilings. **Thermal vari-**
 670 **able:** their temperature predictor is a static reanalysis climatology, whereas
 ours are event-specific satellite thermal observations referenced to their own
 baseline. **Adaptation:** they apply none, whereas the label-blind adaptation
 step is central to our diagnosis. The contrast is therefore between two co-
 herent experimental programmes, not a controlled ablation, and we present

675 it as such.

Why the predictor-response relationship should be reparameterised locally is a question this literature can frame even if our design cannot settle it. Fire is conventionally described not as a continuum but as a small number of distinguishable syndromes. These are pyromes, in the terminology of
680 Archibald et al. [2], differing in the fuel, weather and ignition constraints that limit burning in a given setting, and Pausas and Keeley [30] document how those constraints are shifting under global change. Where burning is fuel-limited, additional dryness in an already sparse system need not produce more of it; where it is drought-limited, the same anomaly is much closer to
685 a sufficient condition. If two regions sit in different limiting regimes, the mapping from an observed thermal anomaly to burning probability differs between them by construction, even where the predictor distributions overlap and the underlying combustion physics is identical. That is precisely the conditional, marginally invisible difference described in Sections 2.3 and 2.4.
690 We offer regime typology as a candidate explanation, consistent with what we observe. We do not offer it as a hypothesis this study tests: with the small number of regions available, a regime difference cannot be distinguished from any other region-specific effect.

Five contributions follow, ordered by weight and stated at the strength
695 the evidence supports.

First, the trade-off itself, which is the paper’s thesis. We quantify, for every ordered region pair, the change in transfer skill attributable to adding the dynamic pre-fire thermal block to the matched baseline, and set that against the within-region increment the same block delivers. The claim

700 is that the block gaining the most locally contributes nothing on average
 between regions, and that its between-region contribution is sign-unstable.
 Whether it helps or harms there is a property of the source-target pair, not
 of the block. Dimarco et al. [8] demonstrate the stationary-predictor half of
 this picture within the same kind of Mediterranean design. Our result is its
 705 complement, not its contradiction. Establishing the portability half requires
 testing whether the predictive skill of pre-fire thermal state itself survives
 strict, label-free train-in-A, apply-to-B transfer. The quantity under test is
 that skill, not the skill of a moisture-retrieval model [23] and not that of
 a pooled multi-region model evaluated locally [9]. To our knowledge that
 710 test has not been performed, but we state this as a gap in the transfer test
 specifically, not as an absence of attention to the transferability of dryness
 predictors: Gelabert et al. come closer than any other study, and the stronger
 wording would be indefensible.

Second, label-free alignment does not recover transfer. We test
 715 region-wise standardisation and covariance alignment [38] as label-blind reme-
 dies and characterise what each does to every direction, including directions
 that already transfer. No application of CORAL or region-wise feature align-
 ment to fire susceptibility, fire occurrence or burned-area prediction was
 found in the literature; work labelled "domain adaptation" in fire research
 720 addresses image-level post-fire tasks. This is a negative result reported with
 a mechanism rather than an absence.

**Third, geographic and bioclimatic similarity are not sufficient
 for transfer.** This is deliberately a sufficiency claim rather than a claim
 about correlation across pairs. It is therefore established by a single strong

725 counterexample rather than by a sample of pairs: Manavgat and Muğla lie in
the same country, burned in the same fire year, sit roughly 200 km apart and
share a bioclimatic setting, about as similar as two independent fire regions
can be, and transfer between them collapses. Notably, Dimarco et al.’s own
730 results point the same way from the other side. They attribute the systemat-
ically weaker transfer to Morocco to anthropogenic drivers, fire management
and data reporting rather than to climate alone, which is itself an admission
that bioclimatic similarity does not account for transfer performance on its
own.

Fourth, a conditional transferability diagnostic. We compute area-
735 of-applicability-style dissimilarity in predictor space [24, 25, 18] alongside
climatic and geographic distance. We then report whether these marginal
indices order the observed transfer outcomes, and supply a conditional al-
ternative: signed reversal in the univariate association between a predictor
and burning, paired with label-free adaptation performance as an operational
740 instrument for separating recoverable from irreducible shift. Shift decomp-
osition in applied remote sensing is not itself new [12]; what is new here is
the fire application, the adaptation-as-instrument formulation, and the di-
rect contrast between a marginal and a conditional diagnostic evaluated on
the same region pairs. This is the constructive contribution, and it is what
745 distinguishes the paper from a purely negative result.

Fifth, within-region replication. We replicate the thermal increment
across independent Mediterranean regions under spatially blocked cross-validation
with spatial-block bootstrap intervals, and characterise its behaviour as blocks
coarsen. This finding is not itself novel, since comparable within-region re-

750 sults exist for these very landscapes [1, 14]. It is included because it supplies the first half of the trade-off in the first contribution.

Alongside these we release the protocol, code and frozen outputs, so that most of this negative transfer result can be re-run rather than taken on trust [45]. Two components fall outside that release and are named in the 755 availability statement of Section 3.13.

We present the third of these as a **live disagreement**, not a settled result. Vesk et al. [41] and Rousseau and Betts [35] align with our direction, having already found that geographic and environmental similarity fail to predict transfer success in species distribution models; Dimarco et al. [8] and Wild- 760 fireGenome [17] run against it, both reporting that similar domains transfer better. Our data support the following reading without proving it. The relationship between domain similarity and transfer success is **predictor-class dependent**: models built on spatially stationary attributes track similarity, while models built on dynamic state need not.

765 3. Materials and methods

3.1. Study regions and temporal windows

Five Mediterranean-basin wildfire regions are analysed. Each is defined as a place-based rectangular area of interest (AOI) in EPSG:4326. Each AOI is defined from place coverage rather than from a fire perimeter, and 770 none is tuned on burned prevalence, on the gate outcome or on any model metric. One AOI choice was label-informed and is stated as such: the North Evia box was extended after the legacy box was found to carry an atypically high burned prevalence, the extended geometry was then defined from place

anchors, and the legacy variant is retained as a sensitivity (Sections 3.16.1
775 and 4.1). The AOIs are deliberately *not* clipped to fire perimeters, so that
unburned cells surrounding each fire constitute the negative class rather than
being excluded by construction.

For each region the analysis is organised around two non-overlapping time
windows. The **predictor window** is the pre-fire period from which every
780 dynamic predictor is composited. It ends the day before the label window
opens. The **label window** is the period in which a burned-area detection
counts as a positive label. Because the predictor window closes before the
first day of the label window, no predictor observation can post-date the fire
it is used to predict. A separate set of **baseline years** supplies the multi-
785 year climatological reference against which thermal anomalies are computed.
These use the same calendar window as the predictor window, transported
to earlier years, with the fire year itself excluded.

North Evia is analysed on an **extended** $0.80^\circ \times 0.60^\circ$ AOI. The legacy
 $0.40^\circ \times 0.40^\circ$ box (23.12, 38.68, 23.52, 39.08; `repo/core/regions.py:67`)
790 shares identical predictor and label windows and is retained **only** as a sensi-
tivity variant (Section 3.16.1).

Sample sizes, burned-cell counts and prevalences per region are reported
in Table 1b, read from each region’s frozen Step 8A dataset statistics (`step8a_
dataset_stats.json`) for all five regions.

795 The region set is constructed to separate two candidate explanations of
transfer failure. Manavgat and Muğla are in the same country, in the same
fire year, roughly 200 km apart, and share a baseline-year set; Bejís is in a
different country, a different year, and about 2500 km away. If transferability

Table 1: Study regions. Every value below is read directly from the experiment registry in `repo/core/regions.py` at commit 48b56e7, which now contains all five regions. AOI constants: Manavgat `regions.py:222`, Bejís :252, Muğla `MUGLA_AOI_BBOX :59`, extended Evia `NORTH_EVIA_EXTENDED_AOI_BBOX :106`, Montiferru `MONTIFERRU_AOI_BBOX :142`. Windows, baseline years and roles from the `EXPERIMENTS` dictionary (:332 to 790; individual entries at :352 Manavgat, :368 Bejís, :391 Muğla, :683 extended Evia, :721 Montiferru). The registry was read by importing `core.regions` and calling `get_experiment()` for each identifier rather than by parsing the file, so the values are the ones the pipeline itself resolves. Muğla and Montiferru no longer require the earlier reconstruction from the 30 m reference-grid transform: both are now declared constants, and their declared corners agree with the previously grid-derived values (Muğla 27.0998, 36.5998, 28.9001, 37.4500 $\rightarrow$ 27.10, 36.60, 28.90, 37.45; Montiferru 8.4497, 40.0499, 8.7502, 40.2700 $\rightarrow$ 8.45, 40.05, 8.75, 40.27) to the rounding shown.

Region	Country	AOI bounding box (W, S, E, N)	Predictor (pre-fire) window	Label window	Baseline years	Role
Manavgat / Antalya 2021	Türkiye	31.05, 36.72, 31.85, 37.35	2021-06-01 $\rightarrow$ 2021-07-27	2021-07-28 $\rightarrow$ 2021-08-31	2017-2020	anchor wildfire
Bejís / Castellón 2022	Spain	-1.05, 39.68, -0.35, 40.15	2022-06-15 $\rightarrow$ 2022-08-14	2022-08-15 $\rightarrow$ 2022-09-30	2018-2021	Mediterranean transfer wildfire
Muğla 2021	Türkiye	27.10, 36.60, 28.90, 37.45	2021-06-01 $\rightarrow$ 2021-07-28	2021-07-29 $\rightarrow$ 2021-09-15	2017-2020	same- country, same-year transfer wildfire
North Evia (Euboea) 2021, extended AOI	Greece	23.05, 38.55, 23.85, 39.15	2021-06-05 $\rightarrow$ 2021-08-02	2021-08-03 $\rightarrow$ 2021-09-30	2017-2020	Mediterranean transfer wildfire
Montiferru (Sardinia) 2021	Italy	8.45, 40.05, 8.75, 40.27	2021-05-25 $\rightarrow$ 2021-07-23	2021-07-24 $\rightarrow$ 2021-08-31	2017-2020	Mediterranean transfer wildfire

were governed by geographic and climatic proximity, the Manavgat↔Muğla
800 pair should transfer best. A **negative control** region (Kozan 2023, Türkiye),
in which the burned area is dominated by agricultural stubble burning rather
than wildfire, exists in the pipeline registry and is used only as a validity
check on the admissibility gate described in Section 3.3. It is never used as
a transfer partner. Its role is described in Section 3.3 and its gate outcome
805 is reported in Section 4.1.

3.2. Burned-area label and the ~ 500 m analysis grid

The target variable is burned/unburned status derived from the MODIS
MCD64A1 Collection 6.1 burned-area product (MODIS/061/MCD64A1), re-
trieved through Google Earth Engine. Collection 6.1 is a reprocessing of the
810 Collection 6 product described by Giglio et al. [10] and validated by Boschetti
et al. [3]. Active-fire detections (FIRMS) are never used as a target.

Grid reconstruction. MCD64A1 is a ~ 500 m product on the MODIS
sinusoidal grid, but it is exported from Earth Engine onto the 30 m EPSG:4326
reference grid used by the rest of the pipeline (VALIDATION_LABEL_EXPORT_
815 SCALE = 30, core/config.py:262), which duplicates each native ~ 500 m
observation across a block of 30 m pixels sharing its value. The analysis grid
is therefore *reconstructed* rather than native: the 30 m reference grid is par-
titioned into non-overlapping square blocks of $\text{round}(500 / 30) = 17 \times$
17 pixels (step8a_prepare_500m_modeling_dataset.py:914 to 917, us-
820 ing STEP8A_MCD64A1_NATIVE_CELL_SIZE_M = 500.0 and STEP8A_REFERENCE_
PIXEL_SIZE_M = 30.0, core/config.py:538 to 539), giving a nominal cell
edge of 510 m.

That cell is square in degrees but not on the ground, and this

is stated here because the paper’s central robustness argument is
825 **about spatial scale.** Every predictor raster is exported at `scale: 30` in
EPSG:4326 (each region’s `predictor_export_metadata.json`), so the ref-
erence pixel is a step in degrees, $30 / 111,319.49 = 0.00026949^\circ$, and the
analysis cell is 17 times that, 0.0045814° , in both axes. On the ground the
cell is about 510 m north to south everywhere, but only 407 m east to west
830 at Manavgat and Muğla, 397 m at Evia, 391 m at Bejís and 390 m at Monti-
ferru. Cell area is 0.199 to 0.208 km² against the MODIS cell’s 0.250 km², so
the analysis cell is about 17 % to 20 % smaller. The derivation is checkable
rather than nominal: dividing each AOI’s span by 0.0045814° and rounding
outward reproduces every frozen cell count exactly, at $175 \times 138 = 24,150$
835 for Manavgat, $153 \times 103 = 15,759$ for Bejís, $393 \times 186 = 73,098$ for Muğla,
 $175 \times 131 = 22,925$ for Evia and $66 \times 49 = 3,234$ for Montiferru. Two
consequences follow. Every block-size label in this paper is the north-south
dimension: a 10-cell block is about 5.1 km by 3.9 to 4.1 km and a 20-cell
block about 10.2 km by 7.8 to 8.1 km, so spatial blocking is systematically
840 weaker in longitude than the labels suggest. And because the cell is smaller
than a MODIS cell and not aligned with it, a typical cell draws on more
than one MODIS cell, which dilates the labelled burned footprint relative
to MCD64A1 and is the quantitative form of the label-geometry concern in
Section 5.11(xiv).

845 The reconstruction is an approximation of the true MODIS sinusoidal
cell in two further respects: it is anchored to the Landsat/EPSC:4326 refer-
ence grid rather than to the MODIS tile grid, and blocks at the AOI margin
are truncated and therefore contain fewer than 289 constituent pixels. This

is stated explicitly because it is a real, and reported, limitation of the label geometry rather than an incidental implementation detail. Each cell is identified by its integer block indices `row_500m = row_offset // 17, col_500m = col_offset // 17` (step8a:920 to 936). These indices are used to construct spatial cross-validation blocks and are never used as predictors.

The 30 m reference grid itself is the Landsat current-period median LST raster produced upstream, and every other continuous predictor is required to match it exactly in width, height, CRS and affine transform. A mismatch is a hard error and the pipeline never silently resamples (step8a:613 to 643). The two exceptions are documented and use nearest-neighbour resampling only, because both layers are categorical: land cover (step8a:479 to 551) and the MCD64A1 burn-date raster itself (step8a:768 to 814).

Label assignment. Within each 17×17 block, only strictly positive burn-date day-of-year values are considered; zeros and no-data are ignored. A cell is unburned by default. If positive values exist, their modal day-of-year is taken as the cell's representative burn date, ties being broken in favour of the smaller value. For the burned/unburned decision that means the conservative direction (step8a:939 to 951). The representative day-of-year is then converted to a calendar date and tested against the region's label window: if it falls inside the window the cell is labelled burned, otherwise the cell is retained as **unburned** and flagged `out_of_window_burndate` (step8a:1217 to 1261). The fraction of positive sub-pixels agreeing with the modal date is recorded as a diagnostic (`burn_date_pixel_agreement_fraction`) but no agreement threshold is imposed. Only the modal date is tested, so the rule is not an "any positive sub-pixel" rule in general: a

block whose positive sub-pixels were, say, one in-window and two out-of-
875 window would be labelled unburned. In this dataset the two rules coincide,
because the exported label raster carries no out-of-window positives at all.
Every region's `label_raster_diagnostics` records `count_positive` equal
to `count_in_label_doy_range` exactly. So for this export, and only because
of that clipping, a single in-window positive sub-pixel is sufficient to label the
880 cell burned.

The label never affects a cell's eligibility for modelling (Section 3.5). Un-
burned cells, all-no-data blocks and out-of-window cells all remain in the
dataset as the negative class.

Pre-label burn exclusion. Optionally, cells that already show a burn
885 detection in the interval between the start of the predictor window and
the start of the label window are excluded from the analysis universe en-
tirely, so that a predictor window cannot be contaminated by an earlier fire
within the same season (`step8a:967` to `1025` reads the exclusion manifest
produced by the gate step, `step8a:3059` to `3072` resolves it for the run,
890 and `step8a:1179` to `1188` applies it per cell). Excluded cells are marked
`analysis_eligible = False`; their raw labels are preserved for audit but
they never enter modelling.

This safeguard is optional in the pipeline and it did not run everywhere,
which the reader needs in order to weigh it. From the frozen dataset statistics,
895 it ran for Muğla (49 cells excluded), Evia (16) and Montiferru (61). For
Manavgat the flag is recorded as false and no cell was excluded. For Bejís the
field is absent from the export altogether, so no status is recorded either way.
The exposure this leaves is bounded by how many cells carry an MCD64A1

detection inside the predictor window in those two regions, and that count has
 900 not been produced; it is listed as an open item in Section 5.11. Two further
 limits belong here. The exported label raster is clipped to the label window in
 every region, so a pre-label detection is not recoverable from the label raster
 itself and the reported count of zero out-of-window burn dates is a property
 of the export rather than an empirical finding. Burning in earlier years is not
 905 screened for any region in the five-region cohort, so a cell that burned in a
 previous year enters the analysis with a baseline climatology computed partly
 over its own post-fire state. The historical-exclusion module is opted into
 exactly once in the registry, for the `mugla_2022_event_relative` experiment
 of Section 3.16.4, where it removes the 2021 scar. Muğla 2021 is the source
 910 of that mask rather than a beneficiary of it, and the 2022 event-relative
 experiment is not one of the five regions of Table 1.

3.3. Burned-landcover admissibility gate

Before any predictor modelling, each region passes through a gate (`step6b_`
`burned_landcover_gate.py`) that answers one question: **of the cells la-**
 915 **belled burned, what land cover dominates them?** The gate reads only
 the burn-date raster and the aligned land-cover raster, and reuses Step8A's
 block size, cell-identity function and land-cover class map verbatim, so its
 cells are identical to the modelling cells.

For each burned cell the modal ESA WorldCover class is taken. Regional
 920 fractions are then formed with the total burned-cell count as denominator: a
 natural-vegetation fraction (tree cover 10 + shrubland 20 + grassland 30)
 and a cropland fraction (class 40). The verdict rule is applied in order
 (`step6b:198 to 219`):

1. fewer than `STEP6_BURNED_LANDCOVER_GATE_MIN_POSITIVES = 30` burned
925 cells $\rightarrow$ *insufficient burned positives*;
2. natural-vegetation fraction $\geq 0.50 \rightarrow$ **wildfire candidate, pass**;
3. cropland fraction $\geq 0.50 \rightarrow$ *cropland-dominated control*;
4. otherwise $\rightarrow$ *mixed or uncertain*.

Thresholds are at `core/config.py:369` to `373`. The gate is diagnostic:
930 a failing verdict does not halt the pipeline, and the gate output always carries
downstream_authorized: False, requiring explicit review. Its role in this
study is to establish that the regions entering the transfer experiment are
genuine forest-fire regions. Per-region verdicts and natural-vegetation frac-
tions are read from each region’s frozen `burned_landcover_gate.json` and
935 reported in Section 4.1.

Negative control. A gate that every candidate region passes carries
no information, so the threshold is calibrated against a region that should
fail it. MCD64A1 does not distinguish the combustion of natural fuel from
the deliberate burning of harvest residue: both are detected as burned area,
940 and a model trained on the latter would be learning agricultural calendar
rather than fire-relevant dryness. Kozan 2023 (Adana province, Türkiye) was
therefore processed through the identical Step 1 to 6 pipeline as a negative
control. It is not a wildfire region: its burned area lies in the Çukurova
agricultural plain and is dominated by post-harvest stubble burning. The
945 gate receives it blind. It sees only the burn-date raster and the land-cover
raster, with no regional label, and the verdict follows from rule 3 above.
The control is reported in Section 4.1 and enters no modelling, transfer or
diagnostic analysis. No Kozan number appears anywhere else in this paper.

3.4. Predictor variables

950 All dynamic predictors are composited over the region’s predictor window; static predictors are time-invariant. Each is produced by a named upstream pipeline stage and delivered on the 30 m reference grid.

Observations behind the composites. The optical and thermal predictors come from one sensor and one collection: Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2). Landsat 9 is not used in any region, 955 including Bejís 2022, where it was operational. Quality screening is applied per pixel from the QA_PIXEL band, masking fill, dilated cloud, cirrus, cloud, cloud shadow and snow, with medium and high confidence bits treated as masked; the water bit is deliberately preserved (step3_landsat_lst.py:67 960 to 98). The ST_QA per-pixel surface-temperature uncertainty band is available in this collection and is not used, so no per-pixel LST uncertainty enters the analysis. The coarse-resolution thermal input is MODIS/061/MOD11A1, the Terra daytime 1 km LST product; Aqua is not used.

A predictor-window composite therefore rests on however many clear acquisitions that window contained, and the pipeline’s floor is low: STEP5_ 965 MIN_CURRENT_VALID_COUNT = 2, so a cell’s median may rest on two clear observations. The number is region-specific and, within a region, position-specific. For Manavgat, the only region with a frozen acquisition inventory, the 57-day window holds seven distinct dates from 14 scenes on two alternating 970 WRS-2 paths, so a cell inside the path overlap is backed by seven dates and a cell outside it by three or four. Section 4.7i shows what this variation does to the within-region increment, and Section 5.11 lists the per-region inventory as an open item, since it exists for Manavgat alone.

Optical greenness (NDVI). Landsat Collection 2 Level-2 surface reflectance, scaled by 0.0000275 with an offset of -0.2 , is used to compute $NDVI = (NIR - Red) / (NIR + Red)$; pixels with a near-zero denominator or values outside $[-1, 1]$ are masked (`core/config.py:89 to 103`). The predictor-window composite is a per-pixel **median**.

Land surface temperature (Landsat). The Landsat Level-2 surface temperature band is scaled by 0.00341802 with an offset of 149.0 K (`core/config.py:86 to 87`) and converted to degrees Celsius. The predictor-window composite is again a per-pixel median (`current_lst`, in $^{\circ}C$). This raster also serves as the 30 m reference grid.

LST anomaly. A z-score of the current-window LST median against the baseline-years distribution for the same calendar window: (`current_median - baseline_mean`) / `baseline_std` (`step5_preprocess_timeseries.py:899 to 905`). The anomaly is set to no-data where the baseline standard deviation is below `STEP5_MIN_BASELINE_STD_CELSIUS = 1.0` $^{\circ}C$, where fewer than `STEP5_MIN_CURRENT_VALID_COUNT = 2` current observations exist, or where fewer than `STEP5_MIN_BASELINE_VALID_COUNT = 3` baseline observations exist. The result is dimensionless.

TVDI and TVDI difference. The Temperature-Vegetation Dryness Index is computed in the LST-NDVI feature space (`step5c_tvdi.py`). The NDVI range $[0, 1]$ is divided into `TVDI_NDVI_BIN_COUNT = 20` bins. Within each bin holding at least `TVDI_MIN_PIXELS_PER_BIN = 30` valid pixels, the wet and dry edges are taken as the 2nd and 98th LST percentiles (`TVDI_WET_EDGE_PERCENTILE = 2.0`, `TVDI_DRY_EDGE_PERCENTILE = 98.0`). TVDI is then $(LST - wet_edge) / (dry_edge - wet_edge)$, clamped to $[0, 1]$. It

is set to no-data where the edge span is below `MIN_TVDI_EDGE_SPAN_C =`
1000 `1.0 °C (core/config.py:162 to 178)`. `tvdi_difference` is the current-
window TVDI minus the baseline-years mean TVDI. A documented caveat
applies to the latter: the baseline TVDI is formed from a single baseline
LST mean raster combined with per-year NDVI, so its inter-annual variation
derives from NDVI alone (`step5c_tvdi.py:480 to 486`).

1005 A second property of TVDI matters for the transfer analysis and is stated
here so that Section 5.2 can be read against it. The wet and dry edges
are percentiles of the LST values that a given scene contains, so they are
fitted per AOI and per window. A TVDI of 0.5 therefore denotes a different
physical moisture state in each region, set by whatever dryness range that
1010 AOI happened to span on those dates, and the AOIs differ widely in size
and relief (Montiferru $0.30^\circ \times 0.22^\circ$ against Muğla $1.80^\circ \times 0.85^\circ$). TVDI
is normalised in a statistical sense, not in a physical one. A common-edge
version of the index, fitted once across the pooled land pixels of all five
regions so that a TVDI of 0.5 denotes the same dryness everywhere, was
1015 computed for this paper and is reported in Section 4.7k. It does not remove
the reversal, so scene-fitted edges are a controlled-for factor here rather than
an open competing explanation.

**Sea water enters the edge fit, and the extent of the resulting
problem was measured rather than assumed.** Because the water bit
1020 is preserved and the AOIs are not clipped to the coastline, sea pixels take
part in the percentile fit. Two of the five AOIs are largely marine: water-
dominant cells are 57.6 % of Evia’s grid and 38.9 % of Muğla’s, against 0.1 %
for the one inland AOI, Bejís. The frozen per-bin edge diagnostics show the

consequence directly. In Evia the three lowest NDVI bins carry dry edges
1025 of 28.8 to 29.9 °C, which is Aegean sea-surface temperature and not a land
dry edge, whereas Bejís’s lowest bins sit near 49 °C. That contamination is
confined to the bins the sea occupies. From NDVI bin 4 upward Evia’s dry
edge runs from 44 to 47.5 °C, in the same range as Bejís’s 41 to 49 °C, and
neither the wet nor the dry edge orders across the five regions by sea fraction:
1030 in the vegetated bins the lowest wet edge belongs to Montiferru, at 7.4 %
water, and the highest to Bejís, at 0.1 %.

The question that matters is whether the modelled population occupies
the contaminated bins, and it does not. Not one cell of the primary natural-
vegetation population in any region has a mean NDVI below 0.15, and 0.01
1035 % or less lies below 0.20; the 5th percentile of that population’s NDVI is
0.376 in Evia and 0.290 to 0.383 elsewhere. Nor is there evidence of the
saturation such a mis-normalisation would produce: the share of primary-
population cells at the upper clamp is 0.5 % in Evia, 0.5 % in Muğla and 0.4
% in Manavgat, against 0.1 % in Bejís, with the distribution well spread in
1040 every region. That check is made on 500 m cell means, so it bounds rather
than excludes the effect on individual 30 m pixels, and it does not license the
all-valid population, where sea cells are present in bulk (Section 4.7f).

The clean test was then run. The edges were refitted on land pixels
alone, and a third set was fitted once over the pooled land pixels of all
1045 five regions, which is the common-edge index promised below. Both re-runs
import the pipeline’s own binning, percentile and clamp functions, so the
only difference between arms is which pixels enter the percentile. The all-
pixel arm reproduces every frozen edge to floating-point noise with exactly

equal bin pixel counts, and its cell-aggregated index reproduces the frozen
1050 `current_tvdi_mean` column to 1.8×10^{-7} , which is float32 storage precision.
Section 4.7k reports the outcome: sea in the edge fit does move the index, by
an amount that scales with each AOI’s sea fraction, and it does not change
any region’s direction.

Downscaled and fused LST. To mitigate Landsat’s sparse thermal
1055 revisit, a MODIS→Landsat downscaling model is trained on the predic-
tor window and applied to the full 30 m grid (`step7c`, `step7d`), yield-
ing `downscaled_lst` in °C, clipped to configured physical bounds. The
downscaling stage is explicitly leakage-guarded: it never sees MCD64A1
or FIRMS labels, and is prevented from ingesting the anomaly, TVDI or
1060 TVDI-difference layers as inputs (`step7d_predict_downscaled_lst.py:1`
to 30). A fused product (`fused_lst`, °C) then combines observed Land-
sat LST with downscaled LST as **gap-fill only**. Valid observed pixels are
never replaced or blended (`step7e_fuse_landsat_downscaled_lst.py:1`
to 26). A companion source mask records, per pixel, whether the fused
1065 value is observed, gap-filled or invalid. The corresponding per-cell fractions
are retained as sensitivity diagnostics and are **never** used as predictors.

Three properties of this step are reported here because a reader cannot
otherwise judge what the two derived channels are. First, the downscaling
model is validated, on spatially blocked 64×64 train/validation/test splits,
1070 each against a MODIS-baseline control. Test RMSE is 2.04 °C (R^2 0.866)
for Manavgat, 1.75 °C (0.795) for Bejís, 1.65 °C (0.956) for Muğla, 1.80
°C (0.866) for Evia and 1.83 °C (0.909) for Montiferru. Second, the down-
scaler’s own inputs include `lon`, `lat`, `row`, `col` and their normalised forms.

Their summed importance is 0.123 (Manavgat), 0.122 (Bejís), 0.097 (Evia),
 1075 0.066 (Montiferru) and 0.035 (Muğla), so `downscaled_1st`, and `fused_1st`
 on its gap-filled share, carry a smooth coordinate-derived component. The
 downscaler never sees a fire label, so this is not label leakage, but Section 3.13
 excludes coordinates from the fire model and this is the one route by which
 a coordinate-derived surface re-enters it. The dominant input also differs
 1080 by region: the MODIS context layer in Manavgat (0.525) and Evia (0.593),
 NDVI in Bejís (0.482) and Montiferru (0.666), and slope in Muğla (0.777),
 where the channel is largely a re-expression of a baseline predictor. Third,
the MODIS input is screened to two different standards across the
cohort. All five regions use single-season predictor-window MODIS sum-
 1085 mary layers, each matched to its own window, as recorded in every region's
`data/modis/modis_metadata.json`. What differs is quality control. Evia
 and Montiferru apply a `QC_Day` bit rule and require at least three valid daily
 observations per pixel, and they write an explicit nodata sentinel of `-9999`.
 Manavgat, Bejís and Muğla apply no quality mask and declare no nodata
 1090 value, so cells with nothing to report are encoded as exact `0.0 °C`. The screen-
 ing was added to the export script on 2026-07-23, after the first three regions
 had already been exported, so the cohort is split by export date rather than
 by design.

The zero-fill was measured for this paper rather than inferred, and it
 1095 is not distributed as the nodata signature alone would suggest. Counting
 exact zeros in each region's MODIS mean layer gives 518 of 6,390 pixels for
 Manavgat (8.11 %), **7,347 of 19,190 for Muğla (38.29 %)** and **none**
at all for Bejís. Both non-zero fractions are above the pipeline's own 5

% suspicious-zero guard (`STEP7B_MODIS_SUSPICIOUS_ZERO_FRACTION`), and
1100 Muğla's is more than four times Manavgat's. The two regions that apply the
QC rule contain no exact zeros, because they carry an explicit `-9999` sentinel
instead. What the zeros are is also settled by the counts: 8.11 % against a
water-dominant cell share of 8.3 % in Manavgat, and 38.29 % against 38.9
% in Muğla, while inland Bejís has 0.1 % water and no zeros. The zero-fill
1105 is therefore the sea rather than missing observation, and it is absent from
the one AOI with no coastline. This matters because `downscaled_1st` and
`fused_1st` rest on these layers, and Section 4.7h reports how much of the
thermal increment those two channels carry. Section 5.11 records it as an
untested candidate explanation.

1110 A note on provenance, because the frozen export contradicts itself here.
Manavgat's Step 7C and 7D metadata describe the same raster as a four-
year summer-mean context layer. That string is a stale literal: at the pinned
commit it is emitted only for the Kozan negative control, and the conditional
that restricts it was committed on 2026-07-10, one day after Manavgat's Step
1115 7C ran. Manavgat is the only region carrying it, and the region's own `modis_`
`metadata.json`, written by the exporting run three hours earlier, states the
single-season predictor window and says in as many words that the layer is
not a multi-year baseline. The `modis_metadata.json` record is the one taken
as authoritative here.

1120 **Terrain.** Elevation (m a.s.l.) and slope (degrees, from `ee.Terrain.`
`slope`) are derived from the Copernicus DEM GLO-30 (ESA, 30 m global
digital surface model; [https://dataspace.copernicus.eu/explore-data/](https://dataspace.copernicus.eu/explore-data/data-collections/copernicus-contributing-missions/collections-description/)
[data-collections/copernicus-contributing-missions/collections-description/](https://dataspace.copernicus.eu/explore-data/data-collections/copernicus-contributing-missions/collections-description/)

COP-DEM, accessed 2026-08-08), accessed as the COPERNICUS/DEM/GL030 Earth
1125 Engine collection. USGS SRTMGL1 v003 is configured as a fallback should
the preferred collection be unavailable (`core/config.py:61 to 80`). The
frozen Step 2B metadata records `used_fallback: false` for every region,
so all reported elevation and slope values come from GLO-30 and the fall-
back was never exercised. Because GLO-30 is a mosaicked collection without
1130 a fixed projection, slope is computed on the DEM's native projection be-
fore reprojection, not after (`step2b_dem.py:148 to 159`). Both bands are
exported at 30 m.

Land cover. ESA WorldCover v200 (10 m native, 2021 epoch), nearest-
neighbour aligned to the 30 m reference grid. The 2021 epoch is applied
1135 unchanged to every region, including Bejís 2022, so for that region the land-
cover layer precedes the fire year by one growing season. Class codes: 10 tree
cover, 20 shrubland, 30 grassland, 40 cropland, 50 built-up, 60 bare/sparse
vegetation, 70 snow/ice, 80 permanent water, 90 herbaceous wetland, 95
mangroves, 100 moss/lichen.

1140 3.5. Cell-level aggregation, validity mask and analysis population

Aggregation. For each continuous predictor and each 17×17 block, the
mean, median, standard deviation, valid-pixel count and valid-pixel fraction
of the finite 30 m values are computed (`step8a:954 to 963`). Modelling
uses the block **mean** of each predictor. For land cover, the modal class
1145 (`landcover_dominant`) and per-class area fractions are computed over valid
land-cover pixels within the block.

Validity. A cell is `valid_for_modeling` if and only if all four of the
following hold (`step8a:1399 to 1422`):

1. it is analysis-eligible, meaning it was not excluded as a pre-label burn;
- 1150 2. at least `STEP8A_MIN_30M_VALID_FRACTION = 0.3` of its 30 m pixels have simultaneously finite NDVI, elevation and slope;
3. its aggregated NDVI, elevation and slope means are each finite;
4. it contains at least one valid land-cover pixel.

Thermal predictors deliberately do **not** enter the validity test, so that
1155 thermal data availability cannot silently reshape the population differently for the two feature sets. The label likewise plays no part in the validity test. That is a design point made explicit in the source, since requiring a valid burn date would collapse the dataset to burned-like cells.

Analysis population. The **primary** population is natural vegeta-
1160 tion: cells that are `valid_for_modeling` and satisfy `burnable_tree_shrub_grass`, defined as a combined tree + shrubland + grassland **area fraction** $\geq$ `STEP8A_BURNABLE_FRACTION_THRESHOLD = 0.50` within the cell (`step8a:1339` to `1342`). Cropland is explicitly excluded from every burnable mask and only ever reported as its own fraction. Restricting to natural
1165 vegetation removes the confound in which agricultural stubble burning and bare-surface thermal contrast could be mistaken for wildfire skill; it is the scientifically defensible population for the transfer question, and, as the analysis shows, the choice materially changes the conclusion. The mixed population of all valid cells (`all_valid`) is retained throughout as a **secondary**
1170 **sensitivity** analysis, and both populations are reported for every headline quantity.

Two definitional asymmetries are noted for transparency: the burnable mask is an **area-fraction** test, whereas the admissibility gate of Section 3.3

uses the **modal** class; and the upstream within-region pipeline’s own configured primary population is `all_valid` (`STEP8B_PRIMARY_POPULATION`, `core/config.py:559`), whereas the transfer analysis reported here takes natural vegetation as primary. All comparisons in this paper are computed on the *same* population and the *same* classifier configuration. This holds for the within-region reference, naive transfer and adapted transfer alike, which is what makes the decomposition of Section 3.12 valid.

`burnable_tree_shrub_grass` is used exclusively as a population mask and is never a feature.

3.6. Feature sets

Two nested feature sets are compared (`step8b_train_baseline_vs_thermal_model.py:121` to `135`):

- **Baseline (4 predictors):** `ndvi_mean`, `elevation_mean`, `slope_mean`, `landcover_dominant`.
- **Thermal (10 predictors):** the baseline set plus `lst_anomaly_mean`, `current_lst_mean`, `current_tvdi_mean`, `tvdi_difference_mean`, `downscaled_lst_mean`, `fused_lst_mean`.

The thermal set is a strict superset of the baseline set, so the difference in performance between them is attributable to the six thermal channels alone. `landcover_dominant` is the only categorical predictor.

Three properties of this table matter for the interpretation and are established elsewhere in this paper. The four baseline predictors are static apart from `ndvi_mean`, which is a predictor-window median composite. The six

Table 2: Feature dictionary. Every modelled predictor, its source product, the pipeline stage that produces it, its unit, whether it is composited over the predictor window or time-invariant, and its feature-set membership. Cell-level values are means over the ~ 510 m cell, except **landcover_dominant**, which is the modal class. Specifications are in Section 3.4.

Predictor	Source product	Stage	Unit	Temporal	Baseline	Thermal
ndvi_mean	Landsat C2 L2 surface reflectance	step4	dimensionless	window median	yes	yes
elevation_mean	Copernicus DEM GLO-30	step2b	m a.s.l.	static	yes	yes
slope_mean	Copernicus DEM GLO-30	step2b	degrees	static	yes	yes
landcover_dominant	ESA WorldCover v200 (2021 epoch)	step2c	class code	static	yes	yes
current_lst_mean	Landsat C2 L2 surface temperature	step3	$^{\circ}\text{C}$	window median	no	yes
lst_anomaly_mean	Landsat C2 L2, against baseline years	step5	z-score	window vs climatology	no	yes
current_tvdi_mean	Landsat C2 L2 LST and NDVI	step5c	dimensionless, [0, 1]	window	no	yes
tvdi_difference_mean	Landsat C2 L2, against baseline years	step5c	dimensionless	window vs climatology	no	yes
downscaled_lst_mean	MOD11A1 downscaled on Landsat predictors	step7c, step7d	$^{\circ}\text{C}$	window	no	yes
fused_lst_mean	Landsat LST, gap-filled from downscaled	step7e	$^{\circ}\text{C}$	window	no	yes

thermal predictors are not six independent measurements: `fused_lst_mean` equals `current_lst_mean` outside a gap-filled share of 0.11 % to 9.70 %, and `downscaled_lst_mean` is a fitted surface whose dominant input differs by region (Section 4.7h). Those same two channels carry a smooth coordinate-derived component inherited from the downscaler's inputs, which is the one route by which coordinates re-enter a feature set that otherwise excludes them (Section 3.4, Section 3.13).

3.7. Classifier and preprocessing

All models are random forests with identical hyperparameters across the within-region, transfer and adaptation experiments (step8b:420 to 429):

```
RandomForestClassifier(n_estimators=300, max_depth=None, min_samples_leaf=3,  
                        class_weight="balanced", random_state=42, n_jobs=-1)
```

with scikit-learn defaults otherwise (`criterion="gini", max_features="sqrt", min_samples_split=2`, bootstrap resampling of training rows). `class_weight="balanced"` compensates for the strong class imbalance (burned prevalence of a few per cent). The configuration is chosen once and never tuned against any evaluation the paper reports. A second, unweighted profile with `min_samples_leaf=2` is retained solely as a sensitivity check.

Preprocessing is a scikit-learn `Pipeline` fitted **inside** each training fold (or, for transfer, on the source region only), so that no imputation or encoding statistic is derived from evaluation data (step8b:449 to 467): numeric predictors are median-imputed; `landcover_dominant` is mode-imputed and one-hot encoded with `handle_unknown="ignore"`, so that a land-cover class present in the evaluation set but absent from the training set produces an

all-zero indicator row rather than an error. This is what makes the one-hot column space consistent between regions in the transfer experiments: it is fixed by the source region. No feature scaling is applied in the within-region models, as the classifier is scale-invariant.

1225 3.8. *Spatial-block cross-validation and block-size robustness*

Within-region evaluation uses **spatially blocked** cross-validation. Contiguous square blocks of $B \times B$ analysis cells are formed at a fixed origin from the grid indices, `block_row = row_500m // B`, `block_col = col_500m // B` (`step8b:277 to 306`), and the resulting block identifier is used
1230 as the grouping variable in `StratifiedGroupKFold` with `n_splits = 5`, `shuffle=True` and `random_state = 42` (`step8b:309 to 414`; `core/config.py:554 to 556`). Blocks are therefore never split between training and test folds, and class stratification is preserved across folds. Fold construction is guarded: at least two test folds must contain a positive, and the pipeline is
1235 explicitly forbidden from ever falling back to a random row-wise split.

Both feature sets are fitted on the **same folds**, so the baseline-versus-thermal comparison is paired by construction. Out-of-fold predicted probabilities are assembled into a single vector per feature set, covering every cell exactly once, and metrics are computed once on the concatenated out-of-fold
1240 vector rather than averaged over folds (`step8b:519 to 690`).

Block size and robustness. The pipeline’s default block size is `STEP8B_SPATIAL_BLOCK_SIZE_CELLS = 2`, i.e. 2×2 cells ≈ 1 km. Because the appropriate blocking scale is not knowable a priori and short-range spatial autocorrelation can survive fine blocking, the whole within-region analysis
1245 is repeated at **$B = 2, 10$ and 20 cells ($\approx 1, 5$ and 10 km)**, on both

populations. A thermal increment that persists as blocks coarsen is evidence of genuine spatial generalisation rather than of neighbour leakage. The trade-off is that coarser blocking also reduces the number of independent groups available for cross-validation and for bootstrap resampling, which widens the intervals.

Metrics. Threshold-free discrimination is reported as ROC-AUC and PR-AUC (average precision); the Brier score is computed as a calibration diagnostic. Threshold-dependent quantities (balanced accuracy, precision, recall, F1) are recorded at a fixed 0.5 cut-off within regions but are not used for the paper’s claims (`step8b:480 to 503`). The primary quantity of interest is the paired difference $\Delta\text{AUC} = \text{AUC}(\text{thermal}) - \text{AUC}(\text{baseline})$ on the same out-of-fold predictions.

3.9. *Spatial-block bootstrap uncertainty*

Uncertainty is quantified by a **spatial-block bootstrap** operating on stored predictions, with no refitting (`step8c_spatial_block_bootstrap_uncertainty.py`). The resampling unit is the spatial block, not the cell: in each replicate, `n_blocks` blocks are drawn with replacement from the `n_blocks` unique blocks, and a block drawn k times contributes all of its cells k times (`step8c:278 to 351`). Resampling cells individually would treat spatially autocorrelated neighbours as independent and produce intervals that are far too narrow. This was an error we made and corrected during this study (Section 3.12).

The number of replicates is 1000 and the seed is 42 (`core/config.py:574 to 575`). Intervals are **equal-tailed percentile** intervals at the 2.5th and 97.5th percentiles. No bias correction or acceleration (BCa) is applied. Repli-

cates in which a resample contains only one class are recorded and excluded from the percentiles rather than silently redrawn. A warning is raised if more than 10% of replicates are invalid.

The bootstrap is **paired**. Within each replicate, both the baseline and
1275 thermal probability vectors are evaluated on the identical resampled index, and the difference is formed inside the replicate. The interval is therefore on the distribution of the paired ΔAUC . The same paired construction is used in the transfer experiments, where a single set of resampled target blocks per replicate is shared by all probability series (within-region reference, raw
1280 transfer, z-scored transfer, CORAL transfer) so that they remain directly comparable.

Following the pipeline’s own reporting policy, no classical p-values or significance tests are reported. A result is described as having **positive bootstrap support** when the 95% percentile interval for ΔAUC excludes
1285 zero, and as **uncertain** otherwise. This is a statement about an interval, not a hypothesis test.

Block sizes used for the bootstrap match the analysis they accompany: $B = 2$ for the transfer scores, B matched to the cross-validation blocking (2, 10, 20) for the within-region robustness analysis, and $B = 10$ (≈ 5 km) for
1290 the univariate concept-shift diagnostic of Section 3.12.

3.10. Cross-region transfer protocol

For each ordered pair of regions (source $\rightarrow$ target), a model is fitted on the **entire** source region, using all cells in the analysis population with no held-out fold, and applied to the entire target region. All ordered pairs among
1295 the completed regions are evaluated in both directions.

Every fitted component is derived from the source region only: the numeric imputation medians, the categorical mode, the one-hot category vocabulary, and the random forest itself. The target region enters the computation exactly once, at `predict_proba`. There is no pooled fitting, no target-derived
1300 imputation, no target fine-tuning, no calibration on the target, and no coordinate or region-identity feature (`step9b_run_cross_region_transfer.py:230 to 236, 370 to 381`). The classifier, hyperparameters, seed and feature contract are imported from the within-region stage rather than redefined, so the transfer model is the same model.

1305 Feature harmonisation across regions requires no separate alignment step: the shared feature list is frozen and validated by an audit stage that fails on any schema mismatch, and the source-fitted one-hot encoder with `handle_unknown="ignore"` guarantees identical column ordering and dimension at target-prediction time. Land-cover classes present in the target but not the
1310 source encode to zero.

Both source and target must contain at least 30 positive and 30 negative cells in the population being evaluated, otherwise the pair is skipped.

Target metrics are threshold-free: ROC-AUC and PR-AUC, with spatial-block bootstrap intervals as in Section 3.9, computed on the target region's
1315 blocks. A ROC-AUC below 0.5 is reported as such and is interpreted as an anti-predictive transfer, in which the source-learned ordering is systematically inverted in the target. It is not folded to $\max(\text{AUC}, 1 - \text{AUC})$. Both the baseline and the thermal feature set are fitted and evaluated under this protocol for every ordered direction, so the per-direction paired baseline-versus-
1320 thermal transfer contrast reported in the Results is a read-only extraction

from these frozen outputs, taking point estimates from the transfer stage’s `baseline_metrics/thermal_metrics`, and paired Δ AUC intervals from the transfer bootstrap’s `delta_roc_auc` field, which evaluates both probability series on identical resampled target blocks as in Section 3.9 (extraction script `paper/baseline_vs_thermal.mjs`; no new models fitted).

3.11. Label-blind domain adaptation

Three transfer variants are compared. All are strictly **label-blind**: the target frame is structurally stripped of the response before adaptation, and an assertion fails the run if a label column is present (`step10b_label_blind_adaptation.py:71 to 79`; `core/step10_shared.py:243 to 250`). No adaptation function accepts labels as an argument.

(a) **Raw.** The source-fitted model applied directly to untransformed target features (Section 3.10).

(b) **Region-wise z-score.** Each region’s numeric features are standardised using **its own** statistics: $z = (x - \mu_{\text{region}}) / \sigma_{\text{region}}$, with the mean and standard deviation computed over non-missing values only and `ddof = 0` (`step10_shared.py:145 to 173`). A near-constant feature ($\sigma < 10^{-12}$) has its divisor set to 1 and the substitution is recorded. Missing values are filled with the region’s own mean, hence map to exactly zero after transformation. The categorical land-cover predictor is left untouched. Source statistics come from source data and target statistics from target data. The two regions are never pooled. The classifier is then refitted on the z-scored source and applied to the z-scored target. This variant removes first- and second-order marginal offsets between regions. It is the simplest possible self-calibration.

(c) **CORAL after region-wise z-score.** Starting from the z-scored features of (b), the source covariance is aligned to the target covariance by the standard CORAL whitening-recolouring map (step10_shared.py:186 to 220):

```

1350 C_s = cov(X_s^z) + lambda*I ,    C_t = cov(X_t^z) + lambda*I
      A    = C_s^(-1/2) * C_t^(1/2)
      X_s* = X_s^z * A

```

with covariances computed at `ddof = 0` and matrix powers obtained by symmetric eigendecomposition with eigenvalues floored at 10^{-12} . The regularisation is `STEP10_CORAL_LAMBDA = 1e-5` (core/config.py:697). Critically, **the transform is applied to the source only**. The target remains exactly as in (b). The classifier is refitted on the aligned source and applied to the unchanged z-scored target.

On the choice of λ . The canonical CORAL formulation adds the identity ($\lambda = 1$) to each covariance, but it is defined for unstandardised features. Here CORAL is applied *after* region-wise standardisation, so each feature already has unit variance and adding the full identity doubles the diagonal, imposing shrinkage strong enough to erase the covariance structure the alignment is meant to exploit. With only nine numeric features and thousands of cells per region, the covariances are well estimated and minimal regularisation is appropriate. Because this is a defensible but not inevitable choice, λ sensitivity was assessed, and the exact scope of that assessment is stated here. Nine λ values were run: 0, 10^{-8} , 10^{-7} , 10^{-6} , 10^{-5} , 10^{-4} , 10^{-3} , 10^{-2} and 10^{-1} . They were run on four transfer directions, Bejís↔Muğla

1370 2021 and Manavgat↔Muğla 2021, in both the baseline and the thermal family. The frozen configuration records this grid and records $\lambda = 10^{-5}$ as the implementation’s canonical value (`drive_new/diagnostics/coral_lambda_sensitivity/b74d643e.../lambda_grid.csv` and `config.json`). The Manavgat↔Bejís directions were not re-run over the grid, and the con-
1375 figuration marks them `contextual_only_not_rerun`. The resulting spread in transfer AUC is at most 0.014 within any direction, and at most 0.008 within the thermal family; the three widest spreads are all baseline-family rows (Section 4.7d).

Why $\lambda = 1$ was not run. The canonical value is the one value the
1380 sweep does not contain, and the omission is deliberate rather than accidental. In the canonical formulation the identity is added to the covariance of un-standardised features, where it acts as a mild ridge relative to feature scale. Here every feature has already been given unit variance by the region-wise z-score of variant (b). Adding the full identity therefore doubles each diag-
1385 onal entry and halves the relative weight of every off-diagonal term. The alignment map is pushed towards the identity and the covariance structure that CORAL exists to transport is largely erased. A $\lambda = 1$ arm would mostly measure how the pipeline behaves when CORAL is effectively switched off. It would not measure how sensitive the reported result is to reasonable reg-
1390 ularisation, which is what the sweep is for.

The $\lambda = 1$ arm was subsequently computed for this paper, and the argument above turns out to overstate the case. The released sweep tool cannot be extended: its λ grid is a fixed token sequence and the module asserts its own fit count at import, so the arm was produced instead

1395 by driving the pipeline’s own primitives (`fit_coral_alignment`, `apply_coral`, the region-wise z-score, and Step 8B’s pipeline builder) in the order `step10b` calls them. $\lambda = 10^{-1}$ was recomputed alongside as a control and reproduces the frozen sweep to 4.4×10^{-9} over all eight rows, seven of them exactly. At $\lambda = 1$ no direction changes side of the chance line, and
 1400 the largest movement is an *improvement*, Muğla→Manavgat thermal from 0.559 to 0.574. Heavy regularisation therefore does not switch CORAL off in this region set, as the scale argument predicted it would. Including $\lambda = 1$ widens the spread over the grid to at most 0.019 in any direction and 0.016 within the thermal family, against 0.014 and 0.008 over the released grid.
 1405 Section 4.7d reports this.

An earlier two-region version of this study, using Manavgat and Bejís alone, ran λ at 10^{-5} , 10^{-3} , 10^{-1} and 1, and there the one direction whose interval sat above chance lost that status at $\lambda = 1$. That analysis is superseded throughout by the five-region analysis, its region set is different, and
 1410 the four directions of the sweep do not include Bejís↔Manavgat, so the two results are not in direct contradiction. Nothing further is drawn from it. The sweep also does not cover the Montiferru and Evia directions, so conclusions for those directions rest on the default $\lambda = 10^{-5}$ alone.

The order of operations for variant (c) is thus: region-wise z-score of
 1415 both regions → CORAL alignment of the source numerics only → the standard median-imputation + one-hot pipeline → random-forest fit on source → prediction on target.

3.12. Transfer-gap decomposition and concept-shift diagnostic

Decomposition. For each ordered pair, the transfer deficit is decomposed against the target region's own within-region performance:

- $\text{total_gap} = \text{AUC_within}(\text{target}) - \text{AUC_raw}(\text{source} \rightarrow \text{target})$
- $\text{recovered} = \text{AUC_adapted}(\text{source} \rightarrow \text{target}) - \text{AUC_raw}(\text{source} \rightarrow \text{target})$
- $\text{concept_remaining} = \text{AUC_within}(\text{target}) - \text{AUC_adapted}(\text{source} \rightarrow \text{target})$

where AUC_within is the target region's own spatially blocked out-of-fold thermal ROC-AUC. All three terms are computed on the **same population, same feature set, same classifier configuration and same block size**, which is what makes them commensurable. A decomposition assembled from components with differing populations or classifier settings would not be interpretable.

The **recovered** term is defined using the **best-performing label-blind method** among (b) and (c), because the question it answers is "how much of the gap *can* label-free alignment close?", whose answer is bounded by the best available label-free method. The weaker method's decomposition is reported as a secondary row rather than discarded, since choosing the weaker method as the numerator would understate the recoverable fraction and overstate concept shift.

Concept-shift diagnostic. To identify the mechanism behind the residual, a **signed** univariate ROC-AUC is computed for every numeric feature against the burned label, independently in each region's analysis population. The AUC is deliberately *not* folded to $\max(\text{AUC}, 1 - \text{AUC})$, so that the direction of association is preserved: a value above 0.5 means higher feature

values are associated with burning, below 0.5 means the opposite. A feature is said to show a **point reversal** when its univariate AUC falls on opposite sides of 0.5 in the two regions, and a **bootstrap-supported reversal** only when the two regions' spatial-block bootstrap intervals are additionally **disjoint**.

The block size for this diagnostic is $B = 10$ cells (≈ 5 km). An earlier version of the analysis used $B = 2$ (≈ 1 km) and, by ignoring short-range spatial autocorrelation, produced intervals roughly four times too narrow, which wrongly promoted several point reversals to bootstrap-supported. The correction is reported here because it materially weakened one of the study's own claims: at the appropriate blocking scale, far fewer reversals survive as statistically supported, and the manuscript reports the conservative result.

The logical role of this diagnostic is specific. Region-wise standardisation and covariance alignment can only translate, scale and rotate the feature distribution; neither can invert the sign of a feature's association with the response. A bootstrap-supported reversal is therefore direct evidence for a shift component that no label-free alignment can repair, and connects the empirical decomposition to a mechanism.

3.13. Leakage control, sensitivity analyses and reproducibility

Leakage control. An explicit forbidden-column set is enforced at every model fit, and its violation raises rather than warns (`step8b:140` to `159`, `265` to `271`, `450`). It contains the response itself and every column carrying label information: `burn_date`, `burn_month`, `burn_day_of_year`, `label_source`, `burn_date_pixel_agreement_fraction` and `out_of_window_burndate`. It also contains `lon`, `lat`, `cell_id`, `row_500m`, `col_500m`, the validity columns,

and the fused-LST provenance columns (`source_mask_majority`, `observed_fraction`, `gapfilled_fraction`, `invalid_source_fraction`). The transfer stage additionally forbids `experiment_id`, `region_key`, `spatial_block_id` and `fold_id`.

Coordinates are excluded because a fire scar is a spatially compact object: with longitude and latitude available, a sufficiently flexible model can memorise the scar’s location instead of learning any relationship with the surface state. Grid indices are used only to construct cross-validation blocks and bootstrap groups. The natural-vegetation mask is used only to define the population. The fused-LST provenance fractions are retained as diagnostics, never as predictors. They are reported per region in Section 4.7h, where the gap-filled share is 0.11 % to 9.70 % of the fused product. A performance sensitivity restricted to cells with a low gap-filled fraction **was run** for this paper and is reported in Section 4.7m. It matters most for Bejís, whose 9.70 % is an order of magnitude above the other four regions, and Bejís is indeed where it moves the increment most, from +0.056 to +0.043, while the increment keeps bootstrap support in all five regions.

Sensitivity analyses. Every headline result is repeated across: two analysis populations (natural vegetation, primary; all valid cells, secondary); three spatial-block sizes for the within-region analysis ($\approx 1, 5, 10$ km); nine CORAL regularisation values from 0 to 10^{-1} on the four directions covered by the sweep (Section 3.11); both feature sets. Where a conclusion depends on one of these choices, the dependence is reported rather than resolved by selecting the favourable setting. Two axes named in earlier drafts are not part of this set and should not be read as covered. The unweighted `min_samples_`

`leaf = 2` random-forest profile was defined as a secondary configuration but no result from it is reported in Section 4 or Section 5, and the CORAL sweep does not include $\lambda = 1$ in the five-region export.

1495 **Reproducibility.** All randomness uses seed 42, covering model `random_`
`state`, cross-validation shuffling and bootstrap generators alike. The boot-
strap uses 1000 replicates throughout. The transfer and adaptation analysis
is executed in a dedicated Python environment separate from the upstream
pipeline environment; because random-forest fits are not bit-identical across
1500 scikit-learn versions, the analysis includes an explicit reproduction check in
which the within-region models are refitted in the new environment and
compared against the frozen upstream outputs, and the independently im-
plemented adaptation is compared against the pipeline’s own implementa-
tion. All numbers reported in this paper were produced under, or verified
1505 against, scikit-learn 1.9.0: two frozen transfer probes reproduce to four deci-
mal places under 1.9.0 but move by +0.021 and +0.026 AUC under 1.7.2 with
byte-identical data, pipeline and seed, so cross-region point estimates carry
an implementation tolerance of roughly ± 0.02 to 0.03 unless the exact li-
brary version is fixed. Within-region AUCs reproduce to ~ 4 decimals across
1510 versions (`paper/sklearn_version_sensitivity.md`; probe script `paper/`
`pairwise_check.py`).

The reproduction check was executed on the final five-region set by the
pipeline author, and its record is archived with this manuscript (`paper/`
`reproduction_check/reproduction_check_5region.json`, SHA-256 `7f7e41f5...09c8be`;
1515 `commit 48b56e7`, Python 3.12.3, scikit-learn 1.9.0, pandas 3.0.2, NumPy
2.4.4, primary population `burnable_tree_shrub_grass`). Two families of

comparison were re-executed against the frozen outputs. The within-region arm refitted `step8b`'s baseline and thermal models on the frozen Step 8A parquet of each of the five experiments and compared ROC-AUC and PR-AUC. All twenty comparisons were produced, with a **maximum absolute difference of exactly 0**. The transfer arm regenerated the label-blind CORAL predictions and re-evaluated them for all twenty directed region pairs, again in both model families and both metrics. That is eighty comparisons over twenty directions, none missing, with a **maximum absolute ROC-AUC difference of 1.6×10^{-7}** (largest case Montiferru→Manavgat, thermal: 0.6060780 against 0.6060778); sixty-nine of the eighty comparisons were bit-identical and the remaining eleven differed by 1.6×10^{-7} or less.

That check was run by the pipeline author, and the script behind it is not in the released repository, so a reader cannot re-execute it (Section 5.11 states this). We therefore repeated part of it independently, using only code a reader does have. The environment was rebuilt from scratch on a different operating system, Windows 11 rather than Linux, pinning the three libraries that govern the fits to the versions the check records (NumPy 2.4.4, pandas 3.0.2, scikit-learn 1.9.0) on Python 3.12.10, and `src/step8b_train_baseline_vs_thermal_model.py` was executed unmodified at commit 48b56e7 against the frozen Step 8A parquet of two regions. Every numeric field of the resulting `step8b_model_comparison_metrics.json` was compared against the archived one: 142 fields for Manavgat with a maximum absolute difference of 6.9×10^{-18} , in a Brier score, and 168 fields for Montiferru with a maximum absolute difference of **exactly 0**. No field differed by more than 10^{-9} , and every reported ROC-AUC agreed to sixteen significant digits, includ-

ing the Table 4 values for Manavgat (baseline 0.8027358197042693, thermal 0.8696419777927898). The within-region results of this paper are therefore reproducible from the released code and the archived data alone, on different hardware and a different operating system, once the library versions are fixed. This does not extend to the transfer arm, whose reproduction-check script remains unavailable.

The tolerance against which these differences are judged is not a criterion chosen for this report. It is the repository’s own pre-existing Step 10C fail-fast reproduction criterion, an absolute difference of $\leq 1 \times 10^{-6}$ defined in `src/step10c_paired_evaluation_bootstrap.py`. It is applied unchanged, and the check record states this explicitly. Both arms fall inside it, the within-region arm by exact equality. The reported **PASS** is a technical-completion status covering cohort resolution, comparison coverage and input-hash agreement. The achieved numerical differences are reported separately, as above, rather than folded into that status.

What this check does and does not establish should be stated plainly. Because it re-executes the pipeline in the same library environment that produced the frozen outputs, the exact-zero within-region agreement demonstrates determinism and the absence of undeclared state, meaning that the recorded numbers are regenerable from the recorded inputs, hashes included. It does not demonstrate robustness to a change of environment. The residual 10^{-7} -scale differences in the transfer arm arise from re-derived rather than re-read intermediate quantities, not from a different library stack. Cross-version behaviour is established separately and is far coarser: the ± 0.02 to 0.03 implementation tolerance quoted above, from the 1.9.0-against-1.7.2

probes. The two statements are complementary and neither substitutes for the other.

Two further properties of the record are stated for completeness. First, the five-region cohort was resolved by three independent routes. These are the experiment registry restricted to canonical records (those carrying no `superseded_by` key, which drops the legacy Evia AOI of Section 3.16.1 and the superseded Muğla calendar-shift record) and with non-cohort roles removed, the ERA5-Land diagnostic’s `DEFAULT_EXPERIMENTS` constant, and the frozen multi-AOI synthesis manifest. All three return the same ordered set (`routes_agree: true`). The two deliberate exclusions by role are Kozan 2023 as `negative_control` and the second Muğla event as `temporal_transfer_wildfire` (Sections 3.1, 3.3, 3.16.4). The roles and the supersession links are as recorded in `core/regions.py`. Second, two unordered region pairs carry duplicate Step 10 namespaces on disk under reversed directory orderings with distinct analysis identifiers (Bejís-Muğla and Manavgat-Muğla). The reference artefact was selected by the frozen synthesis manifest rather than by directory listing, and the check record names both the paths present and the one used for each pair. No file belonging to the upstream pipeline or to previously frozen outputs was modified by this analysis.

Data and code availability. The satellite inputs are all public and are obtained through Google Earth Engine. They are MODIS MCD64A1 Collection 6.1 burned area, MODIS land surface temperature, Landsat Collection 2 Level-2 surface reflectance and surface temperature, Copernicus DEM GLO-30, and ESA WorldCover v200. No data were collected by the authors and no restricted or licensed data were used. The region definitions, date windows

and thresholds given in this section are sufficient to regenerate every input from these sources.

The full processing pipeline (Steps 1 to 10), its configuration constants, and the frozen numeric outputs on which every reported number rests are publicly available at <https://github.com/emrehann17/satellite-thermal-digital-twin>. The repository is the authoritative source for the file and line references cited throughout this section. Analysis code is released under the repository's stated licence (MIT).

Three limits on that release are stated here rather than left for a reader to discover. First, **no digital object identifier is minted and no archival deposit exists**, and the repository carries no tags, so "the version of record" is a commit identifier on a moving branch head rather than a citable snapshot. Readers should cite the repository URL together with commit 48b56e7, which is the state against which every file and line reference in this section was checked. Second, **the code that produced the reproduction check of this section is not in the released tree at that commit**. The check's own record marks `scripts/run_reproduction_check.py` and `src/reproduction_validation/` as untracked, and neither is present at 48b56e7. The check's output artefact is archived with this paper and its 100 recorded values were independently verified against the 20 frozen artefacts they reference, but a reader cannot regenerate it from the repository. Third, **the commit recorded for the few-shot export (19d825b) is not reachable** in the published history, although the module it names, `src/few_shot_recovery.py`, is present at 48b56e7 and predates the run. A fourth, smaller discrepancy is recorded in Section 3.17: the ERA5-Land diagnostic's man-

ifest names a commit at which the diagnostic source does not yet exist, so that production run was made from an uncommitted working tree.

None of these affects a number in this paper, each was verified rather than
1620 assumed, and every affected artefact is archived in frozen form. They do bound the sense in which this negative result is independently re-checkable, and the paper claims no more than that bound allows.

3.14. *Transferability diagnostics versus transfer*

To test whether any pre-transfer measure of region similarity predicts
1625 transfer outcome, twenty candidate diagnostics from four families are each evaluated against the same target quantity under one common correlation framework. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime spatial structure $P(y)$, and conditional feature-response direction $P(y|x)$.

1630 3.14.1. *Rank-correlation framework*

The target quantity is the raw thermal transfer ROC-AUC of Section 3.10 on the natural-vegetation population, over the 20 ordered directions (12 for measures available only on the four-region subset excluding Montiferru). For each diagnostic, Spearman's ρ and Kendall's τ -b (tie-corrected) are computed
1635 against the transfer AUCs. Uncertainty uses a **pair-based bootstrap**: the unordered region pairs (10, or 6 for the subset) are resampled with replacement, and every sampled pair contributes **both** of its ordered directions. The two directions of a pair share geography and data, so they are never treated as independent draws. 2,000 replicates are used with equal-tailed
1640 percentile 95% intervals; replicates yielding degenerate (undefined) rank cor-

relations are excluded and counted. All resampling is seeded from 42 with fixed per-measure offsets and implemented with the mulberry32 PRNG in Node.js. It is fully deterministic, though not bit-identical to NumPy's generator (`paper/regime_correlation.mjs`, `paper/niche_corr.mjs`, `paper/conditional_similarity.mjs`). Because the effective sample is 10 (or 6) unordered pairs that share member regions, power is low. A null result is reported as "not shown to order transfer", never as "shown not to".

Provenance of the candidate set and of the stated expectations. Three things were fixed before the correlations were computed, and each is on record in the project's analysis documents. First, the candidate variants of each family were listed in advance, together with the expected sign of each, and the analysis records state this for the conditional family (`paper/conditional_similarity_transfer.md`), the niche family (`paper/niche_overlap_transfer.md`) and the marginal and regime families (`paper/regime_transfer_correlation.md`). Second, the null expectation for fire-regime distance, that it would not order transfer, was written down before the regime correlation was run and is recorded in `paper/regime_transfer_correlation.md`. Third, the directional hypothesis behind the whole comparison, that conditional measures would order transfer where marginal measures would not, comes from the project's direction document (`paper/POSITIONING.md`, Section 5), which sets it out as the analysis still to perform. These are entries in a project analysis log. They are **not** a formal pre-registration. No independent timestamped public registration exists, and the records entered version control on the same day the diagnostics were computed, 2026-08-08. The claim made here is therefore the narrow

one: the candidates and their expected signs were fixed in advance under the project’s own honesty rule, and they are on record in named files. A reader who wants a stronger guarantee than an internal log should read the regime and conditional results at that reduced strength.

1670 **Multiplicity.** Twenty candidate variants are correlated against the same target quantity. One of them, the signed-AUC vector Spearman restricted to CI-supported features, is not computable, because only two directions retain three or more jointly supported features. Nineteen variants are therefore computed. They share the same transfer AUCs and an effective sample of
1675 ten unordered pairs, or six for the measures available only on the four-region subset. Two intervals exclude zero, both in the conditional family. **No multiplicity correction is applied, and no family-wise error control is claimed.** The two surviving intervals must not be read as significance at a controlled error rate. What can be said in mitigation is limited but
1680 real. The families and their variants were specified in advance rather than searched over, so the twenty are a fixed list and not the survivors of a wider hunt. The conditional-versus-marginal contrast is a directional hypothesis stated before computation, so the two surviving variants are the ones that hypothesis pointed at rather than an arbitrary pair picked after the fact.
1685 Neither point substitutes for an error-rate correction, and none is offered.

3.14.2. Marginal measures and the domain classifier

Six marginal diagnostics are consumed unchanged from the upstream marginal/area-of-applicability audit (12 directed pairs, four-region subset; `drive_new/diagnostics/marginal_aoa_completion/4b2a1c86.../comparison/`
1690 `marginal_diagnostics_with_transfer.csv`): target mean and 95th-percentile

predictor-space dissimilarity, fraction of target cells inside the weighted area
 of applicability, and fraction inside the unweighted support (all directed),
 plus climatic distance and geographic centroid geodesic distance (symmet-
 ric). Marginal separability is additionally measured for all ten pairs by a
 1695 **domain classifier** trained to distinguish source from target cells. It uses the
 same 10-predictor feature contract (`step9_shared_baseline_thermal_v1`),
 the step8b preprocessing pipeline reused unmodified, and the random-forest
 configuration of Section 3.7, evaluated as spatially blocked out-of-fold ROC-
 AUC using `StratifiedGroupKFold` over deterministic 10-cell (≈ 5 km) blocks
 1700 namespaced per region (5 splits, seed 42, strict zero-overlap folds), with a
 paired two-domain spatial-block bootstrap (1,000 replicates, seed 42, blocks
 resampled with replacement independently within each domain). The do-
 main classifier itself never uses burned labels. Design read from `drive_new/`
`diagnostics/domain_classifier_audit/comparison/manifest.json` and
 1705 the per-pair manifests.

One qualification applies to the weighted area-of-applicability rows, and
 the upstream module states it in its own source: the predictor weights are im-
 portances from a random forest fitted on the **source** region's **burned** label, so
 the weighted index is not label-blind in the absolute sense. It is blind to the
 1710 *target's* labels, which is the property that matters for pre-deployment screen-
 ing, and that is the sense in which the marginal family is called label-free in
 this paper. The unweighted support fraction, the dissimilarity quantiles, the
 climatic and geographic distances and the domain classifier use no label at
 all.

1715 3.14.3. Burned-niche overlap

Niche overlap compares the two regions' burned-cell feature distributions, $P(x|y=1)$, using burned cells of the primary population only (`valid_for_modeling & burnable_tree_shrub_grass & burned = 1`; no unburned variant by design). Schoener's $D = 1 - \frac{1}{2}\sum|p - q|$ and Warren's $I = 1 - \frac{1}{2}\sum(\sqrt{p} - \sqrt{q})^2$ are computed in two ways. The first is per numeric feature on 1-D
 1720 histograms with 50 shared bins spanning the **global** range of all five regions' burned cells, averaged over the nine features, and (b) on a 20×20 histogram over the first two components of a global PCA fitted to the pooled, globally standardised (ddof = 0) burned cells of all five regions, with missing
 1725 values imputed by the pooled median. The Mahalanobis distance between standardised burned centroids uses the pooled covariance $((n_a - 1)S_a + (n_b - 1)S_b)/(n_a + n_b - 2)$ over the nine features. All measures are symmetric per unordered pair (`paper/niche_overlap.py`; correlation with transfer via `paper/niche_corr.mjs`).

1730 3.14.4. Conditional sign-agreement index

For each region a nine-dimensional vector of **signed** univariate AUCs (Section 3.12: the probability that a burned cell carries the higher feature value; never folded about 0.5) is taken, with its ≈ 5 km-block bootstrap intervals, from the upstream five-region feature-stability table (`drive_new/diagnostics/multi_aoi_transfer_synthesis/<five-region set>/multi_aoi_feature_stability.csv`, step9g family). Pair-level measures: sign-agreement count and fraction (features whose $AUC - 0.5$ signs match), the Spearman correlation between the two vectors, and the cosine similarity of the $(AUC - 0.5)$ vectors. Each is also computed restricted to **supported** fea-

1740 tures, which are those whose bootstrap interval excludes 0.5 in *both* regions. Consistency is enforced: per-region vectors must be identical across every pair row, and the supported-reversal set must reproduce the upstream **reversal_status** flags exactly (`paper/conditional_similarity.mjs`). This index requires burned labels (or a labelled probe) in **both** regions: it diagnoses
1745 concept alignment and is not a label-free deployment screen.

The same is true of two of the three families it is compared against, and the paper’s taxonomy is stated here once so that no section overstates the contrast. Burned-niche overlap (Section 3.14.3) is defined on the burned cells of both regions, and fire-regime structure (Section 3.14.5) on the target’s
1750 burned map, so both require the target’s labels exactly as this index does. Only the marginal $P(x)$ family, comprising the area-of-applicability rows, the domain classifier and the climatic and geographic distances, is computable before any target label exists. The correct split is therefore **target-label-free** (marginal) against **target-label-requiring** ($P(x|y=1)$, $P(y)$ and $P(y|x)$),
1755 not conditional against everything else. This sharpens rather than softens the comparison in Section 4.4: niche overlap and the conditional index are built from the same labelled information, and only the conditional one orders the transfer matrix. It also means the single family that a practitioner could run before deployment is the family that fails.

1760 3.14.5. *Fire-regime structure*

Burned-area spatial structure, $P(y)$, is summarised by connected components of burned natural-vegetation cells, computed with 8-connectivity on the integer grid indices `row_500m/col_500m`. Per region: component count, largest-component share, and the effective component count, the in-

verse Simpson index of component size shares, $1/\sum s_k^2$. The pair-level regime distance is $|\Delta \log(\text{effective count})|$ (primary) and $|\Delta \text{largest share}|$ (secondary), both symmetric. The implementation was verified to reproduce the upstream burned-pattern audit ("Rejim" table) exactly in all five regions: component counts and largest and second component sizes match identically, and shares and effective counts match to 1×10^{-4} (`paper/burned_components.mjs`).

3.15. Interventions

3.15.1. Pooled multi-region training (leave-one-region-out)

For each held-out target region, the other four regions' primary (natural-vegetation) populations are pooled and a single model is trained on the pool and evaluated on the untouched target, for both feature sets. The classifier is exactly that of Section 3.7, and preprocessing mirrors the transfer stage: numeric medians fitted on the pooled training set, mode-filled land cover expanded to indicator columns from the training vocabulary so that unseen target classes encode to zero. Two variants are run: raw features, and region-wise z-score in which every region, the target included, is standardised with its own mean and σ (ddof = 0, numeric features only), i.e. the label-free definition of Section 3.11(b). Target metrics are ROC-AUC, PR-AUC (reported against its no-skill base, the prevalence) and Brier, with a ≈ 5 km spatial-block bootstrap on the target (blocks `row_500m//10` `col_500m//10`, 1,000 replicates, NumPy `default_rng(42)`, single-class replicates excluded) (`paper/loro_pooled.py`). The analysis was run under scikit-learn 1.9.0 only after two frozen pairwise transfers were reproduced to four decimal places in the same environment (`paper/pairwise_check.py`; Section 3.13).

1790 *3.15.2. Removal of direction-reversing features*

Four thermal-model configurations are compared: full (all 10 predictors, order preserved from the shared feature contract), minus `elevation_mean`, minus `lst_anomaly_mean`, and minus both. Those two features are exactly the bootstrap-supported reversal set under the supported-reversal criterion
1795 of Section 3.14.4, which is why they and no others are removed. The pipeline replicates step8b/step9b verbatim (same imputers, one-hot encoding, classifier and seed), and **hard parity assertions** abort the run on any deviation of the full configuration from the frozen outputs, with a tolerance of 5×10^{-4} against the 20 step9b transfer AUCs and 1×10^{-3} against the step8c
1800 within-region out-of-fold AUCs. The observed deviations were 0.0000 everywhere. Within-region evaluation reuses the spatially blocked out-of-fold protocol of Section 3.8 (2-cell blocks, `StratifiedGroupKFold`, seed 42). All deltas (configuration minus full) are formed inside each replicate of a paired ≈ 5 km-block bootstrap (10-cell blocks, 1,000 replicates, `default_rng(42)`),
1805 so every interval is on the paired difference (`paper/feature_drop.py`; report `paper/feature_drop_transfer.md`).

3.16. Additional sensitivity designs

3.16.1. Evia AOI (legacy versus extended)

North Evia exists in two AOI definitions with identical predictor and label
1810 windows: the legacy $0.40^\circ \times 0.40^\circ$ box and the canonical extended $0.80^\circ \times 0.60^\circ$ box (Table 1), whose $\approx 3\times$ larger area leaves the burned scar essentially unchanged while reducing prevalence substantially (Section 4.1). Both AOIs were processed through the full pipeline, and both transfer arms (baseline and thermal, Section 3.10) were run for every Evia-involved direction un-

1815 der each AOI (`drive_new/cross_region/<pair>/step9b/cross_region_transfer_metrics.json` for the legacy and extended pair folders). The extended AOI is canonical everywhere in this paper. The legacy AOI is retained **only** to test whether AOI extent and the induced prevalence change alter any transfer conclusion.

1820 3.16.2. *Montiferru tree+shrub population*

Montiferru is the weakest admissibility-gate pass (burned cropland fraction 0.274), so a stricter population variant excluding grassland is examined: `burnable_tree_shrub`, present alongside `burnable_tree_shrub_grass` in the frozen within-region bootstrap populations (`drive_new/experiments/montiferru_2021/step8c/step8c_bootstrap_metrics.json`, `bootstrap_ci_by_population`) and in the transfer bootstrap groups of every Montiferru pair (`drive_new/cross_region/montiferru_2021_*/step9c/cross_region_bootstrap_metrics.json`). The within-region thermal ΔAUC and the eight Montiferru-involved cross-region thermal-versus-baseline deltas are compared
1825 between the two populations, reading the frozen outputs only. No new models are fitted for this check.
1830

3.16.3. *Predictor-window closure*

A predictor-timing sensitivity, described here as documented in each region's frozen comparison report (`drive_new/diagnostics/window_closure_region/<region>/<hash>/_production/<region>/compare/report/window_closure_comparison.md` for Bejís, Muğla, Evia-extended and Montiferru;
1835 `drive_new/diagnostics/window_closure_sensitivity/manavgat_2021/compare/report/window_closure_comparison.md` for Manavgat). Three variants

are compared: canonical, closure 7 days earlier, and closure 14 days earlier.

1840 The shifted variants move **both** ends of the predictor window together, so
the window length is preserved. The label window is frozen and identical in
every variant. One exact common cohort (the intersection of analysis-eligible,
primary-population, valid rows of every variant, after removing shared pre-
label-censored cells) and one shared spatial-fold assignment are used by all six
1845 evaluations (two model families $\times$ three variants), with model family, feature
registry, preprocessing, hyper-parameters and seeds held fixed. Uncertainty
is a paired spatial-block bootstrap on the model stage’s own replicate draws
with identical block draws across variants (1,000 replicates, seed 42, resam-
pling unit `spatial_block_id`). The report states the analysis measures the
1850 closure date jointly with its interaction with the fixed MODIS seasonal pro-
duction policy, not the closure date in isolation. Bejís, for scale: 12,814 cohort
rows, 967 positives, 5 folds, 3,641 blocks. The shared folds are blocked at the
pipeline’s default 2-cell edge (≈ 1 km): the analysis assigns block identi-
fiers with `add_spatial_block_id(cohort, spatial_block_size_cells)`
1855 and takes that value from `STEP8B_SPATIAL_BLOCK_SIZE_CELLS = 2` (`repo/`
`src/window_closure_sensitivity.py:8562`, `repo/core/config.py:556`).
The comparison report records fold and block counts but not the block size,
so this was read from the source rather than the report.

3.16.4. Same-geography event-to-event comparison (*Muğla 2021 versus 2022*)

1860 Muğla is the one region for which a second fire event is analysed on
the **identical** AOI and analysis grid (`region_key = mugla_aoi`, the same
bounding box and the same ~ 510 m cell definition), which allows the direc-
tion of feature-label associations to be compared with geography held fixed.

The 2022 experiment is registered as `mugla_2022_event_relative` with windows anchored to its own event: a 58-day predictor window (2022-04-24 → 2022-06-20) closing the day before ignition and a 49-day label window (2022-06-21 → 2022-08-08) opening on it, durations chosen to match the 2021 windows exactly (58 and 49 days).

This is not a clean temporal-transfer design and is not presented as one. The registry records the reason explicitly: the 2022 event ignites roughly five weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is `same_geography_event_to_event`, and that is the claim made here. The entry is deliberately kept out of the canonical five-AOI cohort by its `temporal_transfer_wildfire` role so that a second year of an already-represented AOI cannot be silently enrolled into the spatial comparisons. That exclusion is not merely a convention we observe: it is enforced in code and tested. The ERA5-Land diagnostic’s validator (`scripts/validate_era5_land_regional_diagnostic.py`) carries two relevant checks, and they have different reach. `A08_cohort_is_the_frozen_five` requires the executed cohort to equal the frozen `DEFAULT_EXPERIMENTS` tuple exactly, as an ordered comparison, so *any* addition fails it, including this experiment. `A07_mugla_2022_absent_from_default_analysis` is narrower than its name suggests: it tests literal membership of the string `mugla_2022`, which is the registry’s superseded calendar-shift record, and would not by itself catch the `mugla_2022_event_relative` entry analysed here. Both checks pass on the executed output (Section 3.17), but the guarantee for this experiment rests on A08. The decision to keep this pair out of the 20-direction

transfer matrix is therefore verifiable in the released code rather than resting
1890 on the text of this paper. The pre-label burn exclusion of Section 3.2 ran
for both arms of this pair: cells that burned inside an experiment’s own pre-
dictor window are removed from its analysis universe rather than counted as
unburned. That is the same safeguard used for Evia and Montiferru, and it
is a stricter treatment than the two regions where it did not run (Manavgat,
1895 and Bejís where no status is recorded). It removed 49 cells from the 2021
arm and none from the 2022 arm.

A second and much larger exclusion applies to the 2022 arm alone, and it defines that arm’s population. The registry sets `exclude_historical_burns` for `mugla_2022_event_relative` only, with `mugla_2021`
1900 as the source and a frozen expected count of 3,073. The mask is every cell
with `burned = 1` in the 2021 canonical Step 8A artefact, without any further
restriction by land cover, eligibility or modelling validity. Applied to the 2022
arm it removes 3,073 cells, of which 2,941 belong to the primary natural-
vegetation population. The primary population therefore falls from 41,730
1905 rows in the 2021 arm to 38,790 in the 2022 arm, a drop of 7.0 %, and the
removed cells are the 2021 scar itself: 2,911 of the 2,941 are 2021 burned
cells, with a median elevation of 563 m and a maximum of 1,975 m. The
design intent is to stop the previous year’s burn scar, which carries an altered
surface, from entering the following year’s analysis. The consequence for how
1910 these two transfer directions may be read is set out in Section 4.8, and it is
the reason they are reported separately from the 20-direction matrix rather
than inside it.

Two frozen diagnostics are read for this pair, both produced by the ma-

chinery already described and neither refitting any model. Burned-pattern
1915 structure (component counts, effective component count, component-size and
elevation distributions, land-cover mix) follows Section 3.14.5, computed
on the primary population; outputs in `paper/mugla_temporal_raw/` with
hashes in that directory’s `SHA256SUMS.txt`. Signed univariate direction re-
versal follows Section 3.12: for each of the nine numeric features, the raw
1920 ROC-AUC of the feature against `burned` in each experiment, never folded to
 $\max(\text{AUC}, 1 - \text{AUC})$, so that a value below 0.5 is read as a direction rather
than as weakness. Land cover is excluded from the AUC analysis because its
integer class codes carry no meaningful scalar order, and is reported descrip-
tively instead. Uncertainty is a spatial-block bootstrap on 10-cell (≈ 5 km)
1925 blocks assigned before filtering, 1,000 replicates, seed 42, 2.5/97.5 percentile
intervals. A reversal counts as bootstrap-supported only when both exper-
iments’ intervals exclude 0.5 on opposite sides. Outputs in `paper/step9g_`
`raw/` with hashes alongside. Four pair directories in that export carry a
`_superseded_pre_manavgat_repair` suffix and are superseded. They are
1930 retained for the record and are not read here.

Unlike every other transfer direction in this paper, the two arms of this
pair were not present in the frozen export delivered to us, which contained
no `mugla_2021__mugla_2022_event_relative` directory, so they were com-
puted for this analysis. They were produced by running the project’s own un-
1935 modified `src/step9b_run_cross_region_transfer.py` and `src/step9c_`
`cross_region_block_bootstrap.py` at commit 48b56e7. That is the same
code and the same protocol as Section 3.10 and Section 3.9, with `random_`
`state = 42`, the primary RF configuration, and the 1,000-replicate spatial-

block bootstrap on `spatial_block_id`. The frozen Step 8A modelling datasets
1940 of both experiments were the only inputs, hashed as read (`paper/mugla_
transfer_raw/SHA256SUMS.txt`). Because `repo/` is treated as read-only
and the pipeline resolves its paths from its own location, the run used a
shadow project root containing a copy of `core/`, `src/` and `scripts/` plus
those two inputs, so nothing was written into the pipeline repository or into
1945 the frozen output archive. The environment was Python 3.12.3 with scikit-
learn 1.9.0, pandas 3.0.5 and NumPy 2.5.2, matching the version to which
every other transfer number in this paper is fixed (Section 4.7e). One con-
sequence is stated rather than hidden: the metrics file records `git_commit:`
`null`, because the shadow root is not itself a git repository.

1950 **Independent execution of this pair.** These two directions were sub-
sequently found to have been run by the pipeline author as well, in a separate
environment and before our own run, and that export was supplied to us.
The pair therefore rests on two independent executions rather than on one
performed by the authors of this manuscript. The two runs are separated
1955 in date, machine, code-tree state and library stack. The pipeline author’s
run was produced on 2026-08-09 at commit `a07ea33` under scikit-learn 1.9.0
with pandas 3.0.2 and NumPy 2.4.4, and ours on 2026-08-11 at commit
`48b56e7` under Python 3.12.3, scikit-learn 1.9.0, pandas 3.0.5 and NumPy
2.5.2. Both read the same two frozen Step 8A parquets, identical by SHA-
1960 256 (`c4ab107d...` for 2021 and `7c545f4d...` for the 2022 event-relative
experiment), and both used `random_state = 42`.

Agreement is to floating-point noise rather than to a stated tolerance.
Across both directions and both model families the transfer ROC-AUC and

PR-AUC point estimates agree to $\leq 1 \times 10^{-7}$. The 2021→2022 direction
 1965 is bit-identical in every reported metric, and the largest discrepancy any-
 where in the pair is 9.9×10^{-8} . The 1,000-replicate spatial-block bootstrap
 reproduces to the same order, with the Δ AUC interval bounds agreeing
 to $\leq 3.4 \times 10^{-8}$. The agreement also holds below the metric level: the two
 runs’ per-cell prediction tables have identical row counts and columns, and
 1970 across all 295,402 predicted probabilities the largest absolute difference is
 4.4×10^{-16} , which is double-precision round-off. The prediction files them-
 selves therefore differ in SHA-256, because last-place decimal digits change
 the serialised string length. The human-readable step9b and step9c summary
 files of the two runs are byte-identical. The pipeline author’s copy of these
 1975 outputs is archived alongside ours in `paper/mugla_transfer_raw/emrehan_`
`run_20260809/` with hashes and provenance in that directory’s `SHA256SUMS.`
`txt`. The interval-supported effects reported in Section 4.8 are in any case
 an order of magnitude larger than the ± 0.02 to 0.03 cross-version tolerance
 of Section 4.7e, so the conclusions do not rest on the exact point estimates.

1980 3.17. *Regional meteorological context (explanatory)*

To characterise the meteorological conditions each region actually experi-
 enced, and specifically to test the post hoc explanation that Manavgat’s
 outlying transfer behaviour reflects meteorological extremity (Section 5.7),
 an AOI-level ERA5-Land diagnostic was run outside the modelling pipeline
 1985 (`src/era5_land_regional_diagnostic.py`, with runner and validator in
`scripts/`; outputs in `drive_new/diagnostics/era5_land_regional/<analysis_`
`id>/`). It is explanatory only: it produces a table and exports no raster, and
 its outputs enter no feature set, no model and no transfer arm. The mod-

ule records this status in its own output (`is_model_predictor: false`).

1990 The candidate diagnostic set of Section 3.14 was fixed before any diagnostic-versus-transfer correlation was computed, on the provenance set out in Section 3.14.1, and no meteorological measure is added to it. The reasoning is given in Section 5.7.

Hourly `temperature_2m`, `dewpoint_temperature_2m`, `u_component_of_`
1995 `wind_10m`, `v_component_of_wind_10m` and `total_precipitation_hourly` are read from the ERA5-Land hourly reanalysis [29] as the Earth Engine collection `ECMWF/ERA5_LAND/HOURLY`. Four variables are derived per pixel and per hour, never from window means. They are temperature ($K \rightarrow ^\circ C$), relative humidity as $100 \cdot e_s(T_d)/e_s(T)$ using the ECMWF/Tetens saturation
2000 formula over water for both terms ($T_0 = 273.16$ K, $a_1 = 611.21$ Pa, $a_3 = 17.502$, $a_4 = 32.19$ K), left unclipped and only checked for finiteness; wind speed as $\sqrt{(u10^2 + v10^2)}$; and precipitation depth from the hourly accumulation band ($m \rightarrow mm$), the cumulative `total_precipitation` band never being used, so no differencing of a running total is involved.

2005 Each hourly field is reduced to one AOI value by an explicit pixel-area weighting, $\Sigma(\text{value} \times \text{pixelArea}) / \Sigma(\text{pixelArea})$, with the denominator masked by that variable’s own mask so that a variable undefined over part of the AOI is not credited with that area. Both sums are evaluated on ERA5-Land’s native projection and transform with `bestEffort=False`. An unweighted mean
2010 is not used. Window statistics are then computed over the resulting series of hourly regional means. These are the window mean and maximum for temperature, humidity and wind, and additionally the window total for precipitation. A reported maximum is therefore the most extreme regional hour,

never the most extreme individual pixel.

2015 The observed windows are each region’s own predictor and label windows
taken from the experiment registry (Section 3.1). No date is hard-coded in
the diagnostic, and the registry’s inclusive end date is converted exactly once
to Earth Engine’s exclusive bound. Each statistic is referenced to a clima-
tology built by mapping the same calendar month-day window into the four
2020 reference years 2017 to 2020, computing each year’s statistic independently,
and taking their arithmetic mean and sample standard deviation ($\text{ddof} = 1$).
The standardised anomaly is $(\text{observed} - \text{climatological mean}) / \text{climatolog-}$
 ical SD , written as null, and never as zero or infinity, when the climatological
SD is exactly zero. Every window, observed and climatological alike, must
2025 contain the exact contiguous hourly UTC sequence ($\text{n_days_inclusive} \times 24$
hours). A missing, duplicated, out-of-order, shifted or extra hour fails the
run rather than being sorted, interpolated, padded or dropped. The cohort
is the five canonical regions in a fixed order, Muğla 2022 being deliberately
excluded, and the scientific configuration is hashed into the output names-
2030 pace identifier so that a changed cohort or contract resolves to a different
namespace rather than silently overwriting an existing result.

**Standardised anomalies are computed by the diagnostic but are
not reported in this paper.** Two properties of the climatology make them
unsuitable for the comparative use a reader would put them to. First, it spans
2035 only four years, so each standard deviation carries three degrees of freedom
and is correspondingly unstable. Second, and decisively, the resulting stan-
dard deviations are strongly heterogeneous *between* regions: over the five pre-
dictor windows the climatological SD spans 0.41 to 1.13 °C for temperature,

1.09 to 6.58 % for relative humidity, 0.039 to 0.146 m s⁻¹ for wind speed and
2040 12.6 to 48.0 mm for precipitation total. The ratios are 2.7× to 6.0×, rising
to 7.8× and 11.5× for temperature and precipitation in the label windows.
A standardised anomaly therefore denotes a different physical departure in
each region, and the cross-region comparison it invites is not meaningful.
Anomalies are reported in physical units throughout (Section 4.9). The in-
2045 stability is not hypothetical: Bejís’s label-window temperature and Muğla’s
predictor-window wind speed exceed five standardised units on climatologi-
cal SDs of 0.147 °C and 0.065 m s⁻¹, from physical anomalies of only +0.83
°C and +0.34 m s⁻¹.

Correctness of the output was verified rather than assumed. The four
2050 files were obtained from the pipeline author, and their SHA-256 hashes match
those recorded in the accompanying manifest (`paper/era5_raw/SHA256SUMS.txt`).
The analysis identifier agrees across the manifest, the contract and the
containing namespace. The companion validator (`scripts/validate_era5_`
`land_regional_diagnostic.py`) was then **executed against this output**
2055 **in -mode actual, and all 27 contract checks passed with none failed**
and none skipped. Among them, the climatological mean and sample SD
reproduce from the four retained yearly realisations (A20); observed win-
dows and every climatological realisation are hourly complete (A25, A26).
The standardised anomaly is null exactly when the climatological SD is zero
2060 (A16). The CSV and JSON summaries agree value by value (A17). The sum-
mary contains no infinite or missing quantity and no `Infinity/NaN` literal
(A18, A19). The manifest hashes and byte sizes match (A21). The cohort
is the frozen five with Muğla 2022 absent (A07, A08). The reference years

and `sd_ddof = 1` are as declared (A09, A10). The registry region keys and
2065 window dates agree with `core/regions.py` (A14). The namespace contains
only the four expected files, with no exported raster leaking into it (A24).
The run was performed by the authors on 2026-08-11 under Python 3.12.3
with `earthengine-api` 1.7.39 installed solely to satisfy the module import
chain. The validator opens no Earth Engine session and requires no creden-
2070 tials. The repository was at commit `48b56e7` and the outputs staged in a
scratch namespace outside the repository via `-output-root`.

The manifest names commit `a07ea33`, at which neither the diagnostic
source nor the Montiferru registry entry yet exists. Both were first committed
in `48b56e7`. The production run was therefore made from a working tree
2075 carrying uncommitted changes, and the recorded commit identifies only the
last commit at run time. The code that actually ran can nevertheless be
identified. The output contains Montiferru with its registry windows, which
`a07ea33` cannot supply. `core/regions.py` gains that entry only in `48b56e7`,
and it does so by pure addition (529 lines inserted, none deleted), leaving
2080 the four regions common to both commits byte-identical. `core/paths.py`
and `core/config.py`, the diagnostic's only other internal dependencies, are
unchanged between the two commits. Every free-text semantics string hashed
into the scientific contract matches `48b56e7` verbatim, and the validator's
registry check (A14) confirms that all five regions' keys and window dates in
2085 the output agree with `48b56e7`'s registry. The working tree that produced
this output therefore carried the registry and diagnostic content of `48b56e7`,
which is the version described here and the version pinned as a submodule
of the manuscript repository.

4. Results

4.1. Study regions and data summary

Five Mediterranean regions enter the analysis (Section 3.1, Table 1). Table 3 summarises cell counts, burned-cell counts and prevalence for both analysis populations, together with the burned-landcover gate verdict of Section 3.3.

Table 3: Region summary (recap of Table 1; final numbering at assembly). Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, `burnable_tree_shrub_grass` and `valid_for_modeling == True`, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984 pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991 pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958 pass
North Evia 2021 (extended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945 pass
Monte Ibernia 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723 pass

An earlier version of this table read its TSG columns from the Step 8A fields `burnable_tree_shrub_grass_count` and `burned_count_within_`

each_burnable_mask. Those are legacy fields counted over all grid rows, before the valid_for_modeling filter, and Montiferru's own statistics file labels the first of them "LEGACY field ... Do NOT report it as the modeling population". The counts above are the post-filter ones (burnable_tree_shrub_grass_count_valid_for_modeling and burned_count_within_primary_burnable_mask), cross-checked against the target_row_count and target_burned_count fields of the multi-AOI transfer matrix. The correction changes Manavgat, Muğla, Evia and Montiferru, and moves Montiferru's TSG prevalence from 0.225 to 0.212. The same correction has since been applied to the Burned and Prevalence (all valid) columns, which were still reading the top-level burned_cell_count and burned_rate. Those are also counted before eligibility and validity filtering, so in Muğla, Evia and Montiferru they were being divided by a Valid cells denominator that had already been filtered. The post-filter numerators are pre_label_exclusion.final_modeling_counts_after_predictor_validity.burned: Muğla 3,026 (was 3,073), Evia 2,788 (was 2,803), Montiferru 697 (was 748), each summing with its unburned counterpart to the printed Valid cells value. Montiferru's all-valid prevalence moves from 0.236 to 0.220. Manavgat and Bejís have no pre-label exclusion and are unaffected. **No modelled result changes.** Every model in this paper was already fitted and scored on the post-filter population, so the correction is to the description of that population and not to any number derived from it. Source: paper/referee_round_numbers.md, Tables A4 and A5.

All five regions pass the admissibility gate as wildfire candidates (burned natural-vegetation fraction 0.723 to 0.991). Montiferru is the weakest pass,

with a burned cropland fraction of 0.274; its sensitivity to this composition is examined in Section 4.7b. The negative control behaves as designed: in Kozan 2023 the gate classifies 542 burned cells as 0.017 natural vegetation and 0.983 cropland (533 of 542 burned cells cropland-dominant, 8 grassland, 1 tree cover), returning the verdict *cropland-dominated control*, and the region is excluded from all modelling. The separation is not marginal. The five admitted regions carry natural-vegetation fractions of 0.723 to 0.991 against the control’s 0.017, so the 0.50 threshold falls in an empty interval rather than between neighbouring cases. This establishes that the gate discriminates burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 itself does not distinguish, and that admission of the five study regions is a decision the data supports rather than a selection made by hand.

North Evia is analysed on an extended AOI. Relative to the legacy $0.40^\circ \times 0.40^\circ$ box, the extended $0.80^\circ \times 0.60^\circ$ box ($\sim 3 \times$ the area, identical predictor and label windows) leaves the burned scar essentially unchanged (2,789 $\rightarrow$ 2,803 burned cells) while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of this choice on transfer is reported in Section 4.7a.

4.2. Within-region: the thermal increment replicates in five regions

In every region, adding the six thermal predictors to the baseline increases spatially blocked out-of-fold ROC-AUC. The increment’s bootstrap interval excludes zero in all five regions at 1 km and at 5 km blocking, which are the two scales this design supports as intervals. At 10 km the point estimates hold, from +0.048 to +0.154, but they rest on 6 to 33 positive-carrying blocks

and are reported as indicative rather than as intervals, for the reason given in the table note (Table 4; [Fig. 3]).

Table note (resampling units). The bootstrap resamples spatial blocks, so what bounds an interval’s reliability is the number of blocks that carry at least one burned cell. Those counts fall quickly as blocks coarsen. At 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No bootstrap replicate was invalid at any block size. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage, and Montiferru at 20 cells additionally feeds only 12 groups into a 5-fold grouped split, so its models train on about ten blocks each. **The 20-cell row of this table should be read as indicative rather than as an interval.** The 10-cell row, where every region has 16 to 70 positive-carrying blocks, is the coarsest blocking this design supports properly, and the increment holds there in all five regions. Source: `paper/referee2_numbers.md`, block C, counted from the frozen per-cell prediction tables.

At the default 2-cell blocking, thermal ΔAUC ranges from +0.056 (Bejís) to +0.153 (Evia), with $\Delta\text{PR-AUC}$ from +0.097 (Manavgat, [+0.071, +0.123]) to +0.300 (Evia, [+0.273, +0.329]). Every interval excludes zero. Absolute AUC declines as blocks coarsen, as expected when long-range spatial structure is progressively withheld, but the increment itself does not erode toward zero: at 20-cell (~ 10 km) blocking ΔAUC remains between +0.048 and +0.154 with all intervals above zero. The same holds in the

Table 4: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.8); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

secondary all-valid population (Section 4.7f).

4.3. Cross-region transfer

Transfer is evaluated for all 20 ordered directions among the five regions,
2175 as raw source-only application and under the two label-blind adaptations of
Section 3.11 (Table 5; [Fig. 4]).

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686. Twelve of 20 directions are above chance with CI support, six are *below* chance with CI support (both
2180 directions of Manavgat $\leftrightarrow$ Bejís and Manavgat $\leftrightarrow$ Muğla, plus Bejís $\rightarrow$ Evia and
Evia $\rightarrow$ Bejís), and two intervals span 0.5. Even the best raw transfer (Evia $\rightarrow$ Manavgat,
0.686) remains far below the target’s own within-region thermal performance
(0.870); across all directions the raw deficit against the within-region refer-
ence is 0.184 to 0.592 AUC. Those counts belong to the 2-cell blocking of
2185 Table 5. At the more conservative 10-cell ($\sim$ 5 km) blocking the same points
give 9 above, 4 below and 7 uncertain, with no direction changing side of the
chance line (Section 4.7g, Table 15). Four of the six below-chance directions
keep their support there: Manavgat $\rightarrow$ Bejís, Muğla $\rightarrow$ Manavgat, Bejís $\rightarrow$ Evia
and Evia $\rightarrow$ Bejís. Bejís $\rightarrow$ Manavgat and Manavgat $\rightarrow$ Muğla lose it and carry
2190 no verdict. The qualitative statement is unchanged. The counts should not
be read as exact.

Precision-recall performance must be read against the target’s own prevalence. Prevalence varies by a factor of about 7.5 across the five
targets (0.038 to 0.287; Table 3), so a raw PR-AUC is uninterpretable without
2195 its no-skill line. Table 17 gives raw thermal PR-AUC for all 20 directions
beside that line.

Table 5: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.462]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.513, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.550 [0.524, 0.575]	0.549 [0.525, 0.574]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.582 [0.539, 0.624]

Table 6: Raw transfer PR-AUC, thermal model, TSG population (final numbering at assembly). Target PR-AUC with bootstrap 95% CIs, against the target’s own prevalence as the no-skill value. Read from the multi-AOI transfer matrix (`multi_aoi_transfer_matrix.csv`, `model_family = thermal`, `adaptation_method = raw_source_only`) via `paper/referee_round_numbers.md`, Tables A2 and A4. The adapted arms exist in the same file and are not reproduced here.

Direction	No-skill	Raw PR-AUC [95% CI]	Direction	No-skill	Raw PR-AUC
Manavgat→Bejís	0.072	0.049 [0.043, 0.055]	Muğla→Evia	0.287	0.379 [0.350, 0.
Manavgat→Muğla	0.070	0.063 [0.058, 0.068]	Muğla→Montiferru	0.212	0.214 [0.182, 0.
Manavgat→Evia	0.287	0.343 [0.318, 0.369]	Evia→Manavgat	0.038	0.094 [0.076, 0.
Manavgat→Montiferru	0.212	0.243 [0.202, 0.292]	Evia→Bejís	0.072	0.059 [0.053, 0.
Bejís→Manavgat	0.038	0.034 [0.029, 0.042]	Evia→Muğla	0.070	0.086 [0.078, 0.
Bejís→Muğla	0.070	0.093 [0.085, 0.102]	Evia→Montiferru	0.212	0.283 [0.239, 0.
Bejís→Evia	0.287	0.226 [0.209, 0.243]	Montiferru→Manavgat	0.038	0.042 [0.036, 0.
Bejís→Montiferru	0.212	0.289 [0.236, 0.349]	Montiferru→Bejís	0.072	0.093 [0.077, 0.
Muğla→Manavgat	0.038	0.029 [0.025, 0.033]	Montiferru→Muğla	0.070	0.089 [0.082, 0.
Muğla→Bejís	0.072	0.088 [0.077, 0.102]	Montiferru→Evia	0.287	0.317 [0.293, 0.

The raw thermal PR-AUC interval lies entirely above the target’s prevalence in 11 of 20 directions, entirely below it in 5, and straddles it in 4. This neither rescues nor destroys the transfer picture. It does show that the two metrics do not always return the same verdict. Montiferru→Manavgat is above chance on ROC-AUC at 2-cell blocking, yet its PR-AUC interval straddles the target prevalence (0.042 [0.036, 0.049] against 0.038). Bejís→Manavgat is the same disagreement from the other side, below chance on ROC-AUC at 2-cell blocking with a PR-AUC interval that straddles. Both target Manavgat, the lowest-prevalence region in the set, and both lose their ROC-AUC verdict at 10-cell blocking (Section 4.7g). The PR-AUC intervals here are the 2-cell ones and were not recomputed at the coarser blocking. The absolute values stay small wherever prevalence is small: into the three low-prevalence targets (Manavgat, Bejís, Muğla) the best raw PR-AUC of any direction is 0.094, so none of those directions ranks burned cells well enough for operational triage.

The baseline itself transfers only a little above chance. Over the 20 directions the baseline model’s mean raw transfer ROC-AUC is 0.5371, ranging from 0.3322 (Manavgat→Bejís) to 0.6758 (Evia→Manavgat), with four directions whose interval lies entirely below chance. The thermal model’s mean is 0.5414, ranging from 0.3258 to 0.6858, with six such directions. The mean paired delta is +0.00424. Neither model family is portable in any useful sense. The mean gap between the two families is +0.004, against a mean shortfall of about 0.35 AUC from within-region performance (mean within 0.888 versus mean transfer 0.541; Table 13). Source: `paper/referee_round_numbers.md`, Table A6, from `paper/baseline_vs_thermal_transfer.csv`.

That mean needs an interval, and its interval spans zero. The 20 directions are not 20 independent observations. They are built from five datasets, so each region enters eight of the twenty terms, and the two directions of a pair share geography, data and the same target prediction table. The standard deviation of the 20 paired deltas is 0.0713. Resampling the 10 unordered pairs as clusters gives a 95% interval of $[-0.028, +0.036]$ around the mean of $+0.004$; a t interval on the 10 pair means gives $[-0.034, +0.042]$; a leave-one-region-out jackknife on the five regions gives $[-0.037, +0.046]$. Ignoring the clustering entirely gives $[-0.027, +0.034]$, which is barely narrower, because the width is set by the spread across directions rather than by the count. The pair-cluster interval is the primary one. The point estimate is therefore a small fraction of the width of its own interval, and the correct reading of $+0.004$ is not "a small positive contribution" but "no contribution that this design can distinguish from zero, in either direction". Source: `paper/referee2_numbers.md`, block A, computed from the paired deltas in `paper/transfer_ci_blocksize.json` with 20 000 replicates and seed 42.

The thermal block’s transfer contribution is sign-unstable per direction. Pairing each direction’s thermal model against the baseline model under the same protocol (Table 7) shows that the block worth $+0.056$ to $+0.153$ AUC inside every region is worth $+0.004$ on average across regions: its paired contribution is CI-supported positive in 10 directions, CI-supported negative in 7, and uncertain in 3. That split is the 2-cell one, and it should be read as approximate. At 10-cell blocking it becomes 5 to 6 positive, 3 to 4 negative and 10 to 11 uncertain, and even at 2-cell the boundary between 10 and 11 positive turns on a single bound of -0.00045 (Section 4.7g). What

does not depend on the blocking is the sign instability itself: 12 point deltas are positive and 8 negative at both scales, and no point estimate changes sign. It is the swing factor at the chance line. Adding the thermal block drags three directions from a baseline at or above chance to below it (Manavgat→Muğla 0.508 → 0.470; Bejís→Evia 0.531 → 0.383; Muğla→Manavgat 0.522 → 0.401) and lifts one from below to above (Muğla→Bejís 0.451 → 0.583). In the Manavgat-Muğla pair the baseline transfers at roughly chance and the thermal block pushes both directions below it; in the Bejís-Muğla pair the thermal block carries Muğla→Bejís above chance, while the reverse direction already transferred above chance without it (baseline 0.592 [0.575, 0.609]).

Table 7: Paired baseline-versus-thermal raw transfer contrast (final numbering at assembly). Target ROC-AUC per model; Δ = thermal – baseline computed on identical resampled target blocks (2-cell blocks, 1000 replicates). Full 20-direction table with baseline and thermal CIs in `baseline_vs_thermal_transfer.csv`. The seven CI-supported negative and ten CI-supported positive directions are listed there. Selected rows:

Direction	Baseline	Thermal	Δ [95% CI]	Support
Muğla→Bejís	0.451	0.583	+0.133 [+0.105, +0.158]	positive
Evia→Montiferru	0.549	0.647	+0.097 [+0.054, +0.138]	positive
Manavgat→Muğla	0.508	0.470	−0.038 [−0.051, −0.024]	negative
Muğla→Manavgat	0.522	0.401	−0.121 [−0.146, −0.098]	negative
Bejís→Evia	0.531	0.383	−0.148 [−0.168, −0.126]	negative

Label-blind adaptation compresses the matrix toward chance.

Under region-wise z-scoring the 20 directions span 0.431 to 0.630, and under CORAL 0.443 to 0.624. That is roughly half the raw spread, with no adapted direction exceeding 0.63 against within-region references of 0.859 to

0.918. Adaptation raises the failing directions (e.g. Manavgat→Bejís 0.326 → 0.511 CORAL; Muğla→Manavgat 0.401 → 0.560) and degrades most of the directions that already transferred (e.g. Evia→Manavgat 0.686 → 0.527 CORAL; Bejís→Muğla 0.618 → 0.518 z-score; Muğla→Evia 0.653 → 0.563).

2265 The degradation is the majority case, not the universal one. Of the 12 directions whose raw interval lies above chance, the better of the two adaptations lowers 9 and raises 3, and all three exceptions have Montiferru as their source (Montiferru→Manavgat 0.567 → 0.606, Montiferru→Bejís 0.548 → 0.574, Montiferru→Evia 0.586 → 0.630). Two further directions whose raw interval
2270 spans chance are also raised, and both have Montiferru as their target. Inside the four-AOI subset that carries the decomposition of Section 4.3a, which contains no Montiferru direction, all six above-chance directions are degraded without exception. After the best adaptation per direction the maximum is 0.630 (Montiferru→Evia, z-score), whereas raw transfer reached 0.686, with
2275 three directions at or above 0.647. The compression is therefore the general effect, and Montiferru, the smallest and last-added region, is where it does not hold.

The decomposition ([Fig. 5]) shows both faces of the same behaviour. In the five directions where raw transfer was below chance, the best label-
2280 blind method recovers at most 34% of the gap to the within-region reference. The remaining (concept) fraction is at least 0.66 everywhere. Seven directions show *negative* recovery, meaning that adaptation moves the score away from the within-region reference, in the worst case (Evia→Manavgat) recovering -0.86 of the gap. In six of the seven, raw transfer was al-
2285 ready above chance and adaptation destroyed that advantage; in the seventh

Table 8: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 5. The adapted arms were not recomputed at the coarser blocking of Section 4.7g, which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.550 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

(Manavgat→Muğla), raw transfer was below chance (0.470) and adaptation lowered it further (0.443). Label-blind adaptation therefore does not act as a repair mechanism: it compresses the matrix toward chance, closing a minority of the deficit where transfer fails and destroying performance where transfer works. The compression is the general effect and not a universal one. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began and 6 end further from it. Five of those six involve Montiferru and move upward (Montiferru→Manavgat, Montiferru→Bejís, Montiferru→Evia, Manavgat→Montiferru, Muğla→Montiferru); the sixth is Manavgat→Muğla, which moves downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (x) in Section 5.11: Bejís→Manavgat is counted as compressed on a margin of 0.001, since it moves from 0.444 to 0.555.

4.4. *Transferability diagnostics: only conditional similarity orders transfer*

Twenty candidate diagnostics from four families were each rank-correlated with the same target quantity. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap measures $P(x|y=1)$, fire-regime (label-pattern) structure $P(y)$, and conditional feature-response direction $P(y|x)$. Each was rank-correlated with the same target quantity, the raw thermal transfer AUC over the 20 ordered directions, under a common pair-based bootstrap (Section 3.14.1). Table 9 gives the complete set.

Table note (power): the effective sample is 10 unordered pairs (6 for the 12-direction rows; 8 for the supported-conditional rows). The two directions of a pair are not independent and every pair shares regions with three others. Intervals of width ± 0.5 to 0.8 cannot rule out moderate true correlations;

Table 9: All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95% CIs. Exp. = expected sign. Rows with n = 12 exist only for the four-AOI subset. The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector	P(y x) conditional	+	20	+0.27 [−0.36, +0.77]	no
Spearman, all 9	conditional				
Agreement count, all 9	P(y x) conditional	+	20	+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20	+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20	+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20	+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20	−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20	−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	P(x) marginal	−	20	−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	P(x) marginal	−	103		
Predictor-space p95 dissimilarity	P(x) marginal	−	12	−0.10 [−0.54, +0.43]	no
	P(x) marginal	−	12	−0.08 [−0.59, +0.49]	no

null rows are "not shown to order transfer", not "shown not to".

Table note (count and provenance): twenty variants were specified and nineteen were computed. The twentieth, vector Spearman over supported features, needs at least three jointly supported features and only one pair reaches
2315 that threshold, so it rests on $n = 2$, every bootstrap replicate is degenerate and no interval exists. It is listed so that the count of twenty is honest and it carries no information either way. The six four-region rows exist in the published outputs under two seed offsets of the same pair-based bootstrap, with identical data and identical point estimates; the rows above quote the rendered `all_`
2320 `diagnostics_vs_transfer.md` variant, which is why the weighted-AoA interval reads $[-0.48, +0.59]$ here and $[-0.50, +0.60]$ in the companion CSV. Nothing in the paper depends on the difference.

Of the 20 variants, exactly two have bootstrap intervals excluding zero, and both belong to the conditional family: the sign-agreement fraction over
2325 CI-supported features ($\rho = +0.84 [+0.58, +0.88]$) and the cosine similarity of supported signed-AUC vectors ($\rho = +0.81 [+0.33, +0.88]$) (Section 3.14.4). These indices are computed from signed feature-response directions in *both* regions and therefore require burned labels (or a labelled probe) in the target. So do two of the three families that fail. Burned-niche overlap is defined on
2330 the burned cells of both regions (Section 3.14.3) and fire-regime structure on the target's burned map (Section 3.14.5), so both consume the target's labels exactly as the conditional index does. Only the marginal $P(x)$ family, which includes the area-of-applicability rows, the domain classifier and the climatic and geographic distances, is computable before any target label exists. That
2335 makes the comparison sharper rather than weaker: niche overlap and the

conditional index are built from the same labelled information and only the conditional one orders the matrix. It also means the one family a practitioner could run before deployment is the family that fails.

How strong is the supported-conditional result? It is the paper’s one positive diagnostic finding and it should be read at the size of the design that
2340 produced it. Three things bound it. First, the index is nearly binary: over the 8 pairs on which it is defined it takes three distinct values (0, 0.5 and 1), and five of the eight pairs share the value 1. The tie structure caps the attainable 16-direction Spearman at +0.861, and the observed +0.840 there-
2345 fore sits essentially on its own ceiling, which is why the published interval is narrow. That narrowness reflects heavy ties under pair resampling, not precision. Second, at the pair level, which is the unit the bootstrap resamples, $\rho = +0.866$ with an exact one-sided permutation p of 0.0060 over all 40 320 permutations, and 0.0060 is the *smallest* p this tie structure can produce. A
2350 Bonferroni threshold over the 19 computed variants is 0.0026, so no outcome of this diagnostic could have cleared a family-wise correction on this pair set, whatever the data had been. The design has no headroom, which is a stronger statement than the absence of a correction. Third, the restriction to CI-supported features is a data-dependent selection made from the same
2355 bootstrap intervals, and it is executed once rather than inside each replicate, so the published interval is conditional on a fixed selection rather than propagating it. The unrestricted variant over all nine features reaches $\rho = +0.50$ with an interval spanning zero (Table 9), so the restriction is what separates a positive result from a null one. The finding stands as a mechanism diagnosis.
2360 It does not stand as a validated instrument. Source: `paper/referee2_`

numbers.md, block B.

The remaining families all fail to order the matrix. The domain classifier (Section 3.14.2) is at ceiling for every pair (AUC 0.962 to 0.9999), so marginal shift is essentially total everywhere. A marginal instrument at
2365 ceiling cannot discriminate outcomes ranging from 0.33 to 0.69. The canonical SDM niche-overlap instruments (Schoener’s D, Warren’s I, Mahalanobis distance between burned-cell distributions; Section 3.14.3) span ρ -0.23 to $+0.24$ with all intervals crossing zero. Fire-regime structure (Section 3.14.5) has the *wrong-signed* point estimate ($\rho +0.29$): the most regime-similar pair
2370 (Bejís-Evia, effective component count 1.0000 vs 1.0083) fails in both directions while the most regime-different pair (Bejís-Muğla) transfers above chance. At pair level, niche overlap and sign agreement are empirically distinct (ρ between them -0.36 to $+0.45$, all CIs spanning zero), and in partial rank correlations the conditional index retains its association with transfer
2375 with niche overlap held fixed (partial $\rho +0.82$ [$+0.40, +0.88$]) while niche overlap retains none with the conditional index held fixed (-0.07 [$-0.39, +0.37$]).

Common-subset check. The rows of Table 9 rest on unequal samples ($n = 12, 16$ and 20), so a referee may read the marginal family’s failure as
2380 a power artefact. It is not. Recomputed on the common 12-direction, 6-pair four-region subset, on which every family is defined, the two supported-conditional rows are still the only rows whose intervals exclude zero, and both are slightly stronger there: $\rho = +0.87$ [$+0.65, +0.88$] for the agreement fraction and $+0.85$ [$+0.43, +0.88$] for the cosine. All eighteen other rows
2385 span zero. The same holds on the 16-direction subset for the eleven rows

that can be moved onto it. In the same check all twenty published rows were reproduced at their original sample size to a largest absolute difference of 4.8e-05, below the rounding of the published four-decimal file. Source: `paper/diagnostics_common_subset.md`.

2390 **A high applicability fraction does not protect a direction.** Table 9 asks whether the area-of-applicability fraction *orders* transfer, and it does not. That is a separate question from the one Sections 1 and 2 raise, which is whether a model can sit well inside its nominal area of applicability and still perform at or below chance. That is a sufficiency claim, so a single pair
2395 settles it. Table 18 gives the per-direction audit.

The counterexample pair is at the top and the bottom of the table. Manavgat→Muğla carries the highest weighted applicability of the twelve, 0.8752, and 0.9640 of target cells inside the unweighted support, and it transfers at 0.470, below chance. Its reverse direction, at 0.5305 weighted and the
2400 highest unweighted value in the table, 0.9702, transfers at 0.401, also below chance. Muğla→Bejís sits at the other extreme, with essentially no weighted applicability at 0.0050, and transfers above chance at 0.583. Two of these three keep their verdict at both blocking scales: Muğla→Manavgat stays below chance ([0.353, 0.455] at 10-cell) and Muğla→Bejís stays above it ([0.532,
2405 0.636]). Manavgat→Muğla is supported below chance at 2-cell blocking only and carries no verdict at 10-cell, [0.415, 0.524] (Section 4.7g). The three point estimates do not depend on the blocking. A model may therefore be almost wholly inside its nominal area of applicability and still be anti-predictive, and almost wholly outside it and still work. This is the same sufficiency logic
2410 used for niche overlap in Section 4.5, and it is stated as a counterexample

Table 10: Area of applicability and raw thermal transfer, 12 directions (final numbering at assembly). Weighted AoA = fraction of target cells inside the importance-weighted area of applicability; unweighted support = fraction inside the unweighted predictor-space support. Raw thermal target ROC-AUC repeated from Table 5 with its 2-cell interval. The marginal audit exists only for the four-region subset, so no Montiferru direction appears. Read from `marginal_diagnostics_with_transfer.csv` via `paper/referee_round_numbers.md`, Table A1.

Direction	Weighted AoA	Unweighted support	Raw thermal AUC [95% CI]
Manavgat→Muğla	0.8752	0.9640	0.470 [0.451, 0.488]
Manavgat→Evia	0.7444	0.9425	0.613 [0.593, 0.631]
Evia→Muğla	0.6310	0.9590	0.577 [0.560, 0.593]
Muğla→Evia	0.5810	0.9510	0.653 [0.636, 0.671]
Muğla→Manavgat	0.5305	0.9702	0.401 [0.378, 0.426]
Evia→Manavgat	0.4067	0.8995	0.686 [0.653, 0.716]
Manavgat→Bejís	0.2646	0.9156	0.326 [0.305, 0.349]
Bejís→Manavgat	0.2207	0.7486	0.444 [0.408, 0.480]
Bejís→Muğla	0.1998	0.7638	0.618 [0.601, 0.635]
Bejís→Evia	0.1705	0.6534	0.383 [0.363, 0.402]
Evia→Bejís	0.0729	108 0.8562	0.448 [0.426, 0.470]
Muğla→Bejís	0.0050	0.9076	0.583 [0.561, 0.607]

pair rather than as a correlation. The correlation question is answered separately and negatively in Table 9 ($\rho = +0.22$ $[-0.48, +0.59]$ for the weighted fraction, $n = 12$).

Two limits belong with this table. First, the two applicability quantities
 2415 disagree sharply: the unweighted support fraction is high everywhere (0.65
 to 0.97) while the weighted fraction spans almost the whole unit interval, so
 any claim about applicability must say which is meant. Second, the marginal
 audit was produced only for the four-region subset. No area-of-applicability
 output exists for Montiferru in any direction, so the eight Montiferru direc-
 2420 tions carry no applicability number and none is implied for them here.

4.5. *The contrast pair: highest niche overlap fails, lowest niche overlap works*

The clearest single view of Table 9 is a two-pair contrast ([Fig. 8]; per-
 feature data in `figure_contrast_pairs.csv`). [Production note: the OUT-
 LINE slot reserved for signed univariate AUC per feature with CIs is realised
 2425 by Fig. 8, in two panels of per-feature signed AUC with ~ 5 km-block CIs,
 one per pair, annotated with the overlap and transfer numbers below.]

Manavgat and Muğla are in the same country and fire year, roughly 200
 km apart, and their burned cells occupy the most similar environmental en-
 velope of any pair in the matrix (per-feature D 0.77 to 0.89). Yet five of nine
 2430 feature-response directions point opposite ways. They are elevation (signed
 AUC 0.374 $[0.289, 0.471]$ in Manavgat vs 0.611 $[0.532, 0.690]$ in Muğla, dis-
 joint CIs) and all four absolute thermal channels (e.g. `current_1st_mean`
 0.538 $[0.452, 0.621]$ vs 0.325 $[0.271, 0.382]$). Transfer is below chance in both
 directions at the point estimate, 0.470 and 0.401. Bejís and Montiferru sit
 2435 at the opposite extreme: burned envelopes that barely overlap (per-feature

Table 11: The contrast pairs (final numbering at assembly).

	Manavgat	Muğla	Bejís	Montiferru
Schoener’s $\bar{D}$ (1-D, 9 features)	0.826 — highest of all 10 pairs		0.479 — lowest of all 10 pairs	
Mahalanobis (burned centroids)	2.71 (closest)		8.11 (farthest)	
Sign agreement (9 features)	4/9; jointly supported features 2 (agreement 1/2), with a CI-supported elevation flip		7/9; no jointly supported features — the two regions’ supported sets do not intersect (Montiferru CIs wide)	
Raw transfer, both directions	0.470 and 0.401 — both below chance at the point estimate		0.594 and 0.548 — both above chance at the point estimate	
Interval support, 2-cell (~ 1 km)	both supported: [0.451, 0.488] and [0.378, 0.426]		both supported: [0.560, 0.631] and [0.521, 0.578]	
Interval support, 10-cell (~ 5 km)	Muğla→Manavgat still supported [0.353, 0.455]; Manavgat→Muğla no verdict [0.415, 0.524]		neither supported: [0.467, 0.687] and [0.479, 0.636]	

D 0.23 to 0.77; the pair is the most dissimilar on every overlap measure), yet seven of nine directions agree. The pair has no jointly supported features, because the two regions' supported sets do not intersect, so the supported-agreement index is undefined for it. Transfer is nevertheless above chance in
2440 both directions at the point estimate, 0.594 and 0.548. Where the envelope agrees but the direction reverses, transfer fails; where the envelope disagrees but the direction agrees, transfer works.

The contrast is stated at the point estimate because the point estimate does not depend on the blocking scale and the argument lives there. The
2445 interval support is weaker than the points and is given in full in Table 11. At 2-cell blocking all four directions are supported. At the conservative 10-cell blocking of Section 4.7g only Muğla→Manavgat keeps its support, at [0.353, 0.455] below chance; Manavgat→Muğla widens to [0.415, 0.524] and carries no verdict, and both Bejís-Montiferru directions widen across 0.5
2450 and carry no verdict either. No direction changes side of the chance line at either blocking, so the reversal of ordering between the two pairs survives the coarser bootstrap while the per-direction verdicts do not.

The marginal applicability audit points the same way for the half of the contrast it covers. Both Manavgat-Muğla directions sit deep inside
2455 the nominal area of applicability, at 0.8752 and 0.5305 of target cells inside the weighted region and 0.9640 and 0.9702 inside the unweighted support (Table 18), and both transfer below chance at the point estimate, with Muğla→Manavgat supported at both blockings and Manavgat→Muğla at 1 km only. The Bejís-Montiferru half cannot be given a matching number,
2460 because the marginal audit was never produced for Montiferru. The con-

trast pair is therefore complete on niche overlap and on sign agreement, and one-sided on applicability.

4.6. Interventions: pooling and feature removal obey the same conservation

Two interventions test whether the diagnosis of Sections 4.3 to 4.5 yields a
 2465 remedy. Both show the same pattern: what transfer gains, the within-region model or the direction-aligned pairs pay for.

(a) Pooled multi-region training (leave-one-region-out). Training on the pooled TSG populations of four regions and testing on the held-out fifth (Section 3.15.1) does not rescue transfer (Table 12).

Table 12: LORO pooled training, thermal feature set, target ROC-AUC (final numbering at assembly).

Held-out target	Best Mean pairwise pair- wise (source)	LORO raw	LORO region-z	Within (ceiling)
Manavgat	0.6860.524 (Evia)	0.469 [0.412, 0.529]	0.657 [0.543, 0.754]	0.870
Bejís	0.5830.476 (Muğla)	0.417 [0.369, 0.467]	0.472 [0.428, 0.516]	0.918
Muğla	0.6190.571 (Mon- ti- f- erru)	0.552 [0.475, 0.625]	0.522 [0.476, 0.569]	0.859
Evia (ext.)	0.6530.559 (Muğla)	0.637 [0.592, 0.678]	0.569 [0.520, 0.629]	0.912
Montiferru	0.6470.576 (Evia)	0.601 [0.516, 0.683]	0.523 [0.466, 0.570]	0.883

2470 At the point estimate the pooled thermal model never beats the best single-source pairwise transfer for any target (shortfalls 0.02 to 0.22) and

remains 0.28 to 0.50 AUC below the within-region ceiling. For two targets (Manavgat, Bejís) the pooled model falls below the pairwise mean and below chance, though only Bejís is below chance with interval support (0.417 [0.369, 0.467]); Manavgat’s 0.469 is a point estimate whose interval [0.412, 0.529] spans 0.5. Region-wise z-scoring of the pool raises only the into-Manavgat fold substantially (0.469 $\rightarrow$ 0.657) and leaves Bejís at 0.472 [0.428, 0.516], still below chance at the point estimate but no longer with interval support, while degrading the three folds where raw pooling was least bad (Evia 0.637 $\rightarrow$ 0.569; Montiferru 0.601 $\rightarrow$ 0.523; Muğla 0.552 $\rightarrow$ 0.522). Pooled PR-AUC sits close to its no-skill base in the three low-prevalence folds (Manavgat 0.034 vs base 0.038; Bejís 0.056 vs 0.072; Muğla 0.089 vs 0.070). In the two high-prevalence folds it sits about a quarter above the base in relative terms (Evia 0.355 vs 0.287; Montiferru 0.265 vs 0.212), which is a real but small margin and does not change the ROC picture. No fold achieves operationally useful ranking of burned cells. For two of five targets the pooled *baseline* model beats the pooled thermal one raw (Manavgat 0.563 vs 0.469; Muğla 0.562 vs 0.552); for Montiferru the two are within 0.006 (0.596 vs 0.601); thermal is ahead for Evia (0.637 vs 0.597) and Bejís (0.417 vs 0.389).

(b) Dropping the direction-reversing features. Retraining without the two features with CI-supported reversals (`elevation_mean`; `lst_anomaly_mean`; Section 3.15.2) trades within-region skill for a near-zero mean transfer gain (Table 13).

Dropping both reversal features costs -0.081 mean within-region AUC and changes mean transfer AUC by $+0.014$, gaining one additional CI-supported above-chance direction. The two sides of that statement are

Table 13: Feature-removal trade-off (final numbering at assembly). Means over 5 regions (within, TSG OOF) and 20 directions (transfer); paired ~ 5 km-block bootstrap for per-direction deltas. *Note: the CI-supported above-chance counts in this table use the feature-drop analysis’s ~ 5 km-block bootstrap and are therefore more conservative than the 2-cell-block step10 intervals of Table 5, under which 12 of 20 raw directions are CI-supported above chance. The coarser blocks widen the intervals without changing any point estimate. Section 4.7g recomputes the Table 5 quantities at the same ~ 5 km blocking and independently returns 9 of 20 above chance, so the two counts are the same result at two blocking scales and not a disagreement.*

Config	Mean within AUC	Mean transfer AUC	Directions >0.5 (point)	Directions >0.5 (CI- supported)
Full (reference)	0.888	0.541	14	9
Drop elevation	0.827	0.546	17	9
Drop lst_anomaly	0.875	0.544	14	9
Drop both	0.807	0.556	17	10

not equally well estimated. The within-region cost is bootstrap-supported in every region (largest Bejís -0.114 $[-0.147, -0.079]$) and all five per-region deltas have intervals entirely below zero. The transfer gain is a mean over twenty non-independent directions with SD 0.046 and a range of -0.058 to $+0.102$; clustered to the ten unordered pairs its interval is $[-0.017, +0.045]$ and spans zero, and per direction the outcome is five CI-supported positive, three CI-supported negative and twelve uncertain. No exchange rate between the two is claimed here. The transfer deltas land exactly where the reversal diagnosis points: the three CI-supported gains are Manavgat→Bejís $+0.118$ $[+0.048, +0.203]$, Evia→Bejís $+0.075$ $[+0.023, +0.131]$ and Montiferru→Manavgat $+0.057$ $[+0.017, +0.092]$. The mean delta over the eight Manavgat-involved directions is $+0.025$ against -0.009 over the other twelve. Manavgat is the elevation dissenter (signed AUC 0.374 vs 0.61 to 0.65 elsewhere). Conversely, the five CI-supported losses (Evia→Manavgat -0.078 ; Manavgat→Evia -0.071 ; Muğla→Evia -0.038 ; Montiferru→Bejís -0.036 ; Muğla→Bejís -0.026) occur precisely where elevation’s direction is shared and informative. `lst_anomaly` removal contributes only in the pair where it reverses with support: Bejís→Evia ($+0.046$ alone; $+0.063$ $[+0.026, +0.100]$ combined with elevation). Identifying which pairs cross a reversal requires the target’s signed directions, i.e. target labels; applied label-free, the same removal degrades the five aligned directions.

4.7. Sensitivity analyses

(a) Evia AOI and prevalence. Repeating the raw transfer arms with the legacy (small, prevalence 0.361; TSG prevalence 0.676) versus extended (prevalence 0.122; TSG prevalence 0.287) Evia AOI (Section 3.16.1) leaves

every qualitative conclusion unchanged: thermal raw transfer AUCs move by at most 0.070 (Evia→Bejís 0.378 → 0.448) and no direction changes side of chance (Manavgat→Evia 0.662 → 0.613; Evia→Manavgat 0.670 → 0.686; 2525 Bejís→Evia 0.397 → 0.383; Muğla→Evia 0.630 → 0.653; Evia→Muğla 0.572 → 0.577). The 2.4-fold change in target prevalence does not create or destroy any CI-supported transfer.

The *within-region* increment is a different matter, and the AOI choice moves it. On the same fire, the same predictor and label windows and 2530 the same primary population, the legacy AOI gives a thermal ΔAUC of +0.085 [+0.072, +0.100] against the extended AOI's +0.153 [+0.142, +0.166] (`drive_new/experiments/evia_2021/step8c/step8c_bootstrap_metrics.json` and its `evia_2021_extended` counterpart, `bootstrap_unit: spatial_block_id`). Both intervals exclude zero, so the qualitative within-region conclusion is unaffected in either variant, but Evia carries the top of this paper's 2535 within-region range and that top is specific to the extended AOI. Read the range as +0.056 to +0.153 on the canonical cohort, with the upper end falling to +0.085 for the one region whose AOI was revised.

(b) Montiferru cropland fringe. Within-region, excluding grassland 2540 (tree+shrub population) leaves the thermal increment essentially unchanged: ΔAUC 0.104 [0.075, 0.131] versus 0.101 [0.080, 0.125] for TSG (Section 3.16.2). Cross-region, the thermal-minus-baseline delta on the target agrees in sign and support between the two populations for six of eight Montiferru-involved directions. The two disagreements (Bejís→Montiferru, Montiferru→Muğla) 2545 move from supported to uncertain, not to sign reversal. Montiferru's transfer behaviour is not driven by its cropland/grassland fringe.

(c) **Window closure.** Shifting both ends of the predictor window earlier by 7 and 14 days (window length preserved, fixed common cohort, shared folds; Section 3.16.3) leaves the within-region thermal increment bootstrap-supported in every region and every variant (Table 14). Nowhere does the increment shrink toward zero; in Manavgat and Bejís it increases with earlier closure. That increase has a mechanism, and the window-closure report states it: the closure date is shifted against a fixed production policy, so moving the window changes which acquisition dates enter the median composite and how many clear observations back each pixel. With about seven usable dates in a window, a 7 or 14 day shift is a change of one or two dates. The signature fits: across the same comparison the baseline moves by +0.002 to +0.003 while the thermal family moves by +0.022 ROC-AUC, so the shift acts almost entirely on the thermal channels, exactly as the compositing sensitivity of (i) predicts. This does not weaken the pre-fire reading. Contamination by early fire signal would push the increment the other way, and it does not appear. It does mean the increment is partly a function of how the composite is built, which is the same conclusion (i) reaches from a different direction.

(d) **CORAL regularisation.** The available sweep covers four directions (Bejís↔Muğla, Manavgat↔Muğla) and nine λ values from 0 to 10^{-1} . Over that sweep the CORAL transfer AUC moves by at most 0.014 within any direction, and by at most 0.008 within the thermal family (e.g. Muğla→Manavgat 0.559 to 0.564; Manavgat→Muğla 0.443 to 0.451). The three widest spreads are baseline-family rows, the largest being Muğla→Bejís at 0.014, which the export’s own summary flags as `modest_lambda_sensitivity`. No CORAL-

Table 14: Window-closure sensitivity: thermal ΔAUC (TSG common cohort; final numbering at assembly).

Region	Canonical	Close 7 d	Close 14 d
Manavgat	0.074 [0.062, 0.085]	0.101 [0.088, 0.114]	0.094 [0.078, 0.109]
Bejís	0.058 [0.048, 0.068]	0.077 [0.068, 0.088]	0.079 [0.068, 0.090]
Muğla	0.115 [0.105, 0.125]	0.114 [0.104, 0.124]	0.128 [0.118, 0.137]
Evia (ext.)	0.156 [0.144, 0.170]	0.149 [0.136, 0.162]	0.135 [0.124, 0.147]
Montiferru	0.096 [0.074, 0.118]	0.091 [0.068, 0.113]	0.100 [0.077, 0.122]

dependent conclusion for these directions is sensitive to λ in this range. With nine predictors and thousands of cells, a λ of 10^{-1} is still far too small to bite on the covariance estimate, so a movement of at most 0.014 across the range
2575 establishes numerical stability of the alignment map rather than robustness to regularisation.

The canonical $\lambda = 1$, which the released sweep omits, was computed for this paper by driving the pipeline’s own alignment primitives in the order `step10b` calls them; $\lambda = 10^{-1}$ recomputed the same way repro-
2580 duces the frozen sweep to 4.4×10^{-9} over all eight rows. At $\lambda = 1$ no direction changes side of the chance line. The largest movement is an improvement rather than a collapse, Muğla→Manavgat thermal from 0.559 to 0.574, and the next largest is Bejís→Muğla baseline from 0.564 to 0.574. Including $\lambda = 1$ widens the spread over the grid to at most 0.019 in any direction and
2585 0.016 within the thermal family, against 0.014 and 0.008 over the released grid alone. So heavy regularisation does not erase the effect in this region set, which is the opposite of what the superseded two-region run showed and of what the scale argument in Section 3.11 predicted; note that the sweep’s four

directions do not include Bejís↔Manavgat, where that earlier result sat. The
2590 sweep still does not cover the Montiferru or Evia directions. Source: `paper/
observational_sensitivities.md`.

(e) **scikit-learn version.** With byte-identical data, pipeline and seed, changing only the library version from 1.9.0 to 1.7.2 moves raw transfer AUC by +0.021 (Montiferru→Bejís) and +0.026 (Manavgat→Bejís) on the
2595 two probes tested, which is the same order as some reported effects. Within-region AUCs, by contrast, reproduce to ~4 decimals across environments. All numbers in this paper were produced under, or verified against, scikit-learn 1.9.0. The two probes reproduce to four decimal places in the verification environment (difference 0.0000). Cross-region point estimates therefore carry
2600 an implementation tolerance of roughly ± 0.02 to 0.03 unless the exact library version is fixed. The bootstrap intervals reported throughout are wider than this jitter.

(f) **Analysis population.** In the secondary all-valid population the within-region thermal increment is CI-supported in all five regions, with
2605 smaller deltas than in the primary natural-vegetation population for the three high-increment regions: ΔAUC 0.059 [0.049, 0.068] (Manavgat), 0.048 [0.040, 0.057] (Bejís), 0.072 [0.067, 0.078] (Muğla), 0.053 [0.049, 0.058] (Evia), 0.075 [0.057, 0.094] (Montiferru), against TSG values of 0.067, 0.056, 0.116, 0.153 and 0.101 respectively. Absolute AUCs are higher in the mixed population
2610 (baseline 0.827 to 0.910), consistent with land-cover composition contributing separable but non-thermal discrimination; the within-region conclusion does not depend on the population choice. The nature of that extra discrimination should be stated plainly, because it is not a subtle land-cover

effect in two of the five regions. The AOIs are place-based rectangles and are
 2615 not clipped to the coastline, and sea cells satisfy `valid_for_modeling`, since
 elevation, slope and the vegetation index are all finite over water and per-
 manent water is a valid land-cover class. Counted over the frozen datasets,
 water-dominant cells are 57.7 % of Evia’s all-valid population and 38.9 % of
 Muğla’s, against 8.3 % (Manavgat), 7.4 % (Montiferru) and 0.1 % (Bejís). In
 2620 those two regions most of the extra separability of the all-valid population is
 therefore land against sea rather than any fire-relevant contrast. This is one
 of the two reasons the natural-vegetation population is primary here, and it
 is why the all-valid arm is reported only as a sensitivity.

(g) Blocking scale of the transfer intervals. Table 5 and Table 7
 2625 report 2-cell (~ 1 km) blocks, while Section 3.12 argues that 2-cell blocking
 ignores short-range spatial autocorrelation and gives intervals that are too
 narrow. The transfer quantities were therefore recomputed at 10-cell (~ 5 km)
 blocking from the frozen per-cell predictions, with the resampling rule copied
 from the published code, 1000 replicates and seed 42. The implementation
 2630 was validated first: point estimates reproduce the frozen values to 1.1e-16,
 the recomputed 2-cell bounds match the published ones to a mean absolute
 difference of about 0.001 and a worst single case of 0.0055, and all 20 verdicts
 agree at 2-cell. Point estimates do not depend on the blocking and none of
 them moves. The recomputation covers the raw thermal arm and the paired
 2635 delta only; the z-score and CORAL columns of Table 5 were not repeated at
 the coarser blocking. Source: `paper/transfer_ci_blocksize.md`.

Leading with the conservative blocking: 9 of 20 directions are above
 chance with CI support, 4 are below chance with CI support, and 7 carry

Table 15: Transfer verdict counts at two blocking scales (final numbering at assembly). Level = raw thermal transfer ROC-AUC against chance (Table 5). Delta = paired thermal-minus-baseline contribution against zero (Table 7). Twenty directions in each row.

Quantity	Blocking	Supported one way	Supported the other	Uncertain
Level, versus chance	2-cell (~1 km), pub- lished	12 above	6 below	2
Level, versus chance	10-cell (~5 km)	9 above	4 below	7
Delta, versus zero	2-cell (~1 km), pub- lished	10 positive	7 negative	3
Delta, versus zero	10-cell (~5 km)	6 positive	4 negative	10

no verdict. For the paired delta, 6 directions are CI-supported positive, 4
 2640 CI-supported negative and 10 uncertain. Five directions change verdict on
 the level and seven on the delta, and every one of those changes runs from
 supported to uncertain. No direction moves the other way, from uncertain
 to supported. Intervals widen by a factor of 2.1 to 3.6 (median 2.8) for the
 levels and 1.5 to 3.4 (median 2.3) for the deltas. The "roughly four times too
 2645 narrow" figure of Section 3.12 belongs to the univariate reversal diagnostic
 and is not the right number for these intervals.

The robust statement is the one that matters. No direction changes
 side of the chance line and no point estimate changes sign. The point deltas
 still run from -0.148 to $+0.132$, with 12 positive and 8 negative at both block-
 2650 ings, and the below-chance directions stay far from 0.5 (Manavgat→Bejís
 $[0.266, 0.388]$ at 10-cell). The pattern is therefore robust to the blocking
 choice; the exact counts are not.

The counts should accordingly be read as approximate, because a seed
 sweep over five random streams moves several verdicts. On the level, the
 2655 above-chance count is 9 at every seed and the below-chance count is 3 or 4,
 with Evia→Bejís on the boundary (its 10-cell upper bound lands between
 0.497 and 0.504 across seeds), so that direction is better described as at the
 chance line than below it. On the delta, the honest 10-cell statement is 5 to 6
 positive, 3 to 4 negative and 10 to 11 uncertain, with Manavgat→Muğla and
 2660 Evia→Montiferru on the boundary. The published 2-cell delta split of 10, 7
 and 3 is itself decided by one bound: Bejís→Manavgat has a lower bound of
 -0.00045 , and a different random stream gives 11 positive, 7 negative and 2
 uncertain. Four directions in total should never be quoted as though their

verdict were firm: Bejís→Manavgat at 2-cell on the delta, Manavgat→Muğla
 2665 and Evia→Montiferru at 10-cell on the delta, and Evia→Bejís at 10-cell on
 the level.

(h) **Which thermal channel carries the increment.** The thermal
 block is six predictors, but they are not six independent measurements.
`fused_1st` equals the observed Landsat LST wherever that is valid, and the
 2670 gap-filled share is 2.15% (Manavgat), 9.70% (Bejís), 0.59% (Muğla), 0.90%
 (Evia) and 0.11% (Montiferru), so outside Bejís the two channels are near-
 copies of each other; Table 21 shows the consequence directly, with signed
 univariate AUCs of 0.325 versus 0.325 in one arm and 0.515 versus 0.519 in
 the other. `downscaled_1st` is a fitted surface whose dominant input differs
 2675 by region: the MODIS context layer in Manavgat (importance 0.525) and
 Evia (0.593), NDVI in Bejís (0.482) and Montiferru (0.666), and **slope in
 Muğla (0.777)**, where it is largely a re-expression of a predictor the base-
 line already contains. The pipeline’s Step 8D ablation, run on the primary
 population with the same folds, quantifies the redundancy: a single subgroup
 2680 recovers most of the whole block’s increment in every region.

The dryness pair (`current_tvdi_mean`, `tvdi_difference_mean`) is the
 best subgroup in three of five regions and recovers 96% and 99% of the
 full block in Manavgat and Montiferru. This matters twice. It converts
 "the thermal block" from an opaque bundle into a statement about which
 2685 physical quantity carries the increment, and it means the six reversal tests of
 Section 4.4 are not six independent probes of concept shift. Source: `paper/
 referee2_numbers.md`, blocks D, E and F.

That table reports what each subgroup adds. The complementary ques-

Table 16: Thermal-block ablation, primary population, Δ AUC against the baseline (final numbering at assembly). Source: `drive_new/experiments/<region>/step8d/step8d_ablation_delta_auc_by_population.csv`.

Region	Full thermal block	Best subgroup	Its Δ AUC	Share of the full block
Manavgat	+0.067	TVDI pair	+0.065	96%
Bejís	+0.056	fused + downscaled	+0.040	71%
Muğla	+0.116	TVDI pair	+0.089	77%
Evia (ext.)	+0.153	LST anomaly	+0.125	81%
Montiferru	+0.101	TVDI pair	+0.100	99%

tion is what the block is worth **without** its two coordinate-bearing channels, since `downscaled_lst` is a fitted surface whose own inputs include `lon`, `lat`, `row` and `col`, and `fused_lst` inherits that on its gap-filled share. This is the one route by which a coordinate-derived surface re-enters a feature set from which Section 3.13 excludes coordinates, so the increment ought not to depend on it. It does not. Dropping both channels and re-running the comparison with the pipeline’s own Step 8B and Step 8C leaves Δ AUC at +0.063 [+0.051, +0.074] (Manavgat), +0.046 [+0.038, +0.055] (Bejís), +0.097 [+0.088, +0.107] (Muğla), +0.145 [+0.133, +0.157] (Evia) and +0.105 [+0.082, +0.128] (Montiferru). That is 94 %, 82 %, 84 %, 94 % and 103 % of the full-block value, with every interval excluding zero. The six-channel arm of the same harness reproduces the frozen metrics with a maximum absolute difference of exactly zero in all five regions, so the comparison is not confounded by the re-run. Source: `paper/observational_`

sensitivities.md.

(i) **Landsat compositing.** The current-period LST composite is a median over the clear acquisitions inside the predictor window, and how those acquisitions are weighted is a choice. Three controlled chains exist for Manavgat, on an identical cohort with identical folds and an identical baseline, differing only in that raster: the production scene-weighted reference gives $\Delta\text{AUC} +0.064$ [$+0.052$, $+0.075$], a date-balanced variant gives $+0.084$ [$+0.072$, $+0.098$], and an overlap-harmonised date-balanced variant gives $+0.045$ [$+0.033$, $+0.057$]. The baseline AUC is identical to six decimal places across all three (0.804362), so the entire spread sits in the thermal block, and both paired comparisons have intervals excluding zero. The two are referenced differently, as their source reports define them: date-balanced against the production chain is $+0.021$ [$+0.012$, $+0.031$], and overlap-harmonised against the *date-balanced* chain is -0.040 [-0.050 , -0.029]. Referenced to production instead, the overlap-harmonised difference is -0.019 . The mechanism is documented in the same export: at boundaries where the number of contributing clear acquisitions changes, the residual seam analysis finds excess jumps of 0.850 °C [0.812 , 0.890] in the current-minus-baseline field and 0.526 °C [0.503 , 0.551] in the anomaly z-score, with the final attribution **current_support_dominant**. Manavgat's window is backed by seven acquisition dates from two alternating Landsat paths, so a cell inside the path overlap rests on seven dates and a cell outside it on three or four.

Two limits on this axis are stated by the source reports themselves and are adopted here: the A/B was run for one AOI only, the alternative chains are candidates rather than production, and no non-inferiority claim is made for

them. The conclusion drawn here is correspondingly narrow. **The within-region increment carries a compositing tolerance of roughly ± 0.02 AUC, which is the same order as the scikit-learn tolerance of (e) and wider than the 2-cell interval of Table 4.** It does not approach the increment itself, which stays bootstrap-supported and positive under all three chains. It has been audited in one region, and the other four are unaudited on this axis.

(j) Muğla’s population size. Muğla contributes the largest analysis population of the five regions, 41,730 cells against Manavgat’s 20,511, and it is also the region with the second largest within-region increment and the one that takes part in the transfer directions that work. Whether that is the region or simply its data volume is a question the design can answer, so it was answered rather than argued. The Muğla modelling population was cut to exactly Manavgat’s cell count over 20 deterministic repeats, stratified by 10-cell block and label so that prevalence is preserved (0.0698 full against 0.0701 subsampled) and every block of the full population is retained, which isolates cell count from spatial extent. The full-Muğla block-to-fold mapping is inherited unchanged, so the folds are not re-drawn.

Muğla’s role in transfer does not depend on its size. With Muğla as the source, three of the four direction-and-family combinations put the full-population value inside the subsampling range (Muğla→Manavgat thermal 0.401 against a range of 0.378 to 0.404; Muğla→Bejís thermal 0.583 against 0.548 to 0.608). With Muğla as the target all four lie inside, and tightly (Bejís→Muğla thermal 0.618 against 0.613 to 0.625). The one value outside its range is the baseline family of Muğla→Bejís, where the full-population

0.451 sits *below* the subsampled range of 0.465 to 0.556, so shrinking Muğla would if anything improve that arm rather than explain its failure.

2755 What size does affect is absolute within-region skill, modestly and in the expected direction. The full-population reference lies above the subsampling range for both families, baseline 0.698 against 0.678 to 0.698 and thermal 0.777 against 0.747 to 0.772. **The increment survives the matching:** it is +0.079 at full size and +0.071 at the median size-matched repeat. So Muğla's
2760 larger population inflates its absolute AUC by roughly 0.01 to 0.02 and leaves both the increment and the transfer behaviour substantially intact. Two limits belong with this. The repeats vary only in which cells fill each stratum, so the range describes within-stratum selection variability and is narrower than any sampling distribution; and the positive count is not equalised to
2765 Manavgat's, only the prevalence, so a residual positive-count difference is not separated here. No bootstrap was run and no probability statement is made. Source: `mugla_subsampling` in the frozen diagnostics export, recomputed here from its 20 per-repeat records.

(k) **Sea in the TVDI edges, and a common-edge index.** TVDI is
2770 normalised against wet and dry edges fitted as the 2nd and 98th LST percentiles inside each NDVI bin, over whatever the scene contains (Section 3.4). The AOIs are place-based rectangles that are not clipped to the coastline, so sea takes part in that fit, and two of the five are largely marine. The index is also the one whose internal normalisation was expected to make it portable,
2775 so if its reversal were an artefact of the edges, the paper's mechanism would be weaker than it looks. Three arms were therefore compared: the frozen scene fit; the same fit restricted to land cover classes 10, 20, 30, 40 and 60;

and one set of edges fitted once over the pooled land pixels of all five regions, 26.2 million of them, which puts every region on a common dryness scale.

Table 17: Signed univariate AUC of `current_tvdi_mean` against `burned` under three edge definitions. Primary population; raw AUC, never folded, so a value below 0.5 is a direction; 10-cell (≈ 5 km) spatial-block bootstrap, 1000 replicates, seed 42, at the registered specification of Section 3.12. Bold marks an interval excluding 0.5. Source: `paper/tvdi_land_refit.md`.

Region	Water share	Scene, all pixels	Scene, land only	Pooled, land only
Bejís 2022	0.1 %	0.517 [0.429, 0.595]	0.519 [0.431, 0.597]	0.503 [0.416, 0.578]
Montiferru 2021	7.4 %	0.356 [0.233, 0.499]	0.353 [0.229, 0.500]	0.356 [0.232, 0.500]
Manavgat 2021	8.3 %	0.552 [0.460, 0.641]	0.558 [0.465, 0.646]	0.565 [0.469, 0.656]
Muğla 2021	38.9 %	0.336 [0.276, 0.401]	0.351 [0.288, 0.424]	0.341 [0.280, 0.408]
North Evia 2021 (ext.)	57.6 %	0.362 [0.280, 0.440]	0.379 [0.298, 0.458]	0.378 [0.297, 0.455]

2780 The effect of sea on the index is real and it is small. Removing water moves the signed AUC by +0.017 in Evia and +0.016 in Muğla, the two marine AOIs, against +0.006 in Manavgat, −0.002 in Montiferru and +0.002 in Bejís, so the shift scales with sea fraction as the mechanism predicts. **It changes no region’s direction.** The two regions in which higher TVDI
2785 ranks burned cells stay above 0.5 and the three in which lower TVDI does stay below, with Muğla and Evia keeping intervals entirely below 0.5. One verdict moves, on a knife edge: Montiferru’s upper bound goes from 0.4991 to

0.5004, a shift of 0.0013, which is the count instability recorded as limitation (x) in Section 5.11 rather than a change of finding.

2790 The common-edge arm is the stronger test, because it removes the scene dependence itself rather than only the sea. With one set of edges for all five regions, Manavgat still ranks burned cells by higher TVDI (0.565) while Muğla and Evia still rank them by lower TVDI with bootstrap support (0.341 and 0.378). Putting every region on the same dryness scale does not bring
2795 the directions into agreement. The reversal is therefore a property of the relationship between dryness and burning in these landscapes, not of how the index was normalised. Two limits: only `current_tvdi_mean` was re-derived, since `tvdi_difference_mean` would need the four baseline-year surfaces re-fitted as well; and pooling the current-window LST of five regions observed
2800 on different dates gives a common statistical scale, not a radiometrically harmonised one.

(1) **Label agreement, and whether the reversals are fringe artefacts.** MCD64A1 labels a 500 m cell from its 30 m sub-pixels, and `burn_date_pixel_agreement_fraction` records the share of a cell's positive sub-
2805 pixels that agree with its modal burn date. It is defined for burned cells only, and its median is 1.000 in every region, but its lower tail is not empty: the 25th percentile runs from 0.765 (Muğla) to 0.985 (Montiferru). Cells with low agreement are disproportionately scar-fringe cells, and fringe cells differ systematically in terrain from core cells, so differential fringe contami-
2810 nation is a competing explanation for a reversal in a terrain variable. That is exactly what the paper's sharpest reversal is. The signed univariate AUCs were therefore recomputed with burned cells below an agreement thresh-

old dropped and the negative class left whole, at thresholds of 0.75 and 0.90, which retain 76 % to 88 % and 63 % to 79 % of each region’s burned
 2815 cells. The unrestricted arm reproduces the archived step9g values exactly, to 1.1×10^{-16} .

Table 18: Signed univariate AUC of `elevation_mean` against `burned` by label-agreement threshold. Primary population; raw AUC; 10-cell spatial-block bootstrap, 1000 replicates, seed 42. Bold marks an interval excluding 0.5.

Region	All burned cells	Agreement ≥ 0.75	Agreement ≥ 0.90
Manavgat 2021	0.374 [0.289, 0.471]	0.368 [0.284, 0.462]	0.362 [0.279, 0.460]
Bejís 2022	0.643 [0.558, 0.729]	0.642 [0.560, 0.728]	0.635 [0.550, 0.722]
Muğla 2021	0.611 [0.531, 0.695]	0.606 [0.527, 0.690]	0.602 [0.520, 0.686]
North Evia 2021 (ext.)	0.541 [0.458, 0.632]	0.544 [0.461, 0.636]	0.550 [0.463, 0.636]
Montiferru 2021	0.584 [0.395, 0.762]	0.567 [0.390, 0.748]	0.546 [0.380, 0.729]

The elevation reversal is not a fringe artefact. Manavgat’s interval stays entirely below 0.5 and Bejís’s and Muğla’s stay entirely above it at every threshold, so the disjoint-interval contrast that carries Section 4.4 survives
 2820 the restriction; if anything Manavgat moves further from chance, from 0.374 to 0.362, as thin-evidence cells are removed. The thermal channels behave the same way: Muğla and Evia keep `current_lst_mean`, `downscaled_lst_mean`, `fused_lst_mean` and `current_tvdi_mean` entirely below 0.5 at both thresholds, while Manavgat’s point estimates drift slightly upward and its
 2825 intervals continue to span chance. Two verdicts change, and both are Mon-

tiferru's, both by about 0.005 on an interval bound: `current_tvdi_mean` moves from an upper bound of 0.499 to 0.505 and `tvdi_difference_mean` from 0.497 to 0.503. That is the count instability of limitation (x) in Section 5.11, in the region with the fewest positive-carrying blocks, not a change
2830 of finding. Source: `paper/observational_sensitivities.md`.

(m) Gap-filled thermal cells. `fused_1st` equals the observed Landsat LST outside a gap-filled share, and where that share is high the channel is the downscaled surface rather than an observation. The gap-fill is concentrated rather than diffuse: its median is zero in every region, but 18.8 % of Bejís's
2835 primary-population cells carry some, with a 90th percentile of 0.47, against 2.0 % to 9.1 % elsewhere. The within-region comparison was therefore re-run on cells whose gap-filled fraction is at most 0.10, using the pipeline's own Step 8B and Step 8C on the restricted population.

The increment keeps bootstrap support in all five regions. Bejís moves
2840 the most and in the direction the concern predicts, from +0.056 to +0.043, so part of its increment does rest on gap-filled cells; the remaining four move by at most 0.008, in both directions. The restriction changes the population, so these are not paired comparisons with the full-population row of Table 4. Source: `paper/observational_sensitivities.md`.

2845 *4.8. The same geography, a second fire: direction reversal with place held constant*

Every result above compares different places. Muğla admits a stricter test, because a second fire event occurred inside the identical AOI on the identical analysis grid (Section 3.16.4): the 2021 event, with its 58-day predictor window closing on 28 July, and the 2022 event, whose matched 58-day
2850

Table 19:

Region	Cells retained	Δ AUC, full	Δ AUC, low gap-fill [95 % CI]
Manavgat 2021	95.4 %	+0.067	+0.072 [+0.061, +0.083]
Bejís 2022	85.1 %	+0.056	+0.043 [+0.033, +0.053]
Muğla 2021	99.2 %	+0.116	+0.114 [+0.104, +0.123]
North Evia 2021 (ext.)	98.3 %	+0.153	+0.159 [+0.146, +0.171]
Montiferru 2021	99.7 %	+0.101	+0.109 [+0.087, +0.134]

window closes on 20 June. Region, bounding box, cell definition, feature registry and processing chain are the same. What the design holds fixed is place; what it does not hold fixed is the population. The comparison is therefore same-geography event-to-event, not clean temporal transfer. The 2022 fire
2855 ignites about five weeks earlier in the season, so year and seasonal phase are confounded (Section 3.16.4). Geography, the explanation most often offered for between-region instability, is held fixed by construction.

Two structural properties of this pair must be read alongside its numbers, because they have no analogue anywhere in the 20-
2860 **direction matrix.** First, the two arms are not disjoint samples. The 2022 population is the 2021 population with the 2021 scar removed, so the two share 38,789 of the 2022 arm’s 38,790 cells, and `elevation_mean`, `slope_`

`mean` and `landcover_dominant` are identical to the digit across all 73,098 grid cells of the AOI: only NDVI and the six thermal channels carry new information between the arms. Second, the removal is the target's own positive class. All 2,911 burned cells of the 2021 primary population are excluded from the 2022 arm, so in the 2022 to 2021 direction not one target positive is present in the source training population while 38,789 of the 38,819 target negatives are. Membership of the source training set alone separates the 2021 target's classes at ROC-AUC 0.9996. In the opposite direction all 331 of the 2022 positives were in the 2021 training population, labelled unburned. The 30 target negatives outside the source population are the only internal control available, and they are too few to bound the effect. We therefore cannot demonstrate what this asymmetry does to the two numbers below, only that a transfer direction with this structure is not comparable to the 20 between-region directions, where source and target share no cells. The elevation reversal is the finding; the structural asymmetry is a competing explanation for it that this design cannot exclude. It is worth noting which way the known part of the bias runs: the removed cells are high (median 563 m, maximum 1,975 m) and unburned in 2022, so removing them raises the 2022 elevation AUC and makes the reversal smaller, not larger. The second Muğla event is deliberately kept out of the 20-direction matrix of Section 4.3, and that exclusion is enforced and tested in the released code rather than merely asserted here. The check that carries the guarantee is `A08_cohort_is_the_frozen_five`, which compares the executed cohort against the frozen `DEFAULT_EXPERIMENTS` tuple as an ordered equality, so any addition fails it. The neighbouring check `A07_mugla_2022_`

absent_from_default_analysis is narrower than its name suggests: it tests literal membership of the string `mugla_2022` and would not by itself catch the `mugla_2022_event_relative` entry analysed here. Both pass on the executed diagnostic output, but the guarantee rests on A08 (Sections 3.16.4, 3.17).

The two events are structurally very different fires. Table 20 reports the burned-pattern comparison on the primary population.

Table 20: Burned-pattern structure of the two Muğla events. Primary population (`burnable_tree_shrub_grass` $\wedge$ `valid_for_modeling`), 8-connectivity components (Section 3.14.5). Read from `paper/mugla_temporal_raw/.../multi_aoi_burned_pattern_comparison.csv`.

Quantity	Muğla 2021	Muğla 2022
Burned cells	2,911	331
Connected components	10	2
Largest component (cells, share)	914 (31.4 %)	282 (85.2 %)
Second component (cells, share)	738 (25.4 %)	49 (14.8 %)
Effective component count	4.054	1.337
Component size, median	22.5	165.5
Elevation, median	563 m	187 m
Elevation, q95	1,526 m	437 m
Elevation, maximum	1,975 m	777 m
Land-cover classes observed	7	2
Dominant class (share)	tree cover (0.925)	tree cover (0.979)

The 2021 season burned as a dispersed multi-fire complex, with ten components, an effective count of 4.05 and no single component holding a third of the area. It spanned the region’s full relief from near sea level to 1,975 m.

The 2022 event is one compact scar: two components, an effective count of 1.34, 85.2 % of burned cells in the largest, and confined to the low belt. Its
2900 highest burned cell lies at 777 m, below the 2021 event’s 95th percentile of 1,526 m, and its median elevation of 187 m is a third of 2021’s 563 m. The two fires occupy different parts of the same elevation gradient.

That difference propagates directly into the feature-label relationship. Table 21 gives the signed univariate AUCs.

2905 Elevation reverses with bootstrap support. In 2021 higher cells burn preferentially (AUC 0.611, interval entirely above 0.5); in 2022 lower cells do (0.296, interval entirely below), and the two intervals are disjoint, 0.532 against 0.355, with the difference at -0.317 $[-0.414, -0.220]$. The same predictor, the same region, the same grid, and an association that points the
2910 opposite way in two fires eleven months apart.

The four absolute thermal channels also change side, from bootstrap-supported *lower*-values-burn in 2021 to *higher*-values-burn point estimates in 2022, and each AUC difference is itself interval-supported (for example `current_lst` $+0.188$ $[+0.084, +0.272]$). **These reversals are nonetheless**
2915 **not bootstrap-supported and are not claimed as such.** With 331 burned cells the 2022 intervals are wide, and all four straddle 0.5 (`current_lst` $[0.434, 0.580]$), so the 2022 direction is not established even though the shift from 2021 is. We follow the diagnostic’s own conservative classification: one bootstrap-supported reversal, elevation; four point-level reversals in the ther-
2920 mal block. The two anomaly-referenced channels and the two remaining static predictors do not reverse at all, and NDVI keeps a bootstrap-supported positive direction in both events.

Table 21: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.16.4). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

Transfer between the two events behaves unlike any between-region direction in the matrix, and it does so asymmetrically (Table 22). Both directions stay *above* chance, with thermal intervals of [0.513, 0.604] and [0.654, 0.685], neither touching 0.5. Those intervals use 5 km blocking, and at that blocking four of the twenty between-region directions fall below chance with interval support, six at the 1 km blocking of Table 5 (Sections 4.3, 4.7g). Holding geography fixed removes the collapse.

Table 22: Transfer between the two Muğla events. Primary population, thermal and baseline models, target ROC-AUC with 5 km spatial-block bootstrap 95 % CIs (Section 3.16.4). Within-region references are each target’s own frozen value at 2-cell blocking (Table 4 for 2021; the 2022 figure is its Step 8C point estimate, $\Delta\text{AUC} +0.078 [+0.061, +0.097]$). The baseline and thermal columns are point estimates on the full target. The ΔAUC column is the bootstrap mean reported alongside its interval, so it differs from the difference of the two point estimates in the third decimal (point $\Delta\text{AUC} -0.083$ and $+0.088$ respectively). Both arms of this pair were reproduced independently by the pipeline author to $\leq 1 \times 10^{-7}$ (Section 3.16.4).

Direction	Baseline	Thermal	ΔAUC (thermal – baseline)	Target’s within- region thermal	Gap
Muğla 2021 $\rightarrow$ 2022	0.642 [0.606, 0.674]	0.559 [0.513, 0.604]	−0.082 [−0.127, −0.040]	0.942	0.383
Muğla 2022 $\rightarrow$ 2021	0.581 [0.566, 0.598]	0.670 [0.654, 0.685]	+0.089 [+0.072, +0.104]	0.859	0.189

What does not survive is the thermal block’s contribution, and its failure here is sharper than anywhere else in the paper: **the same six predictors change the sign of their contribution depending on which event is**

the source, and both signs are interval-supported. Carried forward from 2021 to 2022 they cost 0.082 AUC; carried back from 2022 to 2021 they
2935 buy 0.089. This is not a case of a weak signal failing to travel. Within each event separately the thermal block helps and its interval excludes zero, at +0.116 in 2021 (Table 4) and +0.078 in 2022. The block is therefore locally informative in both, and still actively harmful in one direction between them.

The asymmetry follows the elevation reversal. A model fitted on 2021
2940 learned a relationship whose strongest static term points the wrong way for a fire confined below 777 m, and the thermal channels it learned alongside that term were fitted under the same regime; applied to 2022 they subtract skill from a baseline that already transfers at 0.642. In the reverse direction the 2022 model carries a relationship fitted on a narrow low-elevation slice,
2945 which the broader 2021 event contains, and there the thermal block adds. The residual gap to the target’s own within-region performance remains large in both directions, at 0.383 and 0.189, so nothing here rescues transfer; what it shows is that the direction of the thermal block’s contribution is not a property of the predictors but of the pair.

2950 *4.9. Regional meteorological context*

Table 23 characterises the meteorological conditions of each region’s predictor window against its own 2017 to 2020 climatology (Section 3.17). Anomalies are reported in physical units only; standardised anomalies are computed by the diagnostic but are not reported, for the reason given in Section 3.17.
2955 The label window is not characterised here and is used nowhere in this paper: it opens on the ignition date and runs 35 to 59 days into the autumn rains, so it describes conditions during and after the fire rather than the conditions

that preceded it.

Table 23: Predictor-window meteorology against the 2017 to 2020 climatology. ERA5-Land, AOI pixel-area-weighted regional means; temperature, humidity and wind are window means, precipitation is the window total (Section 3.17). Anomaly = observed – climatological mean, in the variable’s own units. Values read from `paper/era5_raw/<analysis_id>/era5_land_regional_summary.json`.

Region	Window (days)	Temp. (°C)	ΔT (°C)	RH (%)	ΔRH (%)	Wind (m s^{-1})	ΔWind (m s^{-1})	Precip. total (mm)
Manavgat	57	22.72	−0.06	54.05	−3.24	1.80	+0.07	48.08
Bejís	61	24.05	+0.92	55.40	−0.65	2.02	+0.02	81.53
Muğla	58	25.28	+0.31	51.61	−5.18	2.97	+0.34	27.70
North Evia	59	25.73	+1.11	59.64	−3.25	2.06	−0.21	19.72
Montiferru	60	22.58	+0.48	64.17	−2.84	2.22	+0.11	14.95

The five predictor windows share a common signature of moderate dry-
ness rather than extremity. Every region is drier than its own climatology in
relative humidity (−0.65 to −5.18 %), and four of five are drier in precipita-
tion, two of them substantially so (Evia −61.6 mm, Montiferru −35.5 mm);
Bejís is the exception, wetter than climatology by +36.3 mm. Temperature
departures are small in absolute terms, spanning −0.06 to +1.11 °C, and
wind departures smaller still (−0.21 to +0.34 m s^{-1}).

The one region-level contrast the analysis was run to test concerns Man-
avgat. Its predictor-window temperature is −0.06 °C from its climatological
mean, while the other four sit 0.31 to 1.11 °C above theirs. It is the only

region at or below its own baseline, and the anomaly is small enough to be
2970 read as *at* the baseline. Its humidity deficit (-3.24%) is mid-range among
the five and its precipitation is within 1.4 mm of climatology, the smallest
precipitation departure in the set. On none of the four variables is Manavgat
the extreme member. The consequence for the interpretation of its transfer
behaviour is taken up in Section 5.7.

2975 *4.10. What target labels cost: the recovery curve*

Everything above measures a failure. This section prices it. If the residual
gap is conditional, and label-free alignment cannot close it, then the missing
resource is target-conditional information, and the direct way to supply it is
target labels. The question is how many.

2980 The frozen few-shot diagnostic answers it for three regions in all six or-
dered directions (Manavgat, Bejís and Muğla; Evia and Montiferru are absent
from that export). The unit of labelling effort is one 10-cell (≈ 5 km) spa-
tial block, which is the same large-block machinery used for the conservative
intervals of Section 4.7g and a plausible unit of survey effort. Budgets are
2985 0, 1, 2, 4, 8, 16 and 32 blocks, nested, drawn under a fixed tier order with
a seed derived from the direction and fold and independent of any result.
Each budget is repeated ten times with different draws. The ceiling is the
target-only model under the same folds and the same blocking, so recovery
fractions are read against a matched ceiling of 0.777 to 0.824 rather than
2990 against the ≈ 1 km within-region values of Table 4. The interval quoted in
the supplement is a selection interval across repeats, not a bootstrap. The
full design, the per-budget table and seven stated limits are Supplementary
S1.

Thirty-two labelled blocks recover 85 to 89 % of the target ceiling in three of the six directions: Manavgat→Bejís 0.326 → 0.772 against a ceiling of 0.824 (89 %), Muğla→Bejís 0.583 → 0.789 (85 %), and Bejís→Manavgat 0.444 → 0.743 against 0.797 (85 %). Two of those three started below chance; Muğla→Bejís started at 0.583, above it. Manavgat→Muğla and Muğla→Manavgat reach 51 and 57 %, and Bejís→Muğla 30 %. The failure this paper documents is therefore expensive rather than structural. It is a shortage of target-conditional information, and target labels are exactly that.

Two qualifications belong with that number and neither is small. **That budget is not modest.** Thirty-two blocks carry 2 700 to 3 000 labelled cells, 7 to 20 % of the target’s natural-vegetation population, and for Bejís the top budget already contains 880 of the region’s 1 100 burned cells. The narrow selection intervals at the largest budgets reflect a nearly exhausted selection pool rather than a well-estimated quantity (Supplementary S1, limit 4). **And a small budget hurts the one direction that already works.** Bejís→Muğla transfers above chance raw (0.618) and is the direction few-shot recalibration helps least: the curve is negative at 1, 2, 4 and 8 blocks (−0.043 to −0.021 AUC), only overtakes raw at 16 blocks, and reaches 30 % at 32, the worst of the six. This is the same asymmetry that label-blind adaptation shows in Section 4.3, arrived at by a different route: where a source model already carries a usable conditional relationship, a small target sample perturbs it before it can replace it.

No label budget is proposed here. Six directions across three regions cannot support a recommendation, and the recovery curve is a price list for

the specific AOIs measured, not a design rule. What it establishes is the
3020 shape of the cost: recovery is achievable, it is slowest where the concept gap
is widest, and at small budgets it is not free.

5. Discussion

5.1. Principal findings

Across five Mediterranean fire regions, six pre-fire thermal predictors
3025 were added to a static and near-static baseline of terrain, fuel and green-
ness, whose only time-varying member is the vegetation-index composite.
Spatially blocked within-region ROC-AUC was raised by +0.056 to +0.153,
with bootstrap support at ~ 1 km and ~ 5 km blocking in all five regions
(Section 4.2). The same models were then applied across regions without
3030 target labels. Transfer AUCs ran from 0.326 to 0.686, and even the best
direction fell 0.184 short of its target’s within-region reference (Section 4.3).

One feature of the transfer matrix is invariant to how the bootstrap is
blocked, and the argument is built on it. The point estimates do not depend
on the blocking, and no direction changes side of the chance line when the
3035 blocking is coarsened. Coarsening only weakens support. At the conservative
5 km blocking that Section 3.12 defends, nine of twenty directions are above
chance with interval support and three to four are below it, with the remain-
ing seven or eight carrying no verdict. The fourth below-chance direction,
Evia to Bejís, sits on the chance line across bootstrap seeds and should be
3040 read as borderline. The finer 1 km blocking used in Table 5 gives twelve
above, six below and two uncertain.

Label-blind adaptation did not repair this. It compressed the whole matrix towards chance, to a range of 0.431 to 0.630. Of the twelve directions that had transferred above chance, nine were degraded and three were raised, all three with Montiferru as their source; inside the four-AOI subset that carries the decomposition there is no exception at all. At most a third of the deficit was recovered where transfer had failed.

Of twenty candidate transferability diagnostics, only two had bootstrap intervals that excluded zero, and both were conditional. They are computed from the *direction* of each predictor’s association with burning in both regions. Every marginal, niche-overlap and fire-regime measure failed to order the matrix (Section 4.4). At the point estimate, the pair with the highest burned-niche overlap failed in both directions while the pair with the lowest overlap transferred in both (Section 4.5); half of that contrast carries interval support at 5 km blocking and half does not.

The trade-off is direct rather than inferred. Paired per direction, the thermal block that adds +0.056 to +0.153 AUC inside every region contributes +0.004 on average to transfer, and that mean carries an interval that spans zero under any resampling unit the design permits (pair-clustered $[-0.028, +0.036]$). What the design supports is not a small positive contribution but no contribution distinguishable from zero. Its sign is unstable across directions, and this too is invariant to the blocking. The paired deltas run from -0.148 to $+0.132$, twelve of twenty positive and eight negative, at either blocking scale. At 5 km blocking the block is CI-supported positive in five to six directions and CI-supported negative in three to four, with ten or eleven carrying no verdict, the range reflecting two boundary directions; at 1 km

blocking the split is ten, seven and three, itself decided by one bound of -0.00045 (Section 4.7g). It is also the swing factor at the chance line: at the point estimate it drags three directions below chance and lifts one above it (Table 7). Dynamic state predictors buy local skill at the cost of portability, and that cost is invisible to the diagnostics the field currently uses to anticipate it.

5.2. *Why the thermal increment is real but local*

The within-region increment is not an artefact to be explained away. It replicates in five independent regions. It survives coarsening of the spatial blocks to ~ 5 km with its interval intact, and its point estimate holds at ~ 10 km. It persists in the secondary all-valid population. It also strengthens rather than weakens when the predictor window is closed earlier (Section 4.7c). The transfer failure is therefore not evidence that the thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The mechanism is visible at the level of single predictors. In the taxonomy of Moreno-Torres et al. [28] this is concept shift. $P(y|x)$ is locally reparameterised, so the same predictor carries a different relationship to burning in different regions, and in the sharpest case an opposite one. Between Manavgat and Muğla, elevation discriminates burned cells in opposite directions with disjoint intervals (signed AUC 0.374 vs 0.611). All four channels describing the absolute pre-fire thermal state reverse as well, for example `current_1st_mean` at 0.538 vs 0.325. A classifier trained on one side of such a reversal does not merely lose skill on the other side. It inverts it. That is why raw transfer lands *below* chance rather than merely towards it. Six of twenty directions are below chance at the point estimate. Three to four of them carry

interval support at the conservative 5 km blocking, and six do at 1 km, with Evia to Bejís the borderline case at the coarser one. Below-chance transfer is the signature that distinguishes concept shift from ordinary covariate-shift degradation, which can only dilute skill towards 0.5.

Two refinements follow, and both are honest. First, in the Manavgat-Muğla pair the reversals concentrate in the absolute channels. The two channels referenced to a local baseline, `lst_anomaly_mean` and `tvdi_difference_mean`, keep a common direction. That is consistent with anomaly-referencing absorbing part of the between-region offset. The protection is not general, however. `lst_anomaly_mean` itself reverses with interval support between Bejís and Evia (Section 4.6b). TVDI reverses alongside the raw LST channels, even though it is internally normalised by construction against scene-fitted wet and dry edges [36]. That was the theoretical portability advantage flagged in Section 2.2, and it does not survive contact with the data.

TVDI’s own construction is a competing explanation for that particular reversal, and it should be stated rather than argued away. The wet and dry edges are percentiles of the LST values a given scene contains (Section 3.4), so they are refitted for every AOI and every window. A TVDI of 0.5 denotes whatever the middle of that AOI’s own dryness range happened to be, and the five AOIs differ by an order of magnitude in area and widely in relief. Two regions can therefore disagree about the sign of the TVDI-burning association without any difference in the underlying dryness-to-burning relationship, purely because the index is anchored to different populations in the two places. We therefore computed the analysis that separates the two explanations, and it does not support the index-artefact reading (Section 4.7k).

Two things were done. Sea water takes part in the edge fit and dominates two of the AOIs, so the edges were refitted on land pixels alone; and because that still leaves each region normalised against its own scene, one set of edges was then fitted once over the pooled land pixels of all five regions, so that a TVDI of 0.5 denotes the same dryness everywhere. The sea does move the index, by an amount that scales with each AOI's sea fraction, +0.017 in Evia and +0.016 in Muğla against +0.002 in inland Bejís. Neither re-run changes any region's direction. Under the common edge Manavgat still ranks burned cells by higher TVDI at 0.565, while Muğla and Evia still rank them by lower TVDI with intervals entirely below 0.5, at 0.341 and 0.378. Since the index's scene dependence was the competing explanation, and removing it leaves the disagreement intact, the reversal belongs to the dryness-to-burning relationship rather than to the normalisation. The honest reading is still narrower than the fully general one: statistical self-normalisation against locally fitted reference values does not guarantee a stable direction of association, and the TVDI reversal is consistent both with concept shift and with the index's own scene dependence. The reversals in the absolute LST channels, which carry no such normalisation, and the elevation reversal, which involves no thermal quantity at all, do not admit this explanation.

Second, the reversal mechanism is not exclusive to the thermal block. The single sharpest supported reversal belongs to elevation, a static predictor. The trade-off claim is about where the within-region gain and the between-region loss co-locate, which is the thermal block. It is not a claim that static predictors are immune to local reparameterisation.

There is an obvious objection to all of this, and it has to be met rather

than deflected. Every reversal cited so far is measured between *different places*. A sceptic can reasonably reply that the relationship was never one relationship. Manavgat and Muğla are distinct landscapes with distinct fuels, terrain and fire histories. On that reading, a predictor that means one thing in one region and another thing in the other is not evidence of instability. It is evidence that two different systems were compared.

The two Muğla events speak directly to this (Section 4.8). Region, AOI, analysis grid, feature registry and processing chain are identical, and the static predictors are identical cell by cell. The design holds place fixed; it does not hold the population fixed, because the 2022 arm is defined by removing the 2021 scar, and that qualification is developed in Section 4.8 and in Section 5.11(ii). Elevation's association with burning nevertheless reverses with bootstrap support. It is 0.611 [0.532, 0.690] in 2021, where higher ground burned preferentially, against 0.296 [0.230, 0.355] in 2022, where lower ground did. The intervals are disjoint and the difference is -0.317 [-0.414 , -0.220]. Holding geography fixed does not stabilise the direction of the relationship.

The mechanism is unusually visible here, and it is physical rather than statistical. The 2021 season burned as a dispersed complex of ten components across the region's full relief, from near sea level to 1,975 m. The 2022 event was a single compact scar confined below 777 m, with a median burned elevation of 187 m against 563 m the year before. The effective component count falls from 4.05 to 1.34, and the observed land-cover classes from seven to two. The two fires simply occupied different parts of the same elevation gradient. A model that learned "high ground burns" from the first would be

actively wrong about the second. This is what concept shift looks like when it can be seen. It is not a subtle distributional drift. It is two events sampling opposite ends of a topographic range that the region contains in full.

3170 This changes how a within-region score should be read, and the implication is worth stating plainly. If a relationship fitted on one event does not hold for another event in the same region, then a within-region ROC-AUC estimates interpolation inside one fire season's footprint. It is not an estimate of a stable susceptibility surface. The 2022 thermal model reaches a
3175 within-region AUC of 0.942, and it does so on 331 burned cells forming essentially one scar. That is the clearest illustration in this paper of how local such a score can be. It also sharpens the practical point of Section 5.10: a within-region AUC prices only half of a predictor block's contribution, and the half it does price is tied to the footprint it was fitted on. This rests on
3180 one region with two events, so it is an observation rather than an established general property.

Two constraints limit how far this carries, and the first is heavier than it first appears. The design is same-geography event-to-event and not clean temporal transfer. The 2022 fire ignites about five weeks earlier in the season, so year and seasonal phase are confounded and the difference cannot be
3185 attributed to elapsed time (Section 3.16.4). What the design does isolate is geography, which is precisely the variable the objection rests on. But seasonal phase is not a neutral nuisance here. The 2022 predictor window closes on 20 June and the 2021 window on 28 July, so a late-spring composite is
3190 being compared with a high-summer one, and in this landscape high ground in late spring is still cool, moist and effectively unburnable while by late July

it is not. A seasonal-phase contrast therefore predicts the observed direction of the elevation reversal, and it predicts the sign change in the absolute LST channels as well, since their baseline climatology is recomputed for the shifted calendar window. The two explanations, a genuine reparameterisation of the elevation-burning relationship and a seasonal-phase artefact, are not separated by this design. A calendar-matched 2022 arm would separate them and was not run; it is the single most valuable follow-up this paper can name and it is listed in Section 5.11. Until it exists, the comparison establishes that *something* reverses the elevation-burning association with geography, grid and pipeline held fixed, and it does not establish that the something is year rather than season. What the confound does not weaken is the narrower reading that the sign of the thermal block’s transfer contribution is a property of the pair rather than of the block, since that holds under either explanation. Second, only elevation reverses with interval support. The four absolute thermal channels move from bootstrap-supported *lower-values-burn* to *higher-values-burn* point estimates, and each shift is itself interval-supported. With 331 burned cells in 2022, however, every one of those intervals straddles 0.5. The 2022 direction is therefore not established, and the thermal reversals are not claimed as supported (Section 4.8).

Transfer between the two events completes the picture and sharpens the trade-off rather than softening it. With geography fixed, transfer no longer collapses. Both directions stay above chance, where six of twenty between-region directions fell below it at the point estimate. The thermal block’s contribution nevertheless *changes sign*. It is -0.082 $[-0.127, -0.040]$ carrying 2021 forward to 2022, and $+0.089$ $[+0.072, +0.104]$ carrying 2022 back

to 2021. Both signs are interval-supported, even though the same block is locally informative within each event separately (+0.116 in 2021, +0.078 in 2022). This is the trade-off at its most explicit. The six predictors that buy
3220 local skill in both events are not merely unhelpful across them; in one direction they subtract from a baseline that would otherwise have transferred at 0.642. Whether a dynamic-state block helps or harms on transfer is a property of the source-target pair, not of the block.

That elevation is the one supported reversal is itself worth noting, because
3225 it is the second time the same predictor has played this role: elevation also carries the sharpest bootstrap-supported reversal in the Manavgat-Muğla pair (signed AUC 0.374 against 0.611). Across a between-region contrast and a within-region between-event contrast, the predictor that most reliably fails to keep its direction is a static, perfectly measured, physically unam-
3230 biguous one. Elevation does not drift between regions and carries no sensor or compositing artefact; what changes is which part of the gradient a given fire occupies. This sharpens the diagnosis in a way that runs against the paper’s own emphasis: the portability problem is not a property of thermal predictors specifically, nor of noisy remote-sensing channels, but of the map-
3235 ping from any landscape variable to burning. The thermal block is where the trade-off is *costly*, because that is where the within-region gain sits; it is not where instability is *worst*.

5.3. Why every similarity-based diagnostic fails

Every diagnostic family that failed measures either where things burn
3240 or what the landscape looks like. None measures which way the response points. Marginal predictor-space measures describe $P(x)$. These are the

area-of-applicability family [24, 25, 18] together with climatic and geographic distance. Burned-niche overlap describes $P(x|y=1)$, through Schoener’s D [37], Warren’s I [44] and burned-centroid Mahalanobis distance. Fire-regime structure describes the spatial pattern of $P(y)$. All are blind to the quantity
3245 that failed here, which is the sign of the conditional association.

They are not, however, all label-free, and the contrast is often drawn too broadly. Only the marginal $P(x)$ family is computable before any target label exists. Burned-niche overlap is defined on the burned cells of both regions
3250 and fire-regime structure on the target’s burned map, so both consume the target’s labels exactly as the conditional index does (Section 3.14.4). This makes the comparison sharper. Niche overlap and the conditional index are built from the same labelled information about where fire happened, and only the one that reads the *direction* of each predictor’s association orders
3255 the transfer matrix. Describing burned cells is not the same as describing the response. It also leaves the practitioner in a worse position than the tidy version of this story would suggest: the single family that could be run before deployment is the marginal one, and it is the family that fails.

The domain classifier makes the blind spot concrete. Source and target
3260 cells are separable at AUC 0.962 to 0.9999 for every pair, so marginal shift is essentially total everywhere. A marginal instrument is therefore at ceiling and cannot discriminate transfer outcomes that range from 0.33 to 0.69.

The applicability numbers themselves say the same thing directly, and they settle a claim made in Section 1.3. Manavgat to Muğla places 0.8752 of
3265 the target’s cells inside the weighted area of applicability and 0.9640 inside the unweighted support. That is the highest weighted value in the matrix.

It transfers at 0.4702, below chance. The reverse direction sits at 0.5305 weighted and 0.9702 unweighted, the highest unweighted value in the matrix, and transfers at 0.4010. Both directions are below chance at the point estimate. Muğla to Manavgat is below chance with interval support at both blocking scales, while Manavgat to Muğla is supported at 1 km blocking and carries no verdict at 5 km. Muğla to Bejís runs the other way. It places 0.0050 of the target inside the weighted area of applicability, the lowest value in the matrix, and transfers above chance at 0.5832 with interval support at both blockings. A model can therefore sit almost entirely inside its nominal area of applicability and still perform below chance, and sit almost entirely outside it and still perform above chance.

Two limits attach to that statement. The marginal audit covers only the four-region subset, so Montiferru has no applicability value in any direction. The Bejís and Montiferru half of the contrast pair described next therefore cannot be given one. The two applicability quantities also disagree sharply with each other across the matrix, so any claim about applicability must say which of them is meant.

The contrast pair makes the same point through burned-niche overlap. Manavgat and Muğla have the most similar burned envelopes of any pair in the matrix (mean Schoener's D 0.826, closest burned centroids), and both directions are below chance, as the applicability figures above record. Bejís and Montiferru have the least similar envelopes (D 0.479, farthest centroids), and both directions are above chance, with interval support at 1 km blocking and no verdict at 5 km. Where the envelope agrees but the direction reverses, transfer fails. Where the envelope disagrees but the direction agrees, transfer

works. At pair level the two quantities are empirically distinct, and in partial rank correlations the conditional index retains its association with transfer when niche overlap is held fixed (+0.82) while niche overlap retains none in the reverse conditioning (−0.07).

This offers a measurable operationalisation of a concept the species-distribution-modelling transferability literature has discussed as "ecological stationarity" [46]: on this evidence, the stationarity that transfer requires is *direction agreement in $P(y/x)$* , not overlap of environmental envelopes. It is also consistent with the SDM findings that environmental and geographic similarity do not predict transfer success [41, 35], now reproduced in a fire application in which the most similar pair falls below chance in both directions. None of this is a criticism of the area-of-applicability construction, which does what it claims for the extrapolation failure mode it was designed for. It is a demonstration that a reassuring marginal diagnostic is not evidence of portability when the predictors are dynamic state variables.

5.4. What the conditional diagnostic is, and what it is not

The constructive result must be stated with its limits attached. The conditional index is the fraction of CI-supported features whose signed univariate association points the same way in source and target. It is the only diagnostic in the programme whose bootstrap interval excludes zero ($\rho = +0.84 [+0.58, +0.88]$; cosine variant $+0.81 [+0.33, +0.88]$).

It is **not label-free**. Signed associations require burned labels in *both* regions. As a pre-transfer instrument it therefore requires either an existing burned-area record or a labelled probe in the target. The marginal diagnostics it outperforms are all label-free, so the comparison is not like-for-like as

an operational tool. The defensible claim is mechanistic: *the failure that is invisible to marginal diagnostics is visible in the conditional structure*. The claim is not that a deployable label-free predictor of transfer has been found.

3320 It is also coarse, and it is undefined for one of the pairs the paper leans on hardest. The supported-feature restriction that gives the index its discrimination leaves six of eight pairs with denominators of one or two features, and the two Montiferru pairs drop out entirely. One of those two is Bejís and Montiferru, which supplies half of the contrast pair of Section 5.3: the
3325 lowest burned-niche overlap in the matrix, and transfer above chance in both directions. The diagnostic this section advances is therefore silent on exactly the pair that carries half of the paper's headline counterexample, and that should be read as a real gap rather than a technicality. The exclusion is not because Montiferru lacks supported features, since it has three. It arises
3330 from set-disjointness between the two regions' supported features, which is the more precise and more defensible statement of the coverage limit. The index is close to a binary flag for "any supported sign disagreement". The power caveat of Section 4.4 also applies to *every* correlation in the diagnostic table, including the two successes: the effective sample is ten unordered
3335 pairs, the two directions of a pair are not independent, and intervals of width ± 0.5 to 0.8 on the null rows cannot rule out moderate true correlations. The null diagnostics are "not shown to order transfer", not "shown not to". The conditional result is a strong ordering on a small pair set, not an established general law.

3340 Multiplicity is not controlled, and no control is claimed. Twenty variants were correlated against one target quantity, nineteen of them com-

putable, on an effective sample of ten unordered pairs. Two intervals excluded zero. No family-wise error correction is applied and none is offered (Section 3.14.1). Two mitigations are genuine, and neither substitutes for an error rate. The families and their variants were listed in advance rather than searched over. The conditional-versus-marginal contrast was a directional expectation stated before computation. The restriction to CI-supported features was pre-specified in the same way, and that matters, because the unrestricted versions did not clear the bar. The best of them, cosine over all nine features, reaches $\rho = +0.50$ with an interval that spans zero. Stating this openly is the defence. Concealing that the unrestricted variant failed would not be one.

A referee may object that the index is close to definitional, and the objection is fair. The model is built from these same features. If a strongly supported feature's association is inverted in the target, the model's ranking must invert with it, so a correlation between the index and transfer is not a discovery. That is precisely the point being made here. The diagnosis is mechanistic rather than predictive: it locates the failure in $P(y|x)$ rather than in $P(x)$. The non-trivial empirical content is that the *marginal* instruments do not see what the conditional one sees, on the same pairs. Nothing beyond that is claimed.

One competing explanation for that contrast has been excluded. The diagnostic rows were not all computed on the same directions. The marginal, applicability, climatic and geographic rows sit on twelve directions, the supported-conditional rows on sixteen, and the rest on twenty, so the marginal family carried the smallest sample by construction and its failure could have been

a power artefact. Every diagnostic was therefore recomputed on a common subset of twelve directions from six unordered pairs. The two supported-conditional rows remain the only ones whose intervals exclude zero, and
 3370 they are slightly stronger there, at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88]. All eighteen other rows span zero. Unequal sample size was not what separated the families. Removing the power confound does not remove the coarseness one: on that same subset the index still takes only three distinct values, and it still requires target labels.

3375 Two further bounds belong here, because they are the ones a sceptical reader would derive independently. The index’s tie structure caps the attainable 16-direction Spearman at +0.861, so the observed +0.840 sits on its own ceiling and the narrow published interval reflects heavy ties under pair resampling rather than precision. At the pair level the exact one-sided permutation
 3380 p is 0.0060, which is the smallest value this tie structure can produce, against a Bonferroni threshold of 0.0026 over the nineteen computed variants. No outcome of this diagnostic could have cleared a family-wise correction on this pair set, whatever the data had been. That is a statement about the design rather than about the result, and it is the reason this section describes a
 3385 mechanism diagnosis rather than a validated instrument (Section 4.4).

5.5. Interventions, and the conservation pattern that runs through them

Every intervention tested obeys the same accounting: what transfer gains, something else pays for.

Label-blind adaptation compresses rather than repairs. Region-wise standardisation and CORAL [38] raise the failing directions part-way towards
 3390 chance and pull the working directions down towards it; seven of twelve de-

composed directions show *negative* recovery, the worst (Evia→Manavgat) recovering -0.86 of the gap. This is what Section 2.4 predicted as a property of the methods: transformations defined on marginal statistics cannot
3395 restore a conditional relationship, and where the relationship has reversed they deliver better-aligned inputs to a decision rule fitted under the opposite one.

Pooled multi-region training does not escape the problem by averaging over it. The leave-one-region-out pooled thermal model never beats the best
3400 single-source pairwise transfer at the point estimate, with shortfalls of 0.02 to 0.22 . It sits 0.28 to 0.50 below the within-region ceiling. For two of five targets it is beaten raw by the pooled baseline, and for a third the two are within 0.006 . In effect, the pooled model resolves conflicting thermal directions by discounting the block that carries them.

3405 Feature removal is not a favourable trade, but its geometry confirms the diagnosis. Dropping the two supported reversal features costs -0.081 mean within-region AUC, supported in every region, and changes mean transfer AUC by $+0.014$, an estimate whose pair-clustered interval spans zero. The gains land exactly where the reversal analysis points. All three
3410 interval-supported gains involve the reversal partners. The mean delta over Manavgat-involved directions is $+0.025$ against -0.009 elsewhere. Removal of `lst_anomaly` contributes only in the one pair where it reverses with support. Knowing *which* features to drop for *which* pair, however, requires the target’s signed directions, that is to say target labels. Applied label-free, the
3415 same removal degrades the five aligned directions. The consistent lesson is the one the adaptation literature reached once the conditional component

was recognised as binding [39, 31]: a small number of target labels is the resource that label-free machinery cannot substitute for. A supervised few-shot recalibration analysis exists in the project diagnostics. Its headline is now in the main text at Section 4.10, its design and limits in Supplementary S1, and Section 5.10 uses it to say what the failure documented here costs to repair.

5.6. *The stated regime hypothesis, reported as it happened*

Section 2.5 offered fire-regime typology [2] as a candidate explanation for the concept shift. The idea was that regions in different limiting regimes should map dryness to burning differently. The expectation that fire-regime distance would *not* predict transfer was written down before the regime correlation was run and before the Evia-extended results were computed. It is an entry in the project’s analysis log rather than a formal pre-registration. No independent timestamped public registration exists, and the record entered version control on the day the diagnostics were computed (Section 3.14.1). The expectation was stated in advance, and that is the whole of the claim made for it. The data confirmed the null. They did so in a way that exposes the intuition behind the hypothesis as wrong, and that is reported plainly rather than as vindication. The regime-distance point estimate has the wrong sign ($\rho = +0.29$). The most regime-similar pair in the set is Bejís and Evia-extended, with effective burned-component counts of 1.0000 and 1.0083, which is as close to identical regime structure as the data allow. That pair fails in both directions with interval support, though the Evia to Bejís interval reaches the chance line at the conservative blocking. The most regime-different pair, Bejís-Muğla, transfers above chance. One

mundane explanation for Muğla’s part in that result can be set aside: Muğla has by far the largest analysis population, but cutting it to Manavgat’s cell count over twenty stratified repeats leaves its transfer behaviour where it was, in both roles and in seven of the eight direction-and-family combinations (Section 4.7j). What the size matching does move is Muğla’s absolute within-region AUC, by roughly 0.01 to 0.02, and the thermal increment survives it. The error was in the hypothesised grouping, not in the data. It was the assumption that Bejís’s single-compact-burn structure placed it in a regime class whose members would behave alike. With ten pairs this cannot refute regime typology as an explanation of concept shift in general, but our data offer it no support in its distance-based form, and the candidate explanation floated in Section 2.5 should be read accordingly.

5.7. *The meteorological-extremity explanation, tested and not supported*

Manavgat is the region whose behaviour most resists the account given above. It supplies the reversal partner in the sharpest contrast pair (Section 5.2), it is where feature removal buys almost all of its transfer gain (+0.025 mean delta over Manavgat-involved directions against -0.009 elsewhere, Section 5.5), and it is the target of the worst negative recovery under adaptation (Evia→Manavgat, -0.86 of the gap). The most natural post hoc explanation is meteorological: that Manavgat 2021 was an exceptionally extreme fire season, so that its thermal predictors were driven by a regional weather anomaly the other regions did not share, and the resulting mapping from dryness to burning was correspondingly idiosyncratic.

We tested that explanation and it was not supported. The ERA5-Land regional diagnostic (Sections 3.17, 4.9) characterises each region’s predictor

window against its own 2017 to 2020 climatology, and Manavgat is not the meteorologically extreme member of the set. Its predictor-window temperature sits $0.06\text{ }^{\circ}\text{C}$ *below* its climatological mean, while the other four regions run 0.31 to $1.11\text{ }^{\circ}\text{C}$ warm. It is the only region at or below its own baseline, and the departure is small enough that the honest reading is simply that Manavgat burned under climatologically ordinary temperatures. Its humidity deficit of 3.24% is mid-range among the five, its wind departure of $+0.07\text{ m s}^{-1}$ is the second smallest, and its precipitation total is within 1.4 mm of climatology, the smallest precipitation departure in the set. On none of the four variables is Manavgat the extreme member; on two it is the least anomalous. Whatever makes its transfer behaviour atypical, regional meteorological extremity in the predictor window is not it.

We report this as a failed prediction rather than as a result, and it carries the same status as the stated regime null of Section 5.6: an expectation on record, tested, and not borne out. The two failures are informative in the same limited way. They remove candidate explanations without supplying one.

The reader may reasonably ask why meteorology, once measured, is not simply added to the diagnostic set of Section 4.4 as a further measure of region similarity. It is not added because the candidate set was fixed before any diagnostic-versus-transfer correlation was computed. Twenty variants across four families were specified, and only two of them, both conditional, ordered the transfer matrix. Appending another measure after seeing the rest fail would be a search over the diagnostic space, and any correlation it returned on ten pairs would be uninterpretable. The measurement is reported

for what it is, namely a descriptive characterisation of the regions and a test of one specific explanation. It is kept out of the ordering analysis by construction.

3495 Two cautions attach to the reading of Section 4.9, and they are the reason
it reports physical units rather than standardised ones. The first concerns
the climatology. It spans four years, and the standard deviations it yields
differ between regions by factors of 2.7 to 6.0 in the predictor windows, so
a standardised anomaly measures a different physical departure in each re-
3500 gion and invites a cross-region comparison that the quantity cannot support
(Section 3.17). The failure mode is concrete rather than theoretical. Be-
jís’s label-window temperature reaches 5.7 standardised units on a physical
anomaly of $+0.83\text{ }^{\circ}\text{C}$. Its four reference years, 19.64, 19.95, 19.94 and 19.91
 $^{\circ}\text{C}$, agree to within a third of a degree and yield a climatological SD of 0.147
3505 $^{\circ}\text{C}$. Muğla’s predictor-window wind speed reaches 5.3 units on $+0.34\text{ m s}^{-1}$
over an SD of 0.065 m s^{-1} . That these are artefacts of a near-degenerate
denominator rather than genuine extremes is settled by comparison: Evia’s
label window closes on the same calendar day as Bejís’s and is twelve days
longer, so any seasonal-composition explanation would apply to it at least as
3510 strongly, yet its comparable $+0.67\text{ }^{\circ}\text{C}$ anomaly yields 1.9 standardised units
against an SD of $0.351\text{ }^{\circ}\text{C}$. The physical anomalies of the two regions are
similar; only their denominators differ.

The second caution is that the label window is not fire weather: it opens
on the ignition date and runs 35 to 59 days into the autumn rains, so it
3515 describes conditions during and after the fire rather than those that preceded
it. Only predictor-window values are used anywhere in this paper, and the

label-window figures quoted immediately above serve solely to demonstrate the instability of the standardised scale.

5.8. *The pre-fire signal is not an early-fire artefact*

3520 The most direct threat to everything above is the possibility that the "pre-fire" thermal composite is contaminated by early fire signal, since each region's predictor window closes one day before its label window opens. The window-closure sensitivity (Section 4.7c) addresses this head-on. Both ends of the predictor window are shifted 7 and 14 days earlier, so that the compos-
3525 ite ends one and two weeks before any labelled burning. The thermal increment stays bootstrap-supported in every region and every variant. Nowhere does the increment shrink towards zero. The within-region signal is therefore a genuine pre-fire dryness signal, not a leakage of the fire itself into the predictors.

3530 One observation from this analysis needs reporting: in Manavgat and Bejís the increment *increases* with earlier closure ($0.074 \rightarrow 0.101$ and $0.094; 0.058 \rightarrow 0.077$ and 0.079). This is unexpected in direction, since contamination by early fire signal would predict the opposite. A mechanism is available and the window-closure report states it. The closure date is shifted against
3535 a fixed production policy, so moving the window changes which acquisition dates enter the median composite and how many clear observations back each pixel. With about seven usable dates in a window, a 7 or 14 day shift adds or removes one or two of them. The evidence fits: over the same comparison the baseline moves by $+0.002$ to $+0.003$ while the thermal family moves by
3540 $+0.022$ ROC-AUC, so the shift acts almost entirely on the thermal channels. That is the same dependence the compositing A/B of Section 4.7i finds by

a different route, where three defensible weighting choices move Manavgat’s increment between +0.045 and +0.084 on an identical cohort. The pre-fire claim is unaffected, and the direction of the effect remains the safe one for it. What this does say is that part of the increment’s magnitude is a function of how the composite was built rather than of dryness alone, and Section 5.11 records the resulting tolerance.

5.9. *The empirical contrast with Dimarco et al. (2026)*

Dimarco et al. [8] and this study form a near-controlled contrast conducted by two independent groups: Mediterranean regions, ~ 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an explicit ordered-pair transfer matrix. Their predictors are spatially stationary attributes of place: terrain, human modification, night-time lights, population density and ERA5-Land long-term climatologies. Every transfer they report exceeds AUC 0.80. Our predictors describe the dynamic state of a particular pre-fire window, and transfer collapses to chance or below. Read together, the two studies are consistent with a predictor-class explanation: stationary-attribute models can learn a spatial ordering of susceptibility that a neighbouring region inherits. Dynamic-state models additionally encode how a given degree of anomalous dryness translates into burning *there, then*, and that mapping is what fails to travel. WildfireGenome’s county-level matrix [17], with its mixture of strong and collapsed transfers, sits between the two poles.

The contrast must not be overdrawn, and three of the bounding differences are stated in Section 2.5: their target is an ignition proxy evaluated against a 1:1 balanced background, ours is burned-area classification at the

true, heavily imbalanced base rate. These are different problems with different achievable ceilings. Their temperature variable is a static reanalysis climatology, ours are event-specific satellite observations referenced to their own baselines. They apply no adaptation, whereas adaptation is central to our diagnosis. The comparison is between two coherent experimental programmes, not an ablation.

A fourth bounding difference is the sharpest, and it must be stated because it bears directly on the reading above. Our own baseline is predominantly stationary, and it does not transfer well either. Across the twenty directions it averages ROC-AUC 0.5371. Four directions are below chance at the point estimate, and at 1 km blocking all four have intervals lying entirely below it. Set against every Dimarco transfer exceeding 0.80, that is not a stationary-versus-dynamic contrast in the abstract. Our baseline carries four predictors of terrain, fuel and greenness. Theirs adds human modification, night-time lights, population density and long-term reanalysis climatologies. The contrast is therefore a thin place-descriptor set plus a dynamic block against a rich place-descriptor set. The predictor-class reading survives, but it survives as a reading of two programmes that differ in predictor richness as well as in predictor class. What the pair of studies supports is that portability may track predictor class, and that the class carrying our within-region gain is the class that fails to port. A stationary predictor set is not on its own sufficient for portability, as our own baseline shows.

5.10. Implications

For practice, the immediate implication concerns regional and "global" fire-susceptibility products. A within-region AUC prices only half of a pre-

dictor block's contribution, even when it is spatially blocked and honestly computed. The portability it consumes appears on no ledger unless transfer is measured. Our results say that for dynamic thermal predictors this cost can be total, and that neither predictor-space applicability screening [24, 18] nor label-free alignment will reveal or repair it. Transferability must be measured, not assumed from regional, climatic or regime similarity. The measurement itself is sensitive to evaluation design [45], down to the library version (Section 4.7e).

What should a practitioner do instead? The results support three concrete answers, and the third is the one this paper adds. First, screen with the right instrument. A marginal applicability check answers "is this target inside the training distribution?" and that question was not the binding one here: Manavgat to Muğla sits at 0.875 weighted applicability and transfers at 0.470, below chance. Where labels exist on both sides, the signed-association comparison of Section 3.14.4 tracked transfer where twenty other candidates did not, so it is the screen to run, with the caveats of Section 5.4 attached. Second, do not spend effort on label-free alignment. Two standard methods, applied carefully, moved 14 of the 20 directions towards chance rather than towards skill, and every direction they improved involves Montiferru, the smallest region in the set. Third, price the labels instead. Section 4.10 gives the shape of that price for three regions: thirty-two labelled 5 km blocks recovered 85 to 89 % of the target's own ceiling in three of six directions, two of them from starting points below chance. That is a real answer to "what do we do", and it is not a cheap one. The same budget is 7 to 20 % of the target's natural-vegetation population, it recovers only 51 to 57 % where the

concept gap is widest and 30 % in the sixth direction, and at small budgets it damages the direction that transfers best without any labels. No general label budget follows from six directions in three regions, and none is claimed
3620 here. What follows is the form of the answer: transfer failure of this kind is bought back with target labels, at a price that scales with the size of the conditional gap, and a campaign that cannot afford the labels should not deploy the model.

For method development, the results point to two resources. The first is
3625 labels, as above: the conditional diagnostic requires a labelled probe in the target, the interventions of Section 4.6 show that knowing where the reversals are is what converts the diagnosis into an action, and Section 4.10 shows what buying that knowledge costs. Only target labels provide it. The second is predictors normalised physically rather than statistically: TVDI’s scene-
3630 internal normalisation and the climatological anomaly referencing both failed to guarantee stable direction, suggesting that portability requires variables whose mapping to fire-relevant state (for example, actual fuel moisture rather than its thermal proxy [47, 23]) is invariant by construction rather than by rescaling. The self-calibrating satellite thermal digital twin that motivated
3635 this project remains future work, and on this evidence its calibration loop will need labelled feedback, not unsupervised alignment.

5.11. *Limitations*

The transfer failure is a finding, not a limitation. The limitations are the boundaries on how far it generalises. (i) No meteorological covariates
3640 (wind, humidity, precipitation) enter the models, so we cannot say how the trade-off behaves for a mixed thermal-plus-weather predictor set. The ERA5-

Land diagnostic of Sections 3.17 and 4.9 characterises the regions but is not a predictor and does not close this gap, and its own four-year climatology limits how firmly its anomalies can be read. (ii) Temporal transfer is measured for one region only, Muğla, and even there year and seasonal phase are confounded by the 2022 event’s roughly five-week-earlier ignition, so the design is same-geography event-to-event rather than clean temporal transfer (Section 3.16.4). Its 331 burned cells also leave the thermal direction reversals unresolved at interval level. The pair holds place fixed but not population: the 2022 arm is the 2021 arm with the 2021 scar removed, so the two share 38,789 of 38,790 cells and three identical static predictors, and in the 2022 to 2021 direction every target positive lies outside the source training population while almost every target negative lies inside it (Section 4.8). We cannot bound what that does to the two reported numbers, so they are not comparable to the 20 between-region directions and the elevation reversal has a structural competing explanation this design cannot exclude. Those two arms are additionally the only transfer directions in this paper computed by us rather than read from the pipeline author’s frozen export, albeit with his unmodified code and the same pinned environment (Section 3.16.4). No other region has a second event. (iii) All labels derive from a single burned-area product, MCD64A1 [10], whose omission and commission characteristics [3] bound every model evaluated here. (iv) The analysis cells approximate, but are not co-registered with, the native MODIS sinusoidal grid (Section 3.2). They are also not square on the ground: about 510 m north to south but 390 to 407 m east to west, giving 0.199 to 0.208 km² against a MODIS cell’s 0.250 km². Two things follow. The block sizes

quoted throughout as about 1, 5 and 10 km are the north-south dimension,
 so blocking is weaker in longitude by roughly a fifth. And the smaller, off-
 set cell dilates the labelled burned footprint relative to MCD64A1, which
 compounds (xiv). (v) Even after the AOI extension, Evia's TSG prevalence
 (0.287) remains the highest of the five regions. It is four to seven times that
 of Manavgat, Bejís and Muğla (0.038 to 0.072), though only modestly above
 Montiferru, whose modelled-population prevalence is 0.212. Prevalence sen-
 sitivity was checked for transfer (Section 4.7a), but Evia remains the most
 imbalance-atypical population. (vi) Each region contributes one fire season,
 so regional concept shift is confounded with event meteorology; distinguish-
 ing them requires multi-year labels. (vii) Cross-region point estimates carry
 an implementation tolerance of roughly ± 0.02 to 0.03 across scikit-learn ver-
 sions (Section 4.7e). All reported numbers are fixed to one verified version,
 but exact reproduction elsewhere requires the archived environment. (viii)
 The diagnostic correlations rest on an effective sample of ten region pairs.
 Both the successes and the failures of Section 4.4 should be read at that
 power. (ix) Manavgat's atypical transfer behaviour remains unexplained. It
 is the region where the conditional diagnosis bites hardest and where fea-
 ture removal recovers most. The one explanation we were able to test, that
 its predictor window was meteorologically extreme, is not supported (Sec-
 tion 5.7). We can say what does not account for it, but we cannot say what
 does. With one fire season per region, the candidates that remain are not
 separable in this design: fuel structure, ignition and suppression history, and
 terrain-driven fire behaviour. (x) The interval-support counts are less stable
 than the point estimates behind them. Several verdicts sit within a thou-

sandth of their reference value. At 1 km blocking the published split of ten positive, seven negative and three uncertain paired deltas turns on a lower bound of -0.00045 for Bejís to Manavgat, and a different random stream gives eleven, seven and two. The point estimates and the sign pattern are stable; the counts are not. Every sentence in this paper that leans on an exact count of supported directions should be read at that precision.

Five further limitations concern the observational layer, and they were added after an internal review found the paper strong on modelling and thin on provenance. (xi) **The within-region increment carries a compositing tolerance.** Three defensible Landsat compositing chains, run on an identical Manavgat cohort with an identical baseline, give increments of $+0.045$, $+0.064$ and $+0.084$ (Section 4.7i). The tolerance is about ± 0.02 AUC, comparable to the cross-version tolerance of (vii) and wider than the 1 km interval of Table 4. It never approaches the increment itself, which stays supported under all three chains, and it was audited for one region only; the other four are unaudited on this axis. The same dependence explains the window-closure behaviour of Section 5.8. (xii) **The thermal block is six predictors but not six independent measurements.** `fused_1st` equals observed LST outside a gap-filled share of 0.11 % to 9.70 %, `downscaled_1st` is a fitted surface whose dominant input is the MODIS context layer in two regions, NDVI in two and slope in Muğla, and a single subgroup recovers 71 % to 99 % of the whole block's increment in every region (Section 4.7h). The six reversal tests of Section 4.4 are therefore not six independent probes of concept shift. Two related concerns have since been tested and neither survives as a threat to the increment. Restricting to cells with a gap-filled

fraction of at most 0.10 leaves the increment bootstrap-supported in all five regions; Bejís moves the most, from +0.056 to +0.043, and the other four by at most 0.008 (Section 4.7m). And dropping the two derived channels, which carry a coordinate-derived component from the downscaler’s own inputs (summed importance 0.035 to 0.123) and are the one route by which coordinates re-enter a feature set from which Section 3.13 excludes them, retains 82 % to 103 % of the increment, with every interval still excluding zero (Section 4.7h). What remains true is the redundancy itself: the block is six predictors but not six independent measurements. (xiii) **The MODIS input behind the two derived channels is quality-screened in two regions and unscreened in three.** Evia and Montiferru apply a QC_Day mask, a three-observation minimum and an explicit nodata sentinel. Manavgat, Bejís and Muğla apply none of these, so cells with nothing to report enter as exact 0.0 °C. Those zeros were measured for this paper: 8.11 % of pixels in Manavgat, **38.29 % in Muğla** and **none in Bejís**, each tracking that AOI’s water share almost exactly, so the zero-fill is sea rather than missing observation (Section 3.4). Muğla’s share is more than four times Manavgat’s and both exceed the pipeline’s own 5 % guard. The split follows export date, not design, so it is confounded with nothing in the study and with everything about when each region was run. Three of the five regions are on the unscreened path, Manavgat among them, and Manavgat is the subject of the unexplained behaviour in (ix). This is a candidate explanation that we have not been able to test, since rebuilding Step 7 on one common contract is upstream of this analysis. An earlier version of this paper attributed the anomaly instead to Manavgat using a four-year summer-mean MODIS layer.

That was read from a stale metadata string and is withdrawn; all five regions used single-season predictor-window layers (Section 3.4). (xiv) **Label noise is spatially structured and its interaction with the predictors is untested.** The cell’s representative burn date is the mode of its positive sub-pixel dates, and because the exported raster holds no out-of-window positives, a single in-window positive sub-pixel labels a cell burned in this dataset (Section 3.2). No agreement threshold is imposed, so the proportion of cells labelled on thin evidence scales with each scar’s perimeter-to-area ratio, which differs sharply across these regions (effective component counts 1.0 to 4.05). Fringe cells also differ systematically in terrain from core cells, which makes differential fringe contamination a competing explanation for a reversal in a terrain variable. The `burn_date_pixel_agreement_fraction` column needed to test this is recorded and hard-excluded from the features, and the sensitivity restricting to high-agreement cells was not run. At the product level, MCD64A1’s omission at 500 m in fragmented Mediterranean terrain is itself correlated with patch size, terrain and land-cover fragmentation, which are the baseline predictors; no second label product was used as a control. (xv) **Two safeguards did not run everywhere.** The pre-label burn exclusion ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat, and has no recorded status for Bejís (Section 3.2). Prior-year burning is screened for no region in the five-region cohort; the only historical-burn exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.16.4. The exposure this leaves cannot be quantified from the archive, and an earlier version of this paper wrongly described doing so as a cheap query. It is not: both the working and

the raw MCD64A1 rasters in the export are clipped to each region's label window, with zero positive sub-pixels anywhere in any predictor window, so a pre-label detection leaves no trace to count. Measuring the exposure needs
3770 MCD64A1 re-exported unclipped, which is an upstream re-run. In the same vein, the per-region acquisition inventory behind the composites exists for Manavgat alone.

Four analyses were named in an earlier version of this section as the ones that would close most of the above. **Two have since been run and**
3775 **are reported here rather than promised.** The common-edge TVDI, fitted once across the pooled land pixels of all five regions, does not bring the directions into agreement, so the index's scene dependence is not the explanation for its reversal (Section 4.7k). The signed univariate AUCs re-computed on high-agreement cells only leave the elevation reversal intact at
3780 every threshold, so differential fringe contamination is not the explanation either (Section 4.7l). Two remain outstanding and both need work upstream of this analysis: the compositing A/B extended to a second region, and a calendar-matched Muğla 2022 arm to separate year from season (Section 5.2), which is the more valuable of the two because it addresses the one confound
3785 the two-event control cannot resolve from the existing export.

6. Conclusions

Pre-fire thermal dryness adds a real and repeatable increment to burned-area discrimination. Across five Mediterranean regions, ROC-AUC was raised by +0.06 to +0.15 over a static and near-static terrain, fuel and greenness
3790 baseline. That baseline's only time-varying member is the vegetation-index

composite. The gain survived spatial blocking at about 5 km, the coarsest scale this design supports as an interval. It also survived a predictor window closed up to two weeks before the first labelled burning.

That skill is local. Paired per direction, the same predictor block adds
3795 nothing distinguishable from zero to cross-region transfer. The mean is
+0.004 over twenty directions and its interval spans zero under every re-
sampling unit the design permits. Its contribution changes sign from one
direction to another. The paired deltas run from -0.148 to $+0.132$, twelve
positive and eight negative, at either blocking scale. At the conservative 5
3800 km blocking, five to six directions are helped and three to four are harmed
with interval support, with the rest carrying no verdict. The block is also
the swing factor at the chance line. Removing the two reversing predictors
costs -0.081 of mean within-region skill, with interval support in every re-
gion, and changes mean transfer by $+0.014$, an estimate whose interval spans
3805 zero. The cost of the trade-off is measured; the return is not.

The failure is conditional. The direction of the link between dryness
and burning changes from one region to another. None of the similarity
diagnostics tested here ordered the transfer matrix: predictor-space distance,
domain separability, niche overlap and regime structure all failed. On ten
3810 effective pairs those null results mean not shown to order transfer, rather
than shown not to. It is also why label-blind adaptation compresses transfer
towards chance instead of repairing it, and why pooled multi-region training
does not escape it.

The practical implication is a change in what is checked before a dynamic-
3815 state fire model is transferred. The question is not whether the target region

lies inside the source’s environmental envelope. In our matrix, the pair with the highest envelope overlap failed in both directions, and the pair with the lowest overlap transferred in both. Both statements are made at the point estimate. The question is whether the signed feature-response directions
3820 agree.

That check requires a labelled probe in the target region, and so, on inspection, do the niche and regime diagnostics. Only the marginal family can be computed before any target label exists, and it is the family that failed. How large the probe has to be is not established here, so no label
3825 budget is claimed. A signed association with a usable interval may need fewer labels than a refitted model, but that was not tested. What is clear is that the supervised recalibration that label-free alignment cannot deliver is not cheap. Section 4.10 prices it for three regions: thirty-two labelled 5 km blocks recovered 85 to 89 % of the target ceiling in three of six directions, 51
3830 to 57 % in two more and 30 % in the sixth, and at its top budget the labelled set already holds most of one target region’s burned cells. Where no target labels exist, the transfer performance of such models should be treated as unknown rather than inferred from similarity.

For model builders, the trade-off should be priced openly. A predictor
3835 block that buys large within-region skill can carry a transfer cost of the opposite sign. Reporting only within-region validation hides that cost completely. Three extensions are left for future work: temporal transfer, meteorological covariates, and physically normalised dryness variables. Each is a step towards the self-calibrating thermal monitoring system that motivated this
3840 study.

CRediT authorship contribution statement

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val-
3845 idation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All satellite inputs are public and are obtained through Google Earth
3855 Engine. The processing pipeline, its configuration and the frozen numeric outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods, together with the three components that fall outside that release. No digital object identifier is minted and no archival deposit exists.

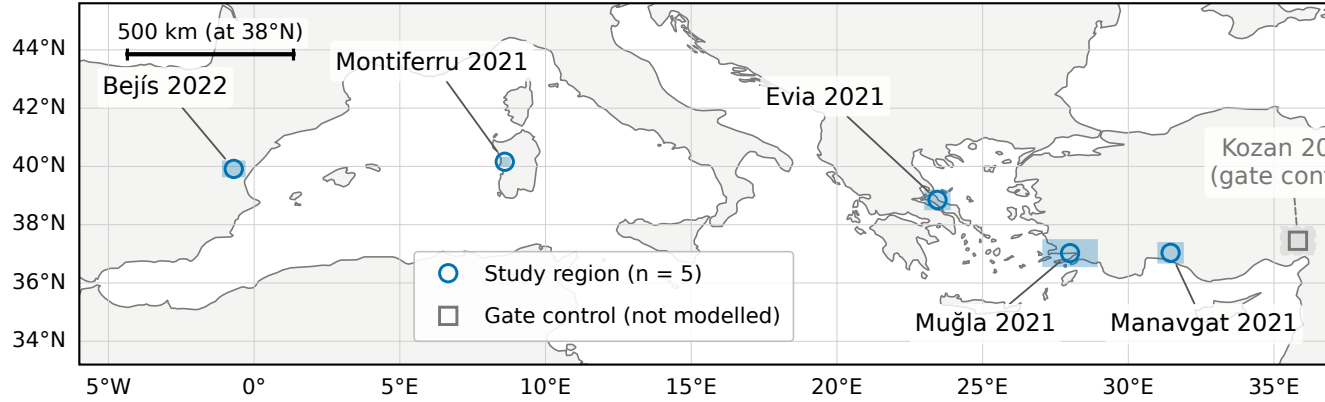


Figure 1: **Study regions and the gate negative control.** Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

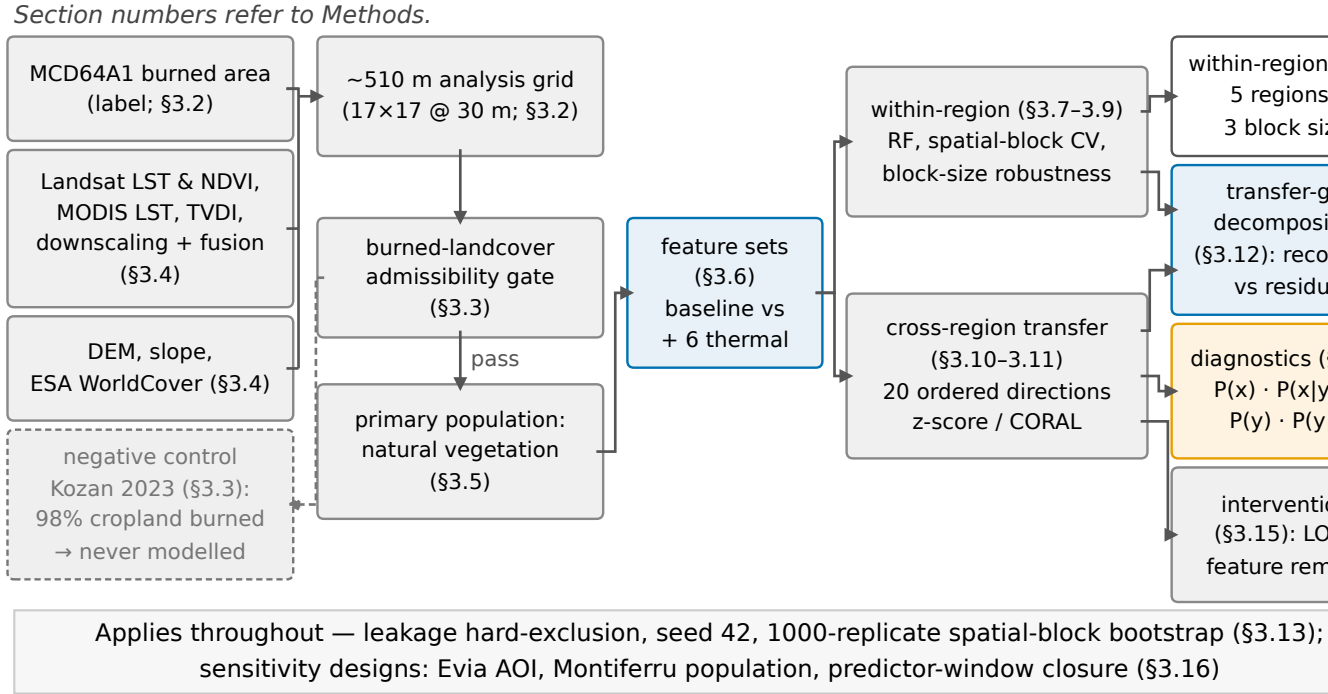


Figure 2: **Processing pipeline and evaluation programme.** Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

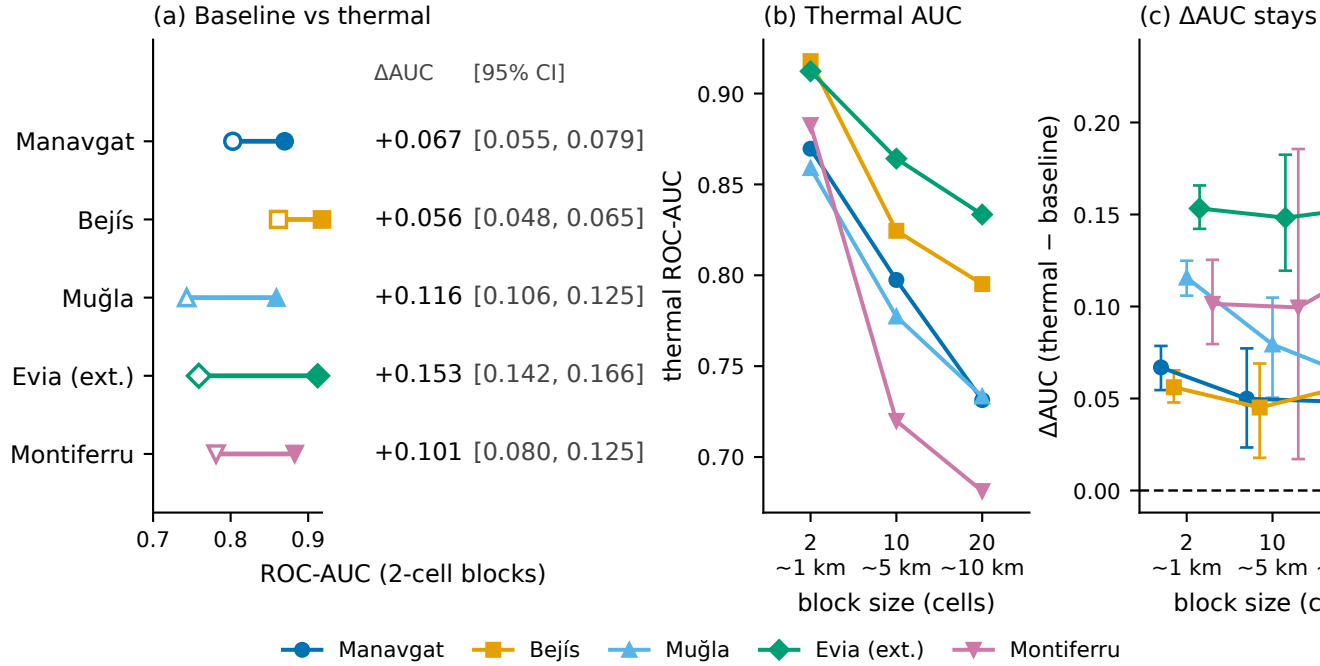


Figure 3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, $\approx 1, 5$ and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 4, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

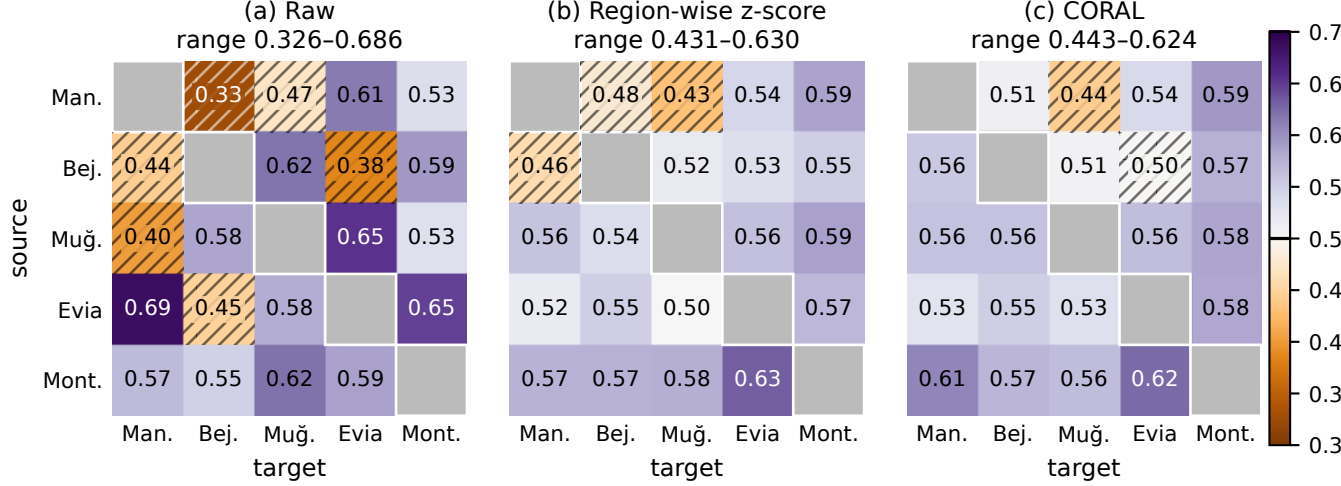


Figure 4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

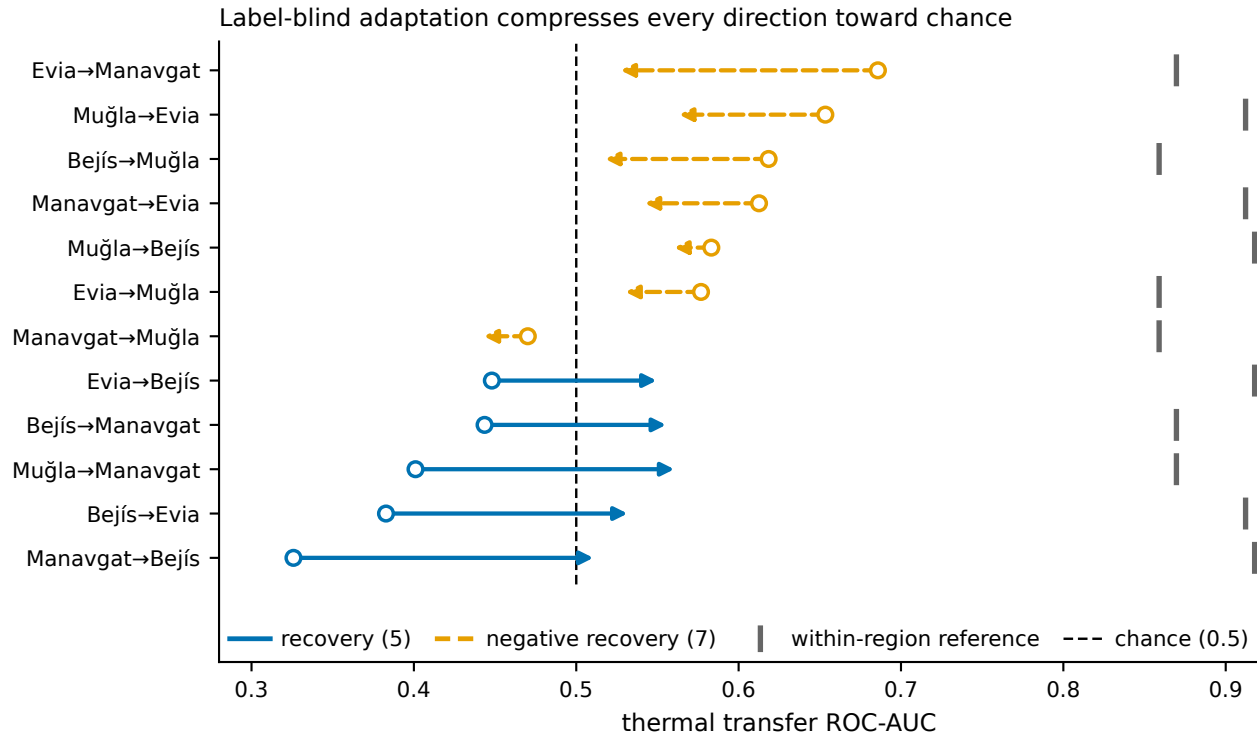


Figure 5: **Label-blind adaptation compresses transfer toward chance rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

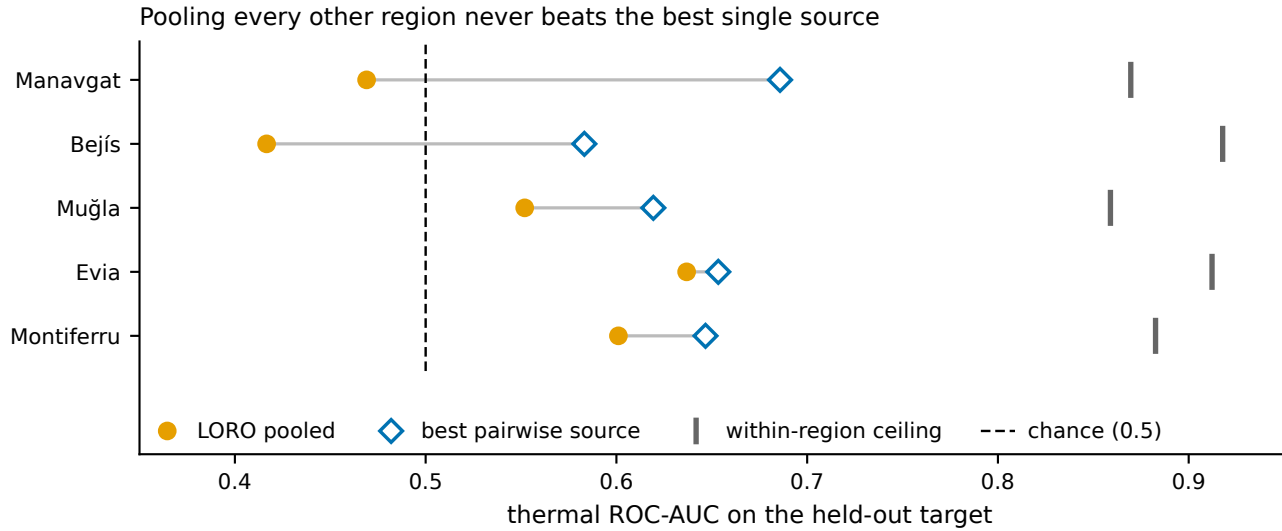


Figure 6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

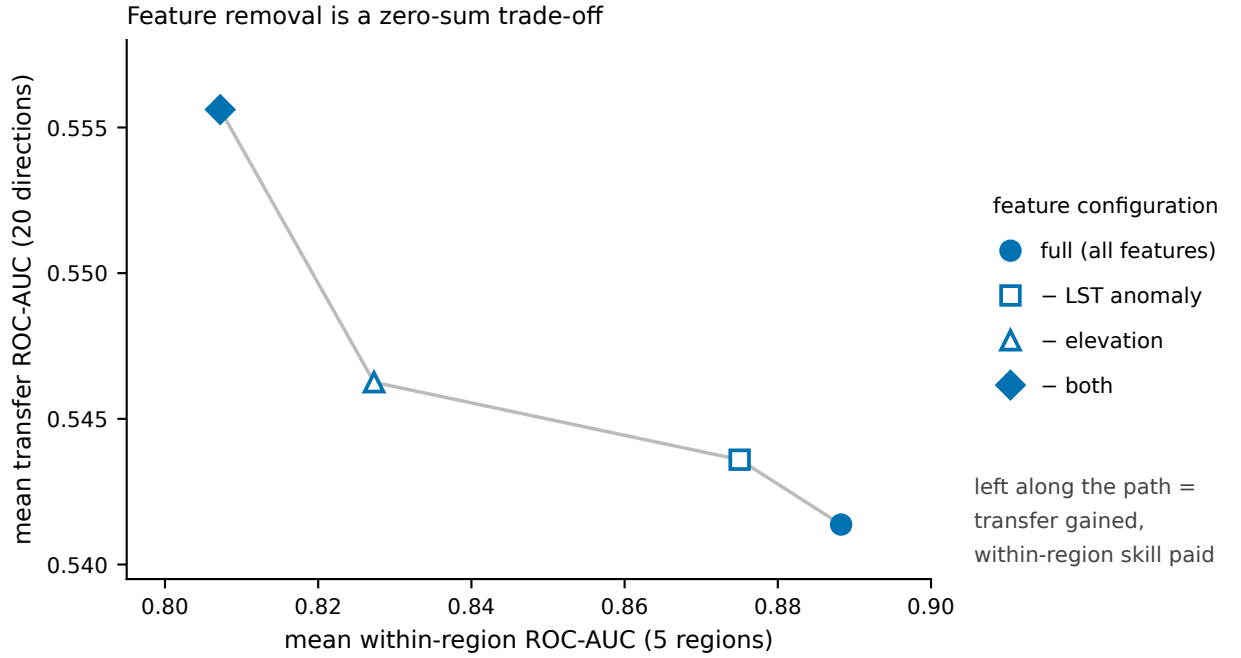


Figure 7: **Removing the direction-reversing features is a zero-sum trade-off.** Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically (0.541 $\rightarrow$ 0.556) while within-region skill falls monotonically (0.888 $\rightarrow$ 0.807). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The transfer gain is 0.014 AUC and the within-region cost 0.081 AUC, about six times larger: what is removed is locally informative and non-portable, and removing it buys portability by giving up skill rather than by finding transferable signal.

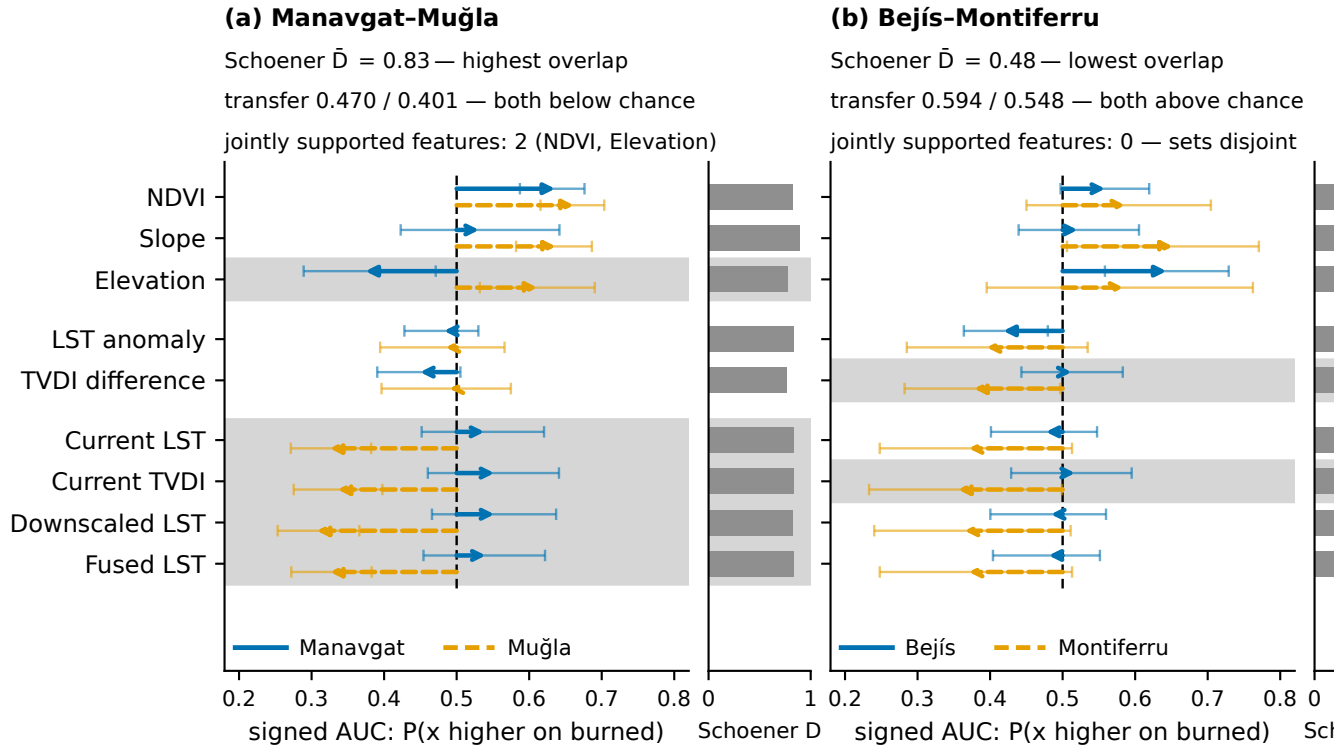


Figure 8: **Burned-niche overlap does not determine transfer: the contrast pair.**

Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

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